

THE DEFENSIVE ARMOUR
AND THE WEAPONS
AND ENGINES OF WAR OF
MEDIÆVAL TIMES, AND OF
THE "RENAISSANCE."

BY
ROBERT COLTMAN CLEPHAN



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1900.

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Fig. 1.—Transitional Gothic Suit at Munich.

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BY

ROBERT COLTMAN CLEPHAN,

OF SOUTHDENE TOWER, GATESHEAD.

With 51 Illustrations from Specimens in his own and in other
English Collections, and also from others in some of
the Great Collections of Europe.

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1900.



PREFACE.

THIS volume has grown out of some “notes” printed in the *Archæologia Æliana* in 1898, and added to as any new facts and lights presented themselves to me. The text is compressed as much as possible, with a view to publishing at a moderate cost; and as a more general interest in arms and armour is decidedly growing, I venture to hope that this volume, however imperfect, may supply a want, and that it does not contain too many manifest errors and inaccuracies. The subject is treated chronologically, and no further detail entered into than seemed necessary for presenting it in a consecutive and concrete form.

All students, myself among the number, owe much to those experts whose original research and delineation of nice points of detail go to make history in the several branches of my subject, and it is to be regretted that more of them do not undertake further comprehensive work.

Defensive armour is the section I am most conversant with, and it is perhaps the one affording the most concrete materials for chronological classification and analysis.

The question of the weapons of the “middle ages” and of the “renaissance,” their chronology, description and classification, is far from being in a satisfactory state. There are no books dealing with the subject as a whole, and many of the “notes” and “papers” I have seen spread over many years are, most of them, very sectional and fragmentary in their scope and character. Technical terms vary exceedingly among the different writers, and some more generally intelligible codification is very desirable. International it cannot be, as Germany naturally has her own terms, while those of England are perhaps as necessarily mixed up with Norman-French.

There are often great difficulties in the way of reasonably approximating the date and nationality of both weapons and armour, owing to causes which will be touched upon later in these pages; but these

apparent inconsistencies must needs be grappled with as far as possible, and herein lies the work of the archæologist. In the case of sword specimens, it very often happens that blades and hilts belong to widely different periods, and even nationalities, and cases of this kind often give rise to much doubt and perplexity; indeed, unless there is evidence that a blade and hilt are contemporaneous, it is always well to consider that they may not be so; for blades were passed down from father to son, and often re-hilted more than once. Hilts also were often re-bladed.

The great question of smiths' marks could only be adequately dealt with in a volume devoted entirely to that subject. This will be seen from the complexity arising from the piracy of marks—such, for instance, as that of the running wolf of Passau, or Scottish blades with the many variations of “Ferrara” impressed upon them. These marks came to be regarded merely as “standards,” and were often used without any intention to defraud—in the sense, in fact, of the name “Wallsend” being applied to express a certain quality in coals. A book dealing comprehensively with this branch of the subject has yet to be written.

While gratefully acknowledging much information and infinite assistance from other writers, I have found many manifest errors, which have been both inherited and perpetuated, handed down, so to speak, through long generations of book-making. I have taken as little as possible from books, especially over the period where actual specimens are available, but have endeavoured, by carefully studying many important collections, both at home and abroad, to compare, as far as possible, the types and fashions prevailing at the different periods dealt with, which, however, greatly interweave, especially among European nations, where easy facilities for interchange existed.

It takes many years and opportunities of study to achieve much in the direction of judging armour, and it is only by a close comparison, not merely of individual pieces, with a careful examination of every detail, but also a knowledge of the makes of steel of the various ages covered, their composition, manipulation, and relative degrees of hardness, that a reasonable amount of certainty can be arrived at. Much ingenuity has been applied in faking up and partially restoring many suits, still, it is obvious to an expert, in most instances, which pieces are of comparatively modern construction, especially in the cases where ornamentation has been applied,

for here the best work of the “renaissance” cannot be adequately reproduced. Many suits, even in national collections, are not only doubtful, but now known to be spurious, while in others the restoration process has left much to be desired. The uninitiated would be surprised if they knew how comparatively few suits are absolutely homogeneous, so many having been set up by dealers, often more or less of pieces of various types and dates.

It is most interesting to trace what may be termed the evolution of arms and armour, and to follow the craft and ingenuity of the armour-smith as pitted against that of the makers of weapons; indeed, all through the history of the armour period this contest has proceeded with varying fortune. Fashion also has always been a potent and arbitrary factor in the direction of change, and hence so many preposterous departures, such as both the extravagantly long and ridiculously wide sollerets of the “Gothic” and “Maximilian” periods respectively. Expansive skirts of steel, which must have been very crippling to the wearers, were used at one time by all cavaliers who had the least pretensions to be considered *à la mode*.

At the risk of the general subject occasionally overlapping, and of some repetition in matters of historical retrospection, I have concluded to divide these pages into two main sections, viz., “Defensive Armour” and “Weapons of War” over the period set forth in the title-page. This has been done in the interests of conciseness and perspicuity, and more especially with a view to an easy reference to any branch of the subject under discussion.

ROBERT COLTMAN CLEPHAN.

SOUTHdene Tower, GATESHEAD,
March, 1900.

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SECTION I.

DEFENSIVE ARMOUR.

PART I.

INTRODUCTORY AND GENERAL.

THE phrases, “the Stone,” “Bronze,” and “Iron Ages” are mere generalizations fast losing their significance, and the purposes of this volume will not permit of any special disquisition on the weapons of these mixed and merging classifications of periods, or even those recorded of the Egyptian, Etruscan, Greek, Roman, and Eastern peoples; beyond what, in some instances, may seem necessary for showing any prototypes or analogies of arms or armour in use during the “Middle Ages” and the “Renaissance.”

The more remote ages of Egypt would have been a blank to us but for the character of the tombs, which preserved so wonderfully the papyri and frescoes we find so valuable, and, above all, the inscriptions and bas-reliefs on stone, affording infinite information concerning the arms of this ancient people and their martial achievements; indeed, we really know more of the weapons of the ancient Egyptians, and even those of the times of Hesiod, Homer, and Cambyzes, than we do of those of the Goths, Vandals, Huns, and Ancient Britons during the centuries immediately following on the final evacuation of Britain by the Roman legions. The vigorous races that had been vanquished by imperial Rome, and those that in their turn had invaded and conquered Italy, inherited much from the earlier Roman wars and domination, more than has been thoroughly understood by historians of the nebulous centuries partly preceding and closely following on the final overthrow of the Western Empire; and the Romans had already gathered together many of the forms of the nations and empires that had preceded them, to say nothing of adaptations from the armament of contemporary tribes and peoples; still, in the main, the Romans had imposed their own methods and civilisation on all the nations they conquered. On a monument

recently brought to light by M. de Morgan at Susa, erected by Naram-Sin about B.C. 3750, is a figure of the king wearing a horned helm, and armed with an arrow in his right hand and a bow in his left; a dagger is thrust into his girdle.

The granite sculptures of Persepolis show the weapons of the Assyrians to have been mainly those perpetuated for many ages and under many degrees of civilisation—viz., the sword, the lance or javelin, the sling and the bow; and in the rusty fragments of solidified iron rings in the British Museum, found at Nineveh, we see the ancestor of the Roman lorica, the bright byrnie of the “Sagas,” and hauberk of the “middle ages.” The same monumental inscriptions clearly indicate to which ancient people the Romans were indebted for their missile-casting engines, for here you have the catapulta and ballista, differing but little from those which were used by the Romans in the third century of our era, and doubtless handed down in their turn principally through the Franks to mediæval times. Strange it is that the principle involved, nay, the very machines themselves, have hibernated, so to speak, again and again!

An antique Greek drawing, representing Amazons fighting, in conjunction with Scythians, against Theseus at Attica, shows the following armament, viz.¹:—Helmets of the Phrygian type; tunics coming half-way down the thighs, fortified with scales; and complete leg armour looking on the drawing like chain-mail, but probably, like the tunics, of small scale armour similar to that found at Æsica, referred to later in these pages. Two of the figures brandish long spears with leaf-formed heads, while the third is in the act of bending a bow, the arrow having a barbed head, and wears a quiver slung over the shoulder. They all have belts, and the tunics are ornamented with a geometrical border. Such long spears were also the weapons of the heavy Greek infantry. We owe, then, the inception of much of the arms and armour of European countries to the ancient civilisations of Asia and Egypt, and much also to the Etrurians, Greeks, and Romans; for, up to the middle of the fifth century, the countries as far as the Danube, in form at least, were still under the domination of Rome, so that Roman influence on armament must still have been very considerable; but with the final break-up of the empire of the West, at the end of the century, the old national and patriotic forms, which were of a more ponderous character, began to reassert themselves. These, again, became much modified, at a

later period, in a considerable revival in the direction of Roman forms among the Franco-Germans, who aimed at a continuation or reconstruction of the traditional Western Empire. Another potent influence in the direction of change and interchange, concerning which we can merely speculate, was the swarming out of Eastern peoples, as well as the constant pressure from the frozen North towards the sunny South.

The analysis of the suits hereinafter presented will be prefaced by a short and concise sketch of mediæval and “renaissance” armour in general, and under its own section, that of the weapons of war, etc. This, no doubt, will be helpful in making the explanations clearer as regards nationality, fashion, and chronology.

During the earlier periods, and in fact throughout the entire time covering the use of defensive armour to its decadence, great difficulties constantly arise regarding the precise antiquity and nationality of specimens preserved, and, consequently, the fashions generally prevailing in a given country at a particular time. This uncertainty is greatly owing to immigration, invasions, and to the importation of foreign artificers, as well as of arms and armour from the more advanced countries to others less forward in mechanical skill, as applied to armour and weapon-making.

Some of the manuscripts, seals, effigies, brasses, and illuminated missals preserved, afford great help in deciding doubtful points; but very little of this kind of evidence goes farther back than the ninth century, besides being sometimes of a more or less fanciful and inaccurate character, and it is only by closely weighing and comparing that some reasonable degree of certainty can be got at.

In English brasses we have the best consecutive representation of armour, extending from that of Sir John Daubernoun, in the reign of Edward I., to that found at Great Chart, near Ashford, Kent, of the reign of Charles II.; but few have been preserved that date from earlier than the fourteenth century, though there are many military effigies. There was formerly a brass in St. Paul’s Church, Bedford, of Sir John Beauchamp (1208), and this would have been the oldest brass known had it been still to the fore. There is now an Elizabethan brass of a knight in this church. The figure on the brass of Sir John Daubernoun (1277), Stoke d’Abernon, near Leatherhead, Surrey, is entirely encased in mail, excepting, of course, the

face. A large number of brasses may be seen in Boutell and Creeny, and you have the best series of effigies in Stothard and the continuation by Hollis. There are, besides, many other books treating both on brasses and effigies. The best German series exists in Hefner's *Trachten*. Some of the foreign brasses are most artistic, but the iconoclast has left us only a couple of hundred, while the English brasses are to be numbered by thousands. The great majority of Continental brasses now left are in Germany and Belgium, while some half-dozen examples cover those of France, and there is only one in Spain. It must be borne in mind that the date on ancient monuments is that of death, so that the armour indicated may be the make of a quarter of a century earlier; besides, it may have been inherited by the defunct. There are also cases where these memorials were executed during the subject's lifetime, or from contemporary models after his death. Suits were also sometimes "restored" by the armourer to correspond with a later fashion, and cases of this kind naturally give rise to some difficulty; and, as in the case of some Egyptian tombs, we have instances of misappropriation in English monuments. A case in point is the memorial of "Vicecomes et Escheator Comitatus Lincolnæ," who died in the reign of Henry VIII. The armour is late fourteenth century, but to whom the monument was originally raised is unknown. Of course, the armour for the back is not shown on brasses and effigies. The Beauchamp effigy at Warwick affords, however, a notable exception, though this is of less importance owing to the fact of there being real armour of that period existing. Another valuable source of information arises from the custom prevailing during the thirteenth and fourteenth centuries, of leaving arms and armour as mortuaries to churches, and several helms and shields have come down to us in this way.

Later in these pages will be found a chapter headed "Details of Defensive Plate Armour." This section deals as fully as a reasonable regard for space will allow with each important piece of armour, as regards its form, history, and chronology. It will serve also, to some extent, as a glossary of terms. It will be seen that there is usually a period of transition between the different well-marked styles of armour, just as is the case in architecture.

PART II.

CHAIN-MAIL AND MIXED ARMOUR.

REMARKABLY little is known of Britain during the centuries immediately following the Roman occupation, and the question as to when real chain-mail was first used in Europe is both difficult and obscure. There is a representation of loricas on the column of Trajan that looks remarkably like chain-mail, and it is almost certain that the Romans used iron chain-mail in Britain. The bronze scales of a lorica, or Roman cuirass, found at Æsica, do not help us;² but interlinked bronze rings of Roman origin have also been found, and if in bronze, why not in iron? This question is adequately answered by the masses of corroded iron rings of Roman times found at Chester-le-Street, and referred to in the report of a meeting held by the Newcastle-upon-Tyne Society of Antiquaries as far back as 1856.³ These rings could hardly be massed together as they are without having been interlinked. The extract from the report of this early meeting of the Society runs thus:—"The Rev. Walker Featherstonhaugh had presented two pieces of chain armour, corroded into lumps, from Chester-le-Street." Similar masses of rings of Roman date have been found at South Shields, and may be seen in "The Blair Collection" at the Black Gate Museum. These are of a date certainly not later than the fourth century. We may then reasonably conclude that these masses of corroded iron rings were once loricas of iron chain-mail. But the Romans were not the first to use chain-mail, for they got it probably, like so much besides, from Asia. In the British Museum are some corroded masses of links brought from Nineveh, similar in character to those found at Chester-le-Street, so it may be taken that this kind of armour is of a remote antiquity.

The Dano-Anglo-Saxon epic poem of "Beowulf," written doubtless during the second half of the eighth century, bears frequent reference to the

hero's arms and armour:—

Beowulf maælde,
On him byrne scan,
Searonet seowed
smipes orpanum.

Beowulf spoke (or sang?),
He bore his polished byrnie,
The war-net sewn
by the skill of the smith.

This poem has been cited as proof that chain-mail was in use in early Saxon England, and by the Vikings also, and there is some supposed confirmation of this idea as regards the latter in the finds of chain armour in the peat mosses of Denmark, which have been freely ascribed to the fifth and sixth centuries; but this mail is of such excellent workmanship, and so similar to that made at a much later period, as to cast grave doubts on this deduction, and there is really nothing whatever to show that it was of so early a date. Every ring of the Danish mail is interlinked with four surrounding rings, and so on throughout the garment. This is the prevailing fashion of all periods, and there is a great variety of mesh. It would seem that the “war-nets” alluded to in “Beowulf” were not chain-mail at all, but leathern or quilted armour with pieces of iron, shaped like the drawn meshes of a net, or steel rings sewn on to it, and that this combination constituted the “bright byrnie”⁴ referred to in the poem, and that the chain-mail found at Vemose, Flensburg, and other places, was made much later. Quite independent of other evidence, the line in the poem, “The war-net sewn by the skill of the smith,” would point to the leathern or quilted tunic being fortified with rings or scales sewn on to the garment; and this was the general method up to and even beyond the time of William the Conqueror.

There are, however, other words in the poem referred to, such as “hand-locen” (hand-locked), and “handum gebroden.” The latter words might well read either twisted or embroidered with hands, while both might point to interlinked mail, so it clearly cannot be affirmed with certainty that there were no instances of real chain-mail in use in Britain at this very early period after the Romans; but if there were any hauberks of the kind it might indicate a much greater continuity from the Roman occupation than the historians of those shadowy times have hitherto imagined. Possibly chain-mail was introduced from Asia, through the Vikings, and that the byrnies mentioned in Beowulf were really made of interlinked rings; but it is

probable that there was no real chain-mail in Northern Europe between Roman times and the ninth or tenth century. That it was in use in the East at an early period is shown by the discovery of a chain-mail tunic in a “barrow” in the Ukraine.⁵

The Arab hordes which were driven back by Charles Martel at the decisive battle of Poitiers in 732 were despoiled of their body-armour, which was of a rich Saracenic character, by the conquerors. This was probably of leather or quilted stuff fortified with small plates or scales; and such armour was henceforth adopted by the Franks, while Charlemagne grafted Roman fashions and traditions on to the armament.

Up to the later middle ages the sizes of the links of chain-mail, which are of hammered iron, vary considerably, extending from one-sixth of an inch to an inch in diameter, and they were soldered, welded, or butted in the earlier times, and often riveted in the later. Most of the earlier Oriental mail is riveted. It is said that the art of wire drawing was discovered by Rudolph of Nuremberg in 1306. At all events its application at this time rendered chain-mail much cheaper and more generally used than when each ring was separately wrought. This discovery was possibly only the revival of an ancient art. Very much was lost during the “dark ages” which followed the disruption of the Roman empire, when so many landmarks were swept away; and the same kind of thing has happened often before in the cycles of obscurity that preceded it. Much was preserved in “Chronicles,” as was also the case in the earlier periods of obliteration, when hieratic writings on stone, papyrus, or parchment restored so much to the newly-awakening times. Double-ringed mail is mentioned by some authorities, but the author has never seen any, and it seems probable that the indistinct drawings on manuscripts, brasses, or tapestry gave rise to the idea—very small ringed mail might easily be taken for double; still, many effigies show what looks very like double-ringed mail.⁶ The Danes of the eighth century generally adopted the Phrygian tunic, reinforced with steel rings, probably obtained through their intercourse with the Byzantine empire; and both Meyrick and Strutt agree that such a tunic was then in use. The paladins of Charlemagne wore jazerant and scale armour of strongly marked Roman characteristics, and, according to the monk of St. Gall, the emperor’s panoply consisted of an iron helmet and breastplate of classic form, with leg and arm armour. This period represents to a certain extent a classic revival, and such forms

were clearly then reverted to. It was under this reign that heavy cavalry attained the pre-eminence which sustained its first check with the successes of the English yeoman with the longbow. Charlemagne adopted the service of the ban, and formed a standing militia of his own vassals.

The real mediæval coat of chain-mail was probably somewhat of a rarity in the tenth century, but that it was in general use by the greater knights late in the eleventh is clear from the testimony of the Princess Anna Comnena, daughter of the Emperor Alexius Comnenus, who says, in describing the body armour of the knights of the first crusade, “it was made entirely of steel rings riveted together.” She further remarks that this kind of armour was unknown at Byzantium up to the time of the first crusade. Mail armour is mentioned by a monk of Mairemoustier (temp. Louis VII., a contemporary of Stephen, 1137), in a description of the armament of Geoffrey of Normandy.⁷

The inception and principles of chivalry were the romantic outcome of the lessons of Christianity as taught in the earlier “middle ages,” though confined to a narrow and privileged class; which class assumed a concrete form under Charlemagne, who did his best to divide society into “the noble” and “the base”; thus promoting the feudal system, the symbol of which became the sword. The earlier stages of the movement were characterised by great fervour and self-abnegation, operating in various ways according to the modes of thought of the different nations brought under its domination. It gradually declined, and by the end of the thirteenth century had degenerated into a fantastic fashion rather than a principle; and culminated, like the church of the period, in licentiousness and frivolity. Froissart alludes to it in this sense. The influence exercised by the laws of chivalry was on the whole beneficent in subjugating the rude passion of combat to some of the limitations of Christian ethics; and the knightly watchword “God and his lady” raised the social status of women of the privileged class. The conquest of England by the Normans, the stirring incidents of the first crusade, when we have the shrewd account of the arms and armour of the crusaders by the Byzantine Princess Anna Comnena, and the general martial spirit of the age, lent an immense impetus in the eleventh and twelfth centuries to warlike equipment of all kinds; but this was more in the direction of improving old forms, rather than in the introduction of new ones.

The Bayeux tapestry—worked, there is little doubt, in the middle of the eleventh century, but whether embroidered in England by order of Matilda for an English cathedral, or in Normandy by noble ladies or hirelings—is of comparatively little moment so long as its authenticity as an approximately contemporaneous monument of the reign of the Conqueror is generally admitted, and this is happily the case. It shows that the Conqueror's chivalry wore conical helms with the nose-guard and hood of mail for protecting the neck, shoulders, and part of the face. The hauberks reached down over the thighs, with a slit in the middle of the skirt for convenience on horseback; and the mail on the arms usually came nearly to the elbows, but sometimes to the wrists; and the continuous coif occurs frequently. The hauberk of this period had no division down the front, but was drawn on over the warrior's head. The Norman knights bear pear-shaped, convex shields with a point at the bottom, secured to the arm by a leathern strap, and large enough to cover the body from the shoulders to the hips; some with a rough device. Some of the shields shown are polygonally formed, with a central spike. The Saxon shields on the tapestry are round or oval, with a central umbo. Maces are shown in the hands of some of the figures. With the exception of William himself, whose legs are encased in chausses, probably of leather, with reinforcing scales or rings, the limbs of his knights are simply swathed in thongs. Probably only the richer knights wore chain-mail, the majority having hauberks of cuir-bouilli (boiled leather) strengthened by continuous rings sewn on to it, side by side or overlapping. Some also had the pieces of lozenge-shaped metal already mentioned, called jazerine or jazerant; or scales, which were occasionally of horn, fixed on to the leather. It is impossible to determine these details absolutely, as all the armour looks very much alike on the tapestry in its present condition, this being especially the case where rings were used; and it is only by careful comparison with other contemporary evidence that any reasonable certainty can be assured. This has naturally given rise to a great diversity of interpretation; and the same difficulty arises with seals. The knights wore no surcoats over their mail. The great seal of William the Conqueror shows him in a hauberk coming down to the knees, with short sleeves and no leg armour. Under the hauberk was the gambeson and tunic. The helm is hemispherical, and fastened under the chin. The Germans were probably before us in the general use of real chain-mail, for the epic poem of Gudrun,

written in the tenth century, states how Herwig's clothes "were stained with the rust of his hauberk."

The panoply of knights was very much the same during the century preceding the Conqueror's time, as shown in the illuminations of a "Biblia Sacra" of the tenth century. Helms with rounded crowns were worn then, and this is all confirmed by the "Martyrologium," a MS. of the same period in the library at Stuttgart.

Defensive armour continued much the same during the reign of Rufus, whose seal shows him in a long-armed hauberk without gloves of mail, and a low conical helm with the nasal; but in the reign of his successor, Henry I. (1100–1135), the reinforcing rings of the hauberk were sometimes oval and set on edgeways, "rustred" mail as it was termed; and this fashion became common in the next reign. The seal of Henry I. shows a conical cap without nasal, and that of Stephen a kite-shaped shield with a sharp spike in the centre. The king wears a hauberk of scales, sewn or riveted on the gambeson. The nasal first appeared in England about the end of the tenth century, and the Bayeux tapestry shows it to have been common among the Normans in the eleventh. Among the seals of the English kings, that of Henry II. is the first to show the hood of mail. The hauberk of the Norman kings was in one piece from the neck. Under Richard I. the hauberk is somewhat lengthened, and armorial bearings become general. The sleeves of the hauberk are lengthened, and terminate in gloves of mail. The first seal of Richard Cœur-de-Lion shows the king on horseback in a hauberk of mail. His spiked shield, shaped like half a pear cut lengthwise and pointed at the bottom, is ensigned with a lion rampant. The arm is mail-clad to the finger tips, and brandishes a simple cross-handled sword; the chausses are of mail, and terminate in a spurred sollet. Over the continuous hood, which is in one piece with the hauberk, he carries a high conical helm without flaps or nasal, bound round with iron bars. On Richard's second seal he bears the great helm with a fan crest, ensigned with a lion; his hauberk is rather longer than in the first seal. The shield on this seal is ensigned with three lions passant guardant, and this is still retained on the royal escutcheon of England, which becomes quartered with the lilies of France in the royal arms of Edward III. Both seals show the plain goad spur. There is a good example of an undoubted suit of chain-mail on an effigy of Robert de Vere (died 1221) in Hatfield Broad Oak Church. This suit was

probably made in the reign of King John. An effigy in Haseley Church, Oxfordshire, of the reign of Henry III., shows a hood somewhat flattened at the crown, hauberk reaching to the knees, and surcoat coming nearly to the ankles.

It is stated that Richard sent home from the crusade numerous suits or rather hauberks of chain-mail. There is a riveted sleeveless shirt of chain-mail, with a fringe of brass rings, dating from the thirteenth century, in the Rotunda, Woolwich; these brass rings are a common feature of the period.

The question as to when coats of arms were first introduced is very uncertain, but it is thought that the custom had its origin in the first crusade, when distinguishing marks among such a motley crowd of warriors were more especially needful. During this crusade the several nationalities taking part in it were distinguished by different coloured crosses sewn on to their garments, each leader displaying his own colour and device; but heraldic bearings first became generally hereditary in the reign of Henry III. His seal shows the king with the fingers of his chain-mail gloves articulated, and wearing the great helm. An early example of a helm with a heraldic device occurs on an effigy of Johan le Botiler about 1300. It is figured in Hewitt. The shield on the brass of Sir John Daubernoun bears a distinctly heraldic device. Heraldry seems to have been most studied, prized, and practised during the fourteenth century. An illumination in the Loutterell Psalter, dating from the middle of the same century, shows heraldic devices spread over the entire person of a knight; being emblazoned over the body, ailette, banner, pennon, saddle, shield, and on the housings of the steed, as well as on the dresses of the ladies of the knight's family. The numerous tournaments of this period encouraged its use and development, mainly in the sense of ostentation and pride of birth. In the Tower collection is a figure on horseback clad entirely in chain-mail. To the hood is attached a fillet of iron round the head. The hauberk has long arms terminating in gloves of mail. A leathern belt with strong iron clasps encircles the waist. Excepting the legs the horse is fully barded with leathern armour, fortified with iron scales. The armour on the figure is labelled "Indian," and the horse "Persian." There are two hauberks at Carlsruhe of riveted chain-mail, hood and tunic in one piece, but the head bears no fillet. On the breast, over nipples and navel, are three small palettes inscribed with Oriental characters; and inscribed clasps at the waist fasten the tunic. These suits are

chiefly remarkable for the presence of the hood, and the date of the mail is about fourteenth century. There are two shirts of mail at Brancepeth Castle, Durham, which are riveted, and probably of early fourteenth century date. It was not uncommon for hauberks to be provided with reinforcements of leathern thongs, which were intertwined through the rings; there is an example of this kind in the Rotunda at Woolwich. This description of reinforced chain-mail is referred to later under the paragraph dealing with “banded” mail. An effigy of a knight in the Temple Church, that of Geoffrey de Mandeville, Earl of Essex (1144), in the reign of King Stephen, engraved by Stothard, shows the warrior armed completely in chain-mail, having a hood of mail over the head and shoulders, surmounted by a cylindrical helmet without nasal. The hauberk is in one piece with the arms and gloves, the last without any articulation; this form of gauntlet is the earliest. Chausses going above the knee, in one web with the demi-poulaine or slightly-pointed shoes; globular triangular shield extending from the shoulder to the hip; and the belt of knighthood above the hips. There is a singular point in connection with this and two other effigies in the church, viz., that the sword is worn on the right side. This peculiarity is noticeable in other figures of the period. The effigy of a knight in the same church, that of William Longespee, Earl of Salisbury (1200–1227), wears mail gloves, the fingers of which are articulated; the sword is on the left side. Both figures wear surcoats. Like most continuous hoods of early thirteenth century date, this example is somewhat flattened at the top. They were usually rounded in the second half of the century, as shown on the Daubernoun brass; and the gloves generally divided into fingers, as may be seen on two of the sleeping guards in Lincoln Cathedral; this form continued well into the fourteenth century; The “Coif de mailles,” or separate hood of chain-mail, followed the same lines as the continuous one, and examples of all may be seen in Stothard’s series, and one of the effigies in the Temple Church shows how they were lapped round the face and fastened. What the separate hood perhaps gained in convenience, it certainly lost in invulnerability, as it left the neck less adequately guarded against a thrust from below. The effigies in the Temple Church are perhaps the most artistic, as well as the most interesting, of any series existing. It is not known that any of them really represented a knight templar, although several of them did crusaders. The only effigy of a knight templar that is known to have existed is that of Jean de Dreux, who was living in 1275.

The figure was unarmed, but bore the mantle of the order. The effigy was formerly in the church of St. Yved de Braine, near Soissons.

A knight in Walkerne Church, Hertfordshire, wears the great helm, rising slightly at the crest, pierced with eye-slits, and showing breathing holes over the mouth.

Coutes or coudières for the elbow are seen but rarely in the thirteenth century; but genouillières (knee pieces) began to appear over mail towards the middle of the century. Examples of both pieces, dating about 1250, may be seen in Stothard. Genouillières occur on the Daubernoun brass (1277), while both pieces appear on that of Sir John D'Argentine (1382). The adoption of these defences and the plastron-de-fer was the first step in the direction of plate armour. Something of the kind had become absolutely necessary by reason of the number of casualties caused by the general use of the deadly battle-axe and mace.

The cuisse and jamb (plate armour for the thigh and shin) are not seen in England before the close of the century. They were first strapped on over the chausses, and only covered the front of the leg. Chain-mail continued in use in the East up to a recent date.

A spirited drawing of a mediæval water ewer of bronze is given in the *Archæologia Æliana*, old series, vol. iv., p. 76, Plate XXII. This ewer, which was found about four miles west of Hexham, represents a knight of the thirteenth century on horseback, wearing chain-mail, and over it a sleeveless chequered surcoat. The figure wears a flat-topped cylindrical helm.

The epoch of chain-mail armour, pure and simple, may be said to close during the reign of Edward I., although in more remote and less advanced countries, such as Ireland and Scandinavia, it was to be met with very much later. There was a revival in the use of scale armour in the fourteenth century, and there are many instances. It was usually applied in pieces such as chaussons, chausses, gauntlets, or sollerets. It is often met with on German monuments. An English example occurs on the brass of Thomas Cheyne, Esquire (1368), at Drayton Beauchamp, Bucks. The mailed horseman continued the main force in every army in the field up to the reign of Edward III.

A good idea of the equipment prevailing towards the close of the century is shown in the will of Odo de Rossilion, dated 1298: he bequeaths “my visored helmet, my bascinet, my pourpoint of cendal silk, my godbert (hauberk), my gorget, my gaudichet (mail shirt), my steel greaves, my thigh-coverings and chausses, my great coutel, and my little sword.”

The surcoat was a device for protecting the armour against wet, and to mitigate the rays of the sun. It is rare towards the close of the twelfth century, when you have an instance in King Sverrer, who wore a rose-coloured surcote (“raudan hiup”). The garment becomes common in the thirteenth and fourteenth centuries, when the ground of the fabric was usually green. There are both sleeveless and sleeved varieties, but the latter did not come into vogue before the second half of the thirteenth century. There is a north-country example referred to in Surtees’s *History of Durham* (vol. iii., p. 155); one on the effigy of an unknown knight in Norton Church; and another in the Temple Church, London. Among the seals of the kings of England this garment first appears on that of John. Chaucer, writing in the reign of Edward III., says:—

“And over that a fin hauberk
Full strong it was of plate,
And over that his cote-armoure.”

The “cote-armoure of Sir Thopas” is the surcoat. There is an admirable example of a thirteenth century surcoat on the figure on the ewer found at Hexham, which has already been referred to. This surcoat is long and sleeveless, with a slit in front. It is embellished by a diamond pattern, interspersed with fleurs-de-lis and stars of six rays. The garment has an ornamental border. A representative example may be seen on the Daubernoun brass. It reaches below the knee, is slit half-way up the front, and is fastened by a cord at the waist. The border is fringed. The surcoat early in the fourteenth century was long, but became gradually shortened and tightened. There are, however, earlier examples of the shorter surcoat, as shown on the Whitworth effigy, which does not reach the knee. The D’Argentine brass (1382) furnishes a good example of the short fourteenth century surcoat, and another may be seen on the effigy of the Black Prince

(1376) in Canterbury Cathedral. It is a sleeveless garment reaching a little below the hips, and was variously fastened, being buttoned, laced, or buckled. On an effigy engraved by Hollis in his Plate II., it is held together by a brooch. The fabrics were rich and costly, and usually ornamented with heraldic devices. The surcoat on the figure of the Black Prince is charged with England and France quarterly, with a label of three points. At this period but little of the trunk armour showed through the “cyclas.” The helm on the figure of the Black Prince was gilt or silvered, and had its scarlet mantling. The surcoat of the fifteenth century presents heraldic devices on the front and arms, both before and behind, indeed it was a “tabard of arms,” and so it continued in the sixteenth century as a herald’s tabard. The garment, of course, gave rise to the term “the coat of arms.” An effigy of Sir John Pechey, figured by Stothard, shows a tabard of arms over the armour; and so does the brass of Sir John Say (1473) at Broxbourne, in Hertfordshire. The short surcoat had almost ceased with the second quarter of the century, although there are still isolated examples, such as the short-sleeved tabard on the Ogle effigy at Bothal, Northumberland, which is early sixteenth century. During the first half of the fourteenth century, English knights wore a garment under the surcoat, called “upper pourpoint”; the true “pourpoint” was the surcoat itself.

A description of the “Ehrenpforte,” written in 1559, gives a representation of the meeting between Henry VIII. and the Emperor Maximilian, which occurred in 1519. The emperor wears a surcoat with slashed sleeves and plaited skirt, obviously suggested by the civil dress of the period, called “bases.” The knightly mantle is but rarely seen on monuments. It was one of the insignia of the Garter, and was usually blue in colour. There is an instance figured by Stothard, Plate LVIII. There were two grades of knights instituted—the banneret and the bachelor. The former had his square banner as well as pennon, and square shield for armorial bearings; his retinue consisted of fifty men-at-arms and their followers. The knight-banneret, so called from having the right to bear a banner, was always a man of large estate, with a great number of retainers. Knight-bannerets first appear during the reign of Philip Augustus, and disappear by ordinance in the reign of Charles VII. The *Gloss du Droit, Fr. de Laurica* defines the etymology of the term “bachelor” as here applied. It does not signify “bas chevalier,” as has often been supposed, but refers to the minimum extent of land that a candidate for the honour must be possessed

of, viz., four “bachelle” of land. The “bachelle” contained ten “max” or “meix” (farms or domains); each of which contained a sufficiency of land for the work of two oxen, during a whole year. It would thus appear that the dignity of knighthood was only conferred on men possessing a suitable estate, and that the two grades were based on the extent of estate; which, of course, implied the number of vassals available for military service. Although the pennon was the ensign of a knight-bachelor, we have the authority of Du Fresne that an esquire could also bear one, always providing that he could ride with a sufficient number of vassals.

Orders of knighthood appear to have originated in France, and were introduced into England probably by the Normans. The most ancient order was the “Gennet,” instituted in 706. It was a military order, but always partook, more or less, of a religious character. The aspirant was usually trained to arms as a page, then he became an esquire, in attendance on a knight. It was unusual to confer the dignity of knighthood before the age of twenty-one had been reached. Knighthood was conferred by the “Accolade,” which appears to have been originally an embrace, but later consisted in the administering of a blow on the neck by the flat of a sword. There was an intermediate grade between a knight and an esquire in the *pursuivant-at-arms*, but the dignity of knighthood was very often conferred on a simple esquire.

Mamillières were circular plates over the paps, with rings affixed. Chains passed through the rings, one being usually attached to the sword and scabbard. These pieces were introduced in the reign of Edward I., and prevailed during the fourteenth century, more especially in the first half. Instances are comparatively rare. There is a beautiful example on an effigy of Otto von Piengenau (1371) in the church at Ebersberg. The chains are attached over the right breast, one fastened to the sword and the other to the dagger. Another on the tomb of Alb. v. Hohenlohe, died 1318. An instance of a mamillière over the left pap, with a thin chain attached to the helmet, occurs on an effigy of Berengar v. Berlichingen, 1377. On an effigy of Conrad von Seinsheim (1369), on his tomb at Schweinfurt, chains connect dagger, sword, and helm. The wood carving in Bamberg Cathedral (1370) affords two remarkable cases, where they directly appear on the almost heart-shaped “*plastron-de-fer*.”⁸ An English example may be seen on the figure of a knight in St. Peter’s Church, Sandwich. This interesting effigy is

also remarkable for skirts of scale-work. The scales are ridged, and are probably of iron. They form the skirt of a garment which is worn between the hauberk of chain-mail and the surcoat. The effigy would appear to date from very early in the fourteenth century. Scale-work frequently occurs on monuments of this century, seldom covering the whole body, but more generally defending the hands and feet. Mamillières are present on an effigy in Tewkesbury Abbey Church, the date of which is doubtless about the middle of the century. A beautiful instance may be seen on an effigy at Alvechurch, Worcestershire (1346), showing clearly the one chain connected with the scabbard and another with the hilt. There is a brass in Minster Church, Isle of Sheppey, which represents an armed figure with only one “mamillière”; it is on the left pap, with the chain going up over the left shoulder—early fourteenth century. The derivation of the word is interesting, being from *mamilla*, the breast. Its origin was a leather band worn by the Roman ladies to support the breasts.

In effigies the knight’s head is usually pillowed on a helm, while a dog or lion crouches at his feet; this latter feature is supposed to be emblematic of fidelity.

There are frequent representations on monuments and in MSS. of a kind of armour that appeared towards the end of the thirteenth century, “banded mail” as it is called; but there has not been any general determination arrived at as to what it really was, and there are no actual specimens for reference. It presents somewhat the appearance of the “rustred” mail of the middle of the twelfth century—that is, of rings set on to the hauberk edgeways. On monuments and drawings these rings frequently appear to be set in continuous rows, whereon the rings turn in a right or left direction alternately; each line of rings being “banded,” or framed with what looks like a rim. Examples of this mail may be seen in Stothard’s series.⁹

We reach the highest point of mediæval culture during the fourteenth century, and broadly the “renaissance” towards its close. Like all periods of transition, it presents many points of interest, especially in armament. It was not before the middle of the century was reached that arms and armour approached to anything like uniformity. In the first moiety the greatest possible irregularity prevailed. Scale armour was still largely used

throughout the century, and splint armour also, though to a less extent. An example of the latter may be seen on the effigy in Ash Church.



PART III.

THE TRANSITION PERIOD.

A COMBINATION of mail and plate armour, the latter strapped on, was in general use in England late in the reign of Edward the Second, when the helm, cuirass, or rather breastplate, and gauntlets were all of plate, and sometimes the cuisse and jamb; but the leg armour was often of cuir-bouilli. Chaucer says; "His jambeux were of cure-buly." An inventory, dated 1313, of the armour which belonged to Piers Gaveston, includes breast and back plates, and two pairs of "jambers of iron"; but most of the monumental figures are still clad in chain-mail and genouillières. These "jambers" were only front plates for strapping on. An effigy of Sir William de Ryther, who died in 1308, shows genouillières of plate on a suit of chain-mail, with the hood covered by a bassinet. This was probably thirteenth century armour, although somewhat early for an example of the bassinet. The earliest brass we have, that of Sir John Daubernoun (1277), exhibits genouillières in a most artistic form. An effigy in Bedale Church, Yorkshire, that of Brian Lord Fitz Alan, wears genouillières over chain-mail, like the Daubernoun brass. He died 1302. Mixed armour continued longer in use in England and Belgium than in Germany, which latter country and Italy always led the way in defensive armour.

An effigy in Hereford Cathedral Church of Humphrey de Bohun, Earl of Hereford and Constable of England (died 1321), engraved by Hollis, wears the camail, a tippet of mail laced to the bassinet, which falls like a curtain over the shoulders, hauberk of mail to the knees, rerebrace, vambrace and gauntlets of plate, the fingers covered with laminated plates, genouillières, jambs with hinges and very slightly pointed sollerets, all of steel, with rondelles to protect the inside of the elbow. Here we have a good example of the transition to full plate armour, as attaching plates are

replaced by rounded ones, fitting round the limbs, but still strapped on. An inventory of the earl's effects (1322) appears in the *Archæological Journal*, vol. ii., p. 349. The bassinet is mentioned as being covered with leather. Other good examples of the lacing of the camail occur on the D'Argentine brass, and on an effigy of a knight of the De Sulney family in Newton Solney Church, Derbyshire. A figure standing in the nave of Hereford Cathedral, that of Sir Richard Pembridge, K.G., who died a year before the Black Prince, wears mixed armour—camail and bassinet with the great helm.

Both the goad and rowel spurs were in use throughout the fourteenth century. The figure of the Black Prince (1376) in Canterbury Cathedral is clad almost entirely in plate, and shows the prince wearing a conical bassinet with camail attached. Breastplate, épaulières, rerebrace, vambrace, coudières, leg armour, and gauntlets, all of plate—his great crested helm has a mantling, or lambrequin, and cap of maintenance, and is surmounted by a gilded leopard; besides the ocularium, it has a number of perforations on the right side in front, in the form of a crown, for giving air. There are gads (knobs) on the knuckles for the *mêlée*, which take the form of small leopards, with the usual spikes on alternate first joints of the fingers. The surcoat is quilted to a thickness of three-quarters of an inch; and this precious relic is the only actual garment of the kind that has come down to us. The material is buckram faced with velvet—lions and fleurs-de-lis embroidered in gold thread. This surcoat is short, and laced at the back. The brass of Sir John D'Argentine, Horseheath, Cambridgeshire (1382), shows a bassinet with camail. The brassards are complete, with articulated shoulder-plates, and gauntlets with finger articulations. The chausses are of studded mail, and jambs, genouillières, and sollerets of steel, while a short surcoat covers the trunk, and the spurs are of the rowel type. Shields disappear from brasses and effigies in this century, the last example on a brass occurs in 1360.^{[10](#)}

A brass in Wotton-under-Edge Church, Gloucestershire, shows a figure in mixed armour of Thomas Lord Berkeley, who died in 1417. The sollerets are “à la poulaine,” though not in the extreme form, and the gauntlets have articulated fingers and a sharp gad over each knuckle. The figure wears a collar of mermaids, the family cognizance. We now get very near to full-plate armour on an effigy of Sir Robert Harcourt, K.G., in Stanton Harcourt

Church, Oxfordshire. The figure wears a horizontally fluted bassinet; a standard of mail; coudières sharply pointed at the elbow; cuirass with lance-rest; laminated taces, and long triangular tuilles; sollerets slightly laminated and pointed. There is a great crested helm with the figure. Sir Robert died in 1471, and the armour was probably made in the first half of the fifteenth century. This is a late example of the use of the standard of mail, but it probably covered a defence of plate, as was often the case. The steel gorget came in with the House of Lancaster. Several of these effigies and brasses have been engraved by Hollis.

It may profitably be mentioned again here that dates on monuments are those of demise. The armour, therefore, may be much earlier, sometimes a generation or so before the date of death; and it was common, nay, usual, for a knight to bequeath his suit or suits to his sons or other persons. For instance, Guy de Beauchamp, who died in 1316, bequeathed to his eldest son his best coat of mail, helmet, etc.; and to his son John, his second suit. It is obvious, however, that many effigies represent the fashion of armour prevailing at the date of demise, or even later. Mixed armour in France went well into the fifteenth century. Broadly speaking, mixed armour was used in England during the last quarter of the thirteenth to the end of the fourteenth century, but nearly full-plate armour began to be seen there in the reign of Richard II. It had, however, been in vogue in Germany and Italy for some decades before it was generally worn by the English, and it is probable that the earlier complete suits in England were imported from Germany or Italy, which countries set the fashion. Studded armour was not uncommon during the second half of the fourteenth century, and even earlier. The effigy of Gunther von Schwarzburg, King of the Romans (1349), shows the body armour to have been of mail, with reinforcing plates for the arms and legs, on which blank and studded lengths are interspersed. He wears the bassinet with camail. The following examples will show to some extent the progress of the evolution in Belgium. A figure in the library at Ghent, of Willem Wenemaer, wears genouillières and jambs of plate, otherwise clad in mail (1325). The sword is covered with a Latin inscription. A brass at Porte de Hal, Brussels, shows John and Gerard de Herre (1398) in mixed armour. On a brass in the Cathedral at Bruges, dated 1452, Martin de Visch has a full armament of plate, excepting the gorget, which is covered by a standard of mail.

This continuous strengthening of armour was clearly rendered necessary by the ever-increasing power and temper of weapons of attack, which was met by a corresponding effort at defence on the part of the armour-smith. We have the same sort of thing to-day in the constant competition between armour-plates and heavy guns. Then, again, weapons were invented to attack some *vif de l'harnois*, or vulnerable place, which was parried in its turn by an alteration or addition in the harness to resist it. The mortality in these days in battle was chiefly on the defeated side, and it took place mostly among the unhorsed combatants.

The crusades exercised a cosmopolitan influence over both arms and armour in Europe, not only in the introduction of new forms from the East, but also in a general assimilation of fashion among the nations of chivalry. The military administration of these two centuries of disastrous warfare, in and towards Palestine, was simply deplorable; and no reasonable provision was made against eventualities; hence plague, leprosy, and famine played havoc among the Christian hosts. The institution of quasi-religious orders of knighthood, however, did much to redeem these ill-starred expeditions from absolute chaos.

The formation of these religious military orders was an outcome of the proselytising zeal of the earlier "middle ages," brought into play by the first crusade. The movement was, to some extent, a fusion of the Church with the military caste for warring against the infidel for the recovery of the Holy Sepulchre. A living faith, boundless devotion, and self-sacrifice characterised these orders in the early stages of their existence, and the principles of charity and humility were strictly enjoined and practised with all men except the infidel, against whom they waged a pitiless war, not only in the East but in Europe also. The Grand Master of the order of St. Lazarus was always chosen from among lepers. The vows of chastity, poverty, and obedience soon, however, became "more honoured in the breach than in the observance;" and as these orders became rich, luxurious, and powerful, they began to nourish ambitions and practices quite at variance with the principles under which they were instituted. As their machinations began to be directed against all authority, and even against thrones and religion itself, they were deprived of many of their privileges, and some were suppressed altogether.

The shoulder-pieces called “ailettes” first appeared in France. They were in use in England late in the thirteenth century, but, as they fell into disuse in the fourteenth, there are not likely to be any actual examples preserved, and they rarely occur on monuments. These pieces assume various shapes, but the usual one is a rectangular figure, longer than it is broad, standing over the shoulders horizontally, perpendicularly, or diagonally, rising either in front or from behind; there are, however, instances of their being round, pentagonal, and lozenge formed. The use of these curious appendages is not very apparent, but the most natural explanation is that they were applied as a defence against strokes glancing off the helmet. They were usually ensigned with a device or crest; and, when worn in front, were often large enough to protect the armpits, instead of palettes or rondelles. They are mentioned in the roll of purchases for the Windsor tournament in 1278. There is an interesting letter in the *Proceedings of the Newcastle Society of Antiquaries*, vol. iv., p. 268, concerning these somewhat puzzling pieces of armour. It is addressed to Dr. Hodgkin, by Captain Orde Browne. The writer refers to the ailettes which he noticed on the effigy of Peter le Marechal, in the cathedral church of St. Nicholas, Newcastle. This highly interesting figure lies immediately behind the monument to Dr. Bruce. Captain Orde Browne mentions examples of ailettes in the churches of Ash, Clehongre in Herefordshire, and Tew in Oxfordshire, and quotes two authorities who state that these three are the only churches in which effigies with these appendages have been found; the names, however, have not been preserved in the letter. At all events, the authorities in question had overlooked the Newcastle example, on the shield of which there seems to be a bend. We refer to this effigy as attributed to Peter le Marechal. Brand believed it to be the effigy of the founder of St. Margaret’s chantry, Peter de Manley, a baron who bore, according to Guillim, “or,” a bend sable. He was associated with the Bishop of Durham, and others, for guarding the East Marches, and died in 1383. His arms therefore correspond with those on the shield of the effigy. The late Mr. Longstaffe, however, ascribes the figure to Peter le Marechal, who died in 1322.

As to the question between Peter de Manley and Peter le Marechal there can be no doubt whatever, as the presence of ailettes, and the general character of the armour, undoubtedly date the figure about the end of the thirteenth century or very early in the fourteenth, and there is an interval of

sixty-one years between the deaths of the two knights. Peter le Marechal was sword-bearer to Edward I., and is buried in St. Nicholas's Church. It appears from the king's wardrobe account that a sword was placed on the body by the king's command. According to M. Viollet le Duc, this innovation, the employment of ailettes, dates from the end of the thirteenth century, but M. Victor Gay cites an example of the employment of ailettes in 1274. There is, however, one of a still earlier date, occurring in a MS. dated 1262, in which is a figure of Georges de Niverlee. This manuscript does not say where this figure is or was. There is an ailette on the right shoulder only, and we may possibly infer that this piece was first used singly. A very interesting example of this kind occurs on an illumination on the psalter executed for Sir Geoffrey Loutterell, who died in 1345; and the single ailette bears his arms, "azure," a bend between six martlets "argent." We see from the roll of purchases made for the tournament of Windsor Park (1278) that the ailettes specified for were to be of leather and carda.¹¹ Ailettes were worn by Sir Roger de Trumpington in the Windsor tournament, but these were of leather; and are figured on his monumental brass rising from behind the shoulders. An incised monumental slab in the church of St. Denis, Gotheim, Belgium, shows a figure of Nenkinus de Gotheim (1296) with these appendages. These are remarkable for their diagonal pose. If any device existed it has been worn off. There is an example of another Gotheim (1307) charged with a rose, and a couple in the Porte de Hal Museum, at Brussels, dated 1318 and 1331 respectively. A very elaborate pair of ailettes appears in the inventory of Piers Gaveston (1313): "les ailettes garniz et frettez de perles." There is a German example on the statue of Rudolph von Hierstein at Bâle (died 1318).

PART IV.

HELMS UP TO THE END OF THE TRANSITION PERIOD.

HELMS with horns were worn by the Vikings, and in all probability the headpiece with these appendages dredged up with a shield in the Thames, and now deposited in the British Museum, is of early Scandinavian origin. Horned helms were probably originally emblematic of the goddess Hathor or Isis, and came to Northern Europe through the Greeks. A helm with horns, about B.C. 3750, found at Susa, has been already referred to in Part I. We have an example of an Etruscan helm with horns, and Meyrick says that such were worn by the Phrygians, though rarely. Diodorus Siculus refers to this form as used by the Belgic Gauls. There are instances of helms with horns as late as the fourteenth and even fifteenth centuries. One occurs on the tomb of Diether von Hael, at Borfe, in the Tyrol, near Moran. This helm has ears as well as horns. The warrior died 1368. Other examples, one on the effigy of Burkhard von Steenberg (died 1379), in the Museum at Hildesheim, and another on that of Gottfried von Furstenberg (died 1341), in the Church of Hasbach; and there is a grotesque helmet in the Tower of London, presented to Henry VIII. by the Emperor Maximilian, with ram's horns; and such appendages were sometimes used on chanfreins of the sixteenth century—there are examples at Madrid and Berlin. The early Anglo-Saxons wore four-cornered helms with a fluted comb-like crest.

The great variety in mediæval and renaissance headgear is somewhat bewildering, but it may all be brought down to a few types with certain salient characteristics, which, however, greatly interweave. The knights of chivalry, or their armour-smiths, seem to have given as great a rein to their fancy and imagination as the constructors of feminine headgear of all time;

still the change and application of weapons of attack played the most important part in the constant modifications of warlike headpieces, as of other defensive armour.

Both Normans and Anglo-Saxons used the word “helm”¹² (of Gothic or Scandinavian derivation) in the eleventh century, as applied to the conical steel cap with the nasal then in use. The equivalent in French was “heaume.” The word “helmet” is of course the diminutive of “helm,” and is specially applied to the close-fitting casques, first used in the fifteenth century, of which more anon. The seal of Henry I. shows that monarch as wearing a conical helm.

The form of helm of the Bayeux tapestry is a quadrilateral pyramid with a narrow strip of iron extending over the nose; but this nasal is but rarely met with after the twelfth century, although it occurs in every century up to the seventeenth. The Norman helm was probably wholly of iron, and sometimes had a neckpiece.

The great helm or heaume, without a movable visor, is of English origin. It first appeared about the middle of the twelfth century, and was worn over a hood of mail, which was then found inadequate to resist either the lance or a heavy blow from a battle-axe or mace, or even a stroke from the then greatly improved sword. The helm had the effect of distributing the force of the blow, and to a certain extent parried it. The second seal of Richard I. shows him in a great helm, which is either flat-topped or conical, with the nasal, and is obviously derived from the antique. The cylindrical or flat-topped variety came into vogue towards the end of the twelfth century. There is an example of the conical form in the Museum of Artillery at Paris, and one of the nearly flat-topped variety, rising very slightly towards the centre, in the Tower of London. The great helm is often represented as a pillow for the head in effigies.

The next form, which is in great variety, the knight's early tilting helm, was used pre-eminently for jousting; the visored bassinet being worn generally in battle. It was introduced to resist the heavy lance charge. This form was hemispherical, conical, or cylindrical, with an aventail to cover the face,¹³ and ocularia or slits for vision, and sometimes a guard for the back of the neck. Breathing holes first appear early in the reign of Henry III. It formed a very heavy single structure, sometimes with bands of iron in

front constituting a cross; and in the earlier forms the head bore the whole weight; but later it was constructed to rest on the shoulders, and the crossbands disappeared. It was fastened to the saddle-bow when not in use. The movable aventail appears on the second seal of Henry III. An excellent example may be seen on the male effigy in Whitworth Churchyard, which is described in the *Proceedings of the Newcastle-upon-Tyne Society of Antiquaries*, vol. iv., p. 250. This monument shows two recumbent figures—male and female. We are concerned with the male effigy, and have the authority of Mr. Longstaffe that it represented a member of the family of Humez of Brancepeth. The character of the armour would indicate a date in the second quarter of the thirteenth century. The helm is cylindrical and flat-topped. There are two other north-country effigies of about the same date, one at Pittington, the helmet of which is round-topped, and the other at Chester-le-Street (both in the county of Durham). The round-topped helm appeared late in the thirteenth century. A very early thirteenth century helm may be seen on an effigy in Staunton Church, Nottingham, and a flat-topped cylindrical specimen surmounts the figure on the curious water ewer shown in Plate XXII. of *Archæologia Æliana*, vol. iv. (O.S.). There are instances of this form as early as the last quarter of the twelfth century.

De Cosson gives drawings of several of these helms in his *resumé* of the specimens exhibited in 1880 (for which see *Proceedings of the Royal Archæological Institute*). That on the seal of Henry III. has breathing holes, and that of Edward II. shows his helm to have been cylindrical, with a grated aventail. Helms at this period were sometimes made of brass. The helm formerly hanging over the tomb of Sir Richard Pembridge, K.G., in the nave of Hereford Cathedral, and now in the possession of Sir Noel Paton,¹⁴ is a good example of the reign of Edward III. This helm has been minutely described by De Cosson in his catalogue of the helmets already referred to. The great jousting helm of the fifteenth century will be described later. The bassinet, lined with leather, basin-shaped as its name implies, was lighter and close fitting; and in England usually provided with staples for a camail. It was often used under a crested helm of large size, but, as mentioned before, when the bassinet became visored it was worn heavier, and then largely superseded the great helm. The bassinet was generally worn in England in the fourteenth century and late in the preceding. This helmet is more fully described later.

The chapel-de-fer is an iron helmet of the twelfth century, with or without a broad brim. It was often holed for a camail, and was worn sometimes under a hood of mail. The one without brim is often termed a chapeline, and is, we take it, the small bassinet. Illustrations of two great helms at the Zeughaus, Berlin, are given in [Fig. 2](#).



PART V.

PLATE ARMOUR.

It was late in the reign of Edward II. when considerable progress was made in the direction of full “plain” armour in England, but, as shown in the section headed “Chain-mail,” etc., the use of the standard of mail survived until the beginning of the fifteenth century and even later. It is, in fact, impossible to lay down any arbitrary dates, or anything like a clear line of demarcation in respect to the relative proportions of chain and plate armour in use by English men-at-arms up to the beginning of the fifteenth century; but the fortunate preservation in our churches of the remarkable series of effigies and monumental brasses helps us greatly. There is, however, very little evidence of this kind before the middle of the thirteenth century. Breastplates, as distinguished from the old *plastrons-de-fer*, were to be met with early in the reign of Edward II., but the general rule was still a *hauberk* of mail, with *épaulières*, *coudières* of plate, and some splint plates on the arms, all fastened with straps and buckles; the legs were still generally encased in mail, with, of course, *genouillères* at the knees.



Fig. 2.—Great Helms at Berlin.

1250–1300.

1350–1400.

The long reign of Edward III. (1327–77) saw great strides towards the general use of full plate armour. An illumination on the psalter of Sir Geoffrey Loutterell (died 1345) furnishes an interesting example of the time. The knight is on horseback, sheathed in plate; he wears the pointed bassinet, a rectangular ailette on his right shoulder. His coat-of-arms (“azure,” a bend between six martlets “argent”) is repeated wherever possible: on the ailette, helm, pennon, shield, and housings; and again on the dress of a lady who is handing up the helm. Another lady holds the shield: her dress impales “azure,” a bend “or,” a label “argent” for Scrope of Masham. The saddle is the “well,” and the spurs rowelled. The lance-rest (an adjustable hook of iron for supporting the lance shaft) was introduced about 1360. A brass of Sir John Lowe, at Battle, Sussex, gives a good idea of the armour prevailing late in the reign of Richard II. and in that of Henry IV. The surcoat is omitted, so that in this instance the whole front panoply is exposed to view, though the garment continues to appear occasionally on monuments well into the fifteenth century, as shown on the brass of Sir William de Tendering in Stoke-by-Nayland Church (1408). The bassinet becomes less acutely pointed than on the effigy of the Black Prince. Épaulières show articulations, and gauntlets are articulated at the fingers. This is the case on the brass of Sir John Lowe, where the armpits are protected by rondelles, and the now visible taces of steel hoops form a skirt of from six to eight laminations. The cuisse is articulated, and the sollerets are “à la poulaine,” though not in the extreme form. The spurs are of the rowel type, and the figure is armed with sword and dagger.

Full plate armour was used in Germany and Italy earlier than in England. There is ample evidence of this, but care must be taken in sifting the testimony of old “Chronicles.” In the “Tristan and Isolde” MS., by Godfrey of Strasburg, of the second half of the thirteenth century, the German men-at-arms are represented in “white” armour; helms with the bevor attached to the cuirass, the upper part of the face open, jambs of plate and sollerets “à la poulaine.” Their horses appear with bards. A statute of Florence of the year 1315 is remarkable for the following statement, viz.:—“Every knight to have a helm, breastplate, gauntlets, cuisses, and jambs, all of iron!”

These manuscripts, however, must not be taken as conclusive. On the contrary, they really represent what is now considered to be a late stage of mixed armour. An Italian example figured in Hewitt (Plate XXVII.) shows the statue of a knight in a church at Naples (1335). He wears a hauberk of mail, with rondelles at the shoulders and elbows, rounded plates strapped over the

upper arm, and jambs of iron. The sollerets are in chain-mail. The heavy horsemen of the "middle ages" are often referred to as "knights," but of course there could only be a very small percentage of them enjoying that degree. Presumably many were eligible for the honour of knighthood for marked bravery in the field.

* * * * *

Before the use of gunpowder in warfare the baronial fortress was almost impregnable, but cannon turned the tables on the feudal nobility, dealing a severe blow at extreme feudalism, of which these castles were the invariable centres.

The reason for the introduction of the cuirass proper was the exceeding weight of the hauberk of chain-mail, in conjunction with the heavy plates often riveted on to it, and the quilted gambeson, etc., underneath; and also by reason of the inefficient protection it afforded against the lance in full career, or strokes from the greatly improved and heavier swords, or blows from the deadly battle-axe; indeed, it often happened that a portion of the chain-mail itself was driven into a wound. It was, however, far from uncommon early in the fifteenth century for a hauberk of chain-mail to be worn under the cuirass, with a gambeson next the body, and another between the mail and the cuirass; but this multiplicity of garments was far too heating, heavy and cumbersome, and at least one of them, and generally two, were discarded on the full introduction of plate armour. These cast-off garments were, however, utilised by the lighter troops. The gambeson is a quilted tunic, often worn in battle in early times without other armour, having been made tough enough to turn a sword stroke; but when plate armour became general it was of quilted linen, fortified with rings under the arms and breastplate. There is a most interesting gambeson of the kind in the national museum at Munich, an example of late fourteenth century date, and the only one known as surviving; it also covers the legs, and is strengthened with mail over the knees. There is a specimen at Munich, thought to be unique, of the familiar horizontal belt one sees on effigies of the fourteenth and early fifteenth centuries. The underclothing varied greatly at the different periods, and there is often some confusion of terms among the "Chroniclers" regarding these garments. Chaucer calls the gambeson a "haketon," the habergeon of his day being a shirt of chain-mail. He says:—

“Next his shirt an haketton,
And over that an habergeon,
And over that a fin hauberke,
Full strong it was of plate.”

There is a fine specimen of a fifteenth century habergeon in the Porte de Hal Museum, Brussels. A MS. of this period says that esquires were not allowed a sautoir (stirrup) to their saddles. The order had a distinct status, even to its costume. The esquire was the auxiliary and companion of the knight. His duty consisted in carrying the knight's arms, breaking-in and seeing to his horses, and generally looking after him; he fought at his side and guarded his prisoners. The spurs of the knight were of gold, those of the esquire of silver. To “win his spurs” and be dubbed a knight, he was required to have performed some valiant deed. There was an intermediate grade between a knight and an esquire in the pursuivant-at-arms. There was a varied and costly elaboration of ornament used by the more courtly cavaliers of the fourteenth century and later times. The figure of the Black Prince in Canterbury Cathedral is highly decorated. The knightly belt has a blue enamel ground, with bosses of gilt leopards' heads. The bassinet bears a coronet embellished with precious stones. The sword scabbard is inlaid with lapis-lazuli, and the spurs are gilt. Inventories of the period often divulge items such as rich velvet and embroideries, gold and silver. Pearls and carbuncles among gems were especially affected for decorative purposes. The inventory of Piers Gaveston (1313) has been already referred to as mentioning “les alettes garniz et frettez de perles.” Mr. Hewitt mentions the inventory of Louis Hutin, temp. 1316, which has “Item, cote, bracières, housse d'escu, et chapel de veluyan, et couvertures a cheval des armes du Roy, les fleurs de lys d'or de Chypre broudées de pelles [pearls]. Item, picières et flanchières de samit [satin] des armes le Roy, les fleurs de lys d'or de Chypre. Item, uns gantelez couvers de velveil vermeil.” Such portable and valuable adjuncts induced a deal of looting among the fallen champions after a battle, and many wounded lost their lives from this cause who would otherwise have been put to ransom. Stringent sumptuary laws were very rife at this time, but these severe enactments were found very difficult to enforce, and were much evaded; indeed, this has always been the case. Single feathers were worn in the fourteenth century; but in the fifteenth and sixteenth great plumes, drooping gracefully behind, were the rule.

The degradation of a knight under King René d'Anjou was a very elaborate ceremony: he was stripped of his armour, which was broken to pieces before him, and his spurs were thrown on a dunghill; there was also much besides. In later times, the knight's spurs were hacked off by the king's master-cook.

Early representations of bards are very rare; they probably originated in the twelfth century, when they were most likely of fortified leather. They did not become general in England until towards the close of the thirteenth century. Wace says that the horse of William Fitz-Osbert was housed in chain-mail at the battle of Hastings, but this is incredible.

As already mentioned, German men-at-arms appear with barded horses in the second half of the thirteenth century, but it was towards its close, or at the beginning of the fourteenth, that they became common. The earliest English official mention occurs in the statute of 27 Edward I., when bards were of chain-mail, leather, or quilted material. In the inventory of the armour of Louis X. occurs, "item, a chanfrein." Nothing like a full equipment in steel plate for horses was attained before the second quarter of the fifteenth century, when, according to a picture in the imperial arsenal at Vienna, "Der Ritter sitzt auf seinem, bis auf die Hufe, verdeckten Hengst." The material differs very much in the fifteenth century, being of full plate, fortified mail, quilted cloth, or cuir-bouilli.

Bards comprised the chamfron or chanfrein, for the face, worn sometimes with a crest; picière, breast; flanchière, flanks; croupière, hinder parts; estivals, legs. The crinet, neck, appears first in England on the seal of Henry V. The horses were gaily caparisoned. The emblazoned housings were often made of costly material, such as satin embroidered with gold or silver. Examples are given in Figs. 3 and 24.

The horsemen of late in the twelfth, thirteenth, and fourteenth centuries consisted of men-at-arms or heavy cavalry,¹⁵ hobilers and armati, or common horse troops. The infantry consisted of spear and billmen,—that is, men armed with long-handled weapons,—crossbowmen and archers. Hobilers were light cavalry taken from the better class of yeomen. The "hobby" horse was a much lighter steed than that used by a knight or man-at-arms, clad in his armour of proof. Part of the light cavalry consisted of bowmen. The gynours had charge of the catapultæ, ballistæ, and other siege engines.

Grose, in his *Military Antiquities*, vol. i., p. 278, cites an old Latin MS., giving the numbers of the army of King Edward III. in Normandy and before

Calais, in the twentieth year of his reign, with their several stipends, as follows, viz.:—

		At per Diem.		
		£	s.	d.
	My Lord the Prince	1	0	0
	Bishop of Durham	0	6	8
13	Earls, each	0	6	8
44	Barons and bannerets	0	4	0
1046	Knights	0	2	0
4022	Esquires, constables, centenary, and leaders	0	1	0
5104	Vintenars and archers on horseback	0	0	6
335	Paunceners	—		
500	Hoblers	—		
15,480	Foot Archers	0	0	3
314	Masons, carpenters, smiths, engineers, some at 12 ^{d.} , 10 ^{d.} , tent-makers, miners, armourers, gunners, and artillery men, 6 ^{d.} and 3 ^{d.} per diem			
4474	Welch foot, of whom 200 vintenars at the rest at	0	0	4
		0	0	2
700	Masters, constables, mariners, and pages	—		
900	Ships, barges, balingers, and victuallers			
	Sum total of the aforesaid men, besides Lords	31,000—294		
Of whom some men from Germany and France, each receive for their wages 15 florins a month.				

It would appear from this “establishment” that King Edward’s main force consisted of foot archers, and that the predominance of this item largely accounts for the English victories of the time, against greatly superior numbers on the side of the French. It will be observed that gunners and artillerymen are mentioned in this MS., but they were probably for serving siege-guns before Calais.

The institution of feudalism, which was in direct opposition to the Roman system, exercised an immense influence on the form and constitution of the armies of Northern and Central Europe during the “middle ages” and later. The inauguration of the movement proceeded mainly from the division of lands by Clovis among his followers; but it was the policy of Charlemagne that gave it form and substance in the direct creation of a martial and a sacerdotal aristocracy. Europe then became dotted over with seignories and strong places, erected originally with a view to save the countries from being overrun and enslaved by barbarous hordes; and by these means the invaders were compelled to confine their depredations mainly to the sea-coast regions, which they ravaged without mercy. Each vassal swore fealty to his liege-lord in the ceremonial “homage-lige.” The vassal was bound to fight under the banner of his liege-lord for a continuous term of from twenty to sixty days when called upon, and to assist him in many other ways; and as long as his duties were faithfully and diligently performed he remained master of his fief, and was also permitted to infeudate or sub-feudalise it. The seignior on his part extended his protection to his vassals, and was bound to render them full justice; and in cases of default an appeal to the suzerain of the seignior was provided for. This was the theory, but the practice too often meant an organised system for the oppression of the weaker classes, and so on down to the lowest rung of the feudal ladder. The church itself united in exercising a feudal as well as a spiritual jurisdiction, and bishops wielded this double power over the seignior in their bishoprics.

The rise of the third estate, and especially that of communal government, brought about modifications of the system as time moved on. These causes, with their influence on military matters, will be lightly touched upon in these pages as they arise; but it must be borne in mind that though feudalism was the same in principle everywhere, it differed in its application in the various countries it dominated, according to the characteristics and circumstances of the peoples.

The principle of the ban or feudal levy was that those holding land should contribute to the king’s army in war time a certain fixed proportion of retainers, according to the acreage of their holding; but in cases of great national peril the levy, the *arrière-ban*, was much larger, and there was often an arrangement under which actual service might be compounded by a money payment called “scutage.” The *arrière-ban* or the *ban-fieffé* dates from the sixth century. It summoned the vassals, which the suzerain alone had a right to

command. The increasing number of mercenary troops employed steadily diminished the importance of the ban, and “scutage” became more general.

The battle of Courtray, fought in 1302, was the turning point in the greater estimation of the use of infantry combinations, when the French chivalry was so completely routed by the Flemish guild-bands, armed with the goedendag, which, whatever its form really was, then proved a most effective weapon against a rush of horsemen. About six thousand of the chevaliers were killed, a heavy blow struck at the nobility of France. The object-lesson thus afforded showed, even at this early period, that heavy horsemen charging with the lance, or striking with the mace or battle-axe, had ceased to be “the strength of the battle.” This experience was amply confirmed at a later period at the decisive battles—Granson, Morat, and Nancy. After the death of Charles the Bold at Nancy, in 1477, a victory won by the Swiss infantry with staff weapons, the “chivalry” of battle became much discredited, and the extreme feudalism which had hitherto dominated the military systems of Europe underwent its first serious check in the diminished importance of the mailed horseman, and the growing power of the third estate, which henceforward became a more weighty factor in warlike tactics and combinations. This process, which had been growing for some time in the gradual enfranchisement of the communes, developed from the motley swarms of yeomen and peasants at length into a communal militia. To these were now added “condottieri” and other free companies, such as stradiots, routiers, brabançons, and tard-venus, and with these more stable elements of an army, tactics and generalship, which had hitherto been of the most elementary character, soon made great strides. There are, however, early instances of the addition of “mercenary bands” to armies in the field. William the Conqueror’s army at Hastings contained a large proportion of these troops, which were placed in the first division during the battle. The Plantagenets also used them very freely. Mercenary troops, however efficient in action, had many drawbacks in campaigning. They were not unfrequently known to change sides at a critical moment, such as on the eve of, or even during an engagement. A notable instance of this may be cited in the case of the battle of Pavia, in 1525, when Francis I. was made prisoner.

The growing power of the Hanseatic Bund did more than anything else in Germany towards the enfranchisement of the towns from the galling fetters of feudalism. This mighty organisation, in the heyday of its power, consisted of over a hundred of the most important towns, scattered over Germany and Northern Europe, and extending as far as Wisby in the Gulf of Bothnia, and

even to Novgorod in Russia. Its power became so great that even the Emperor exercised but a nominal supremacy over the German cities enrolled. Almost the entire commerce and banking of the time in Northern Europe centred in this powerful association, fenced in its walled towns. It supplied the sinews of war, and the equipments for nearly every campaign; often indeed for both the opposing armies. Its power and monopolies in England, where it had stations, especially in London, were immense.¹⁶ Feudalism thus became greatly banished to the country districts, which constantly underwent a depletion of able-bodied men by a rush of serfs towards freedom under the syndics. Soon the standards and war-cries of the great seigniors ceased to cause confusion in the ranks.

The equipment of each man-at-arms in the fifteenth century was two archers with two mounted followers; and a little later a sixth man and horse were added. An army of fifteen hundred “complete lances” required a contingent of at least five thousand mounted archers.

It was not uncommon for armour to be imported from Italy during the fourteenth century. Froissart states that Henry IV., when Duke of Hereford, sent messengers to Milan asking Duke Galeazzo to forward him a harness. The Duke complied with the request, sending four Italian armourers with the suit.

Broadly, the period of full plate body armour is reached in England early in the fifteenth century, when the mentonnière, rondelles, cuirass, taces and tuilles, garde de reine, épaulières, gauntlets, cuisse, genouillères, jambs and sollerets were all of plate. The ingenious application of overlapping or lobster-tail plates, first applied to the solleret and rerebrace, had now extended to the shoulders and taces, and we find this system gradually developing towards the fine ridged and scalloped armour, which originated in Italy in the second quarter of the fifteenth century. Effigies of the first quarter of the fifteenth century are characterised by the bassinet, standard of mail, and beautiful fan-shaped coudières pointed over the elbow-joint. The skirt of mail shows itself beneath the taces, with an scalloped fringing. Articulated épaulières prevailed until towards the middle of the century, when pauldrons began to displace rondelles over the armpits; an early example of which may be seen on a brass in Arkesdon Church, Essex. Pauldrons are, however, exceptional until the “Maximilian” period. Examples of most of the features of the period may be seen in the series of plates published by Stothard, Hollis, Greeny, and others. We pass now out of the period during which we have been mainly indebted to effigies, brasses, and pictorial representations for our knowledge of armour,

and enter on much surer ground, when there are actual and contemporaneous specimens to deal with. Still there is but too frequently ground for doubt and perplexity, as comparatively few suits are quite homogeneous; in many cases some of the parts are often restorations, faulty enough, as most restorations are. Pieces sometimes belonged to other suits, and not unfrequently to widely different periods. New tactics in battle had to be parried by the armour-smith with changes and modifications in armour; for instance, at the battle of Crécy the English men-at-arms fought for the first time in foot formation, and they adopted the same tactics at the battle of Poitiers on the 19th September, 1356. This innovation having been copied by the French, the armourer had to meet the occasion, and different harnesses began to be made for foot-fighting and horseback; and somewhat later additional pieces were added to screw on to the other armour, for further protection in tilting and in battle. These pieces were devised for the protection of the more vulnerable places, on the principle that energy always takes the line of the least resistance. Besides this, at various periods when defence was stronger than attack, improvements in the arms then in use took place; and new weapons were devised with a view to the attack of weak points in armour. Before the battle of Poitiers the French men-at-arms were ordered to shorten their lances to five feet, and to take off their spurs; and the lances were similarly shortened at the battle of Auray in 1364. The great helm was now rarely used, giving place to the visored bassinet, the visor to be raised or lowered at pleasure. The bassinet was in its turn superseded by the sallad in the first quarter of the fifteenth century, and the latter towards its close by the armet, followed closely by the burgonet. A monument in the cathedral at Posen gives a good idea of the armour in use in Germany in the first half of the fifteenth century—it is a figure of Lucas de Corta, who died in 1475. The armament consists of a mentonnière of several laminated plates to be raised or lowered, cuirass with rondelles, taces of five or more overlapping plates, going right across the lower body, but no tuilles, cuisse with genouillières and hinged jambs; laminated rerebraces, and large pointed coudières. The fingers of the gauntlets are articulated, with a sharp gadling over each knuckle, and sollerets “à la poulaine.” This monument doubtless represents armour of the first half of the century. A brass in the church at Altenberg gives a figure of Gerart, Duke of Gulich, who died in 1475, with a similar armament excepting that he wears an early form of armet, and the tuilles are attached to the taces. The armour of this period, with its pretty shell-like ridgings, is both graceful and practical, and also lithe and supple.

The armour of the second half of the fifteenth century, which is usually styled “Gothic,” it is impossible to say why, is by far the most graceful of all the periods, combining beauty of form and contour with excellence of material and workmanship; together with an admirable adaptability for defence against the then existing weapons of attack. The main features of this remarkable period are the scalloped and shell-like form of some of the pieces, and especially the presence of *tuilles*. The *coudières* are excessively large, sometimes preposterously so, and channelled with a view to the lance glancing off them. The breastplate is rendered both stronger and more elastic by being made in two and even three laminated plates. *Sollerets* are “à la poulaine.” The helmet of this armour is the *sallad* with the *mentonnière*. An excellent English example may be seen on the Beauchamp effigy at Warwick (1454); and another on the brass of Sir Robert Staunton at Castle Donnington (1458). There is a very instructive series of monumental effigies at Meissen, engraved by Hollis, of successive dukes of Saxony, showing the continuous advances in armour. Albert, who died in 1500, wears the *armet*, *pauldrons* with *pikeguards*,¹⁷ and broad *sollerets*. Another duke, who died seventeen years later, shows *tassets* of five *lames*, and “bear-paw” *sollerets*. The armour of Duke Frederick, who died in 1539, shows *mitten gauntlets* of numerous narrow *lames*.

Gothic armour is the most perfect of all. It is more “mobile” than any of the later schools, and was made to fit almost like a glove; and as the details of suits are no longer obscured by the surcoat on effigies, we have these representations to guide us, as well as actual specimens. The steel, which looks as if it had an admixture of silver, is stronger in texture, brighter and tougher than that of any other period. Sad it is that there are so few perfect specimens of this armour left to us, for most of the armour wrought up to the middle of the century has become the prey of rust, the iconoclast, and the melting pot. The suits at Sigmaringen, Munich, Nuremberg, Vienna, and Berlin are among the most homogeneous the author has seen.

Armour made at Milan was already famous at the end of the fourteenth century, and many suits were ordered there at that time for English account; and later in Germany, for it took a considerable time before the wave of the “renaissance” reached the more northern country. The famous Milan armour-smiths, the *Missaglias* and *Negrolis*, and in Germany, the *Kolmans* of Augsburg, *Hans Grünewald* of Nuremberg, and the *Seusenhofers* of Innsbruck, all turned out work of the highest character and finish; as also did many of the

later masters, such as Anton Peffenhauser of Augsburg, Lucio Piccinino of Milan, and Georgio Ghisi of Mantua. Both armour and weapons of a high quality were produced in other towns in Italy, such as Florence, Brescia, Lucca, Pisa, and Pistoja. The work of the armour-smith, pure and simple, seems generally to have reached its highest point of excellence during the second half of the fifteenth century, the force of the “renaissance” expending itself more on ornamentation.

Until comparatively recently very little was known concerning the great armour-smiths and their coadjutors of the fifteenth, sixteenth, and even seventeenth centuries; but much has now been accomplished in this direction by Dr. Wendelin Boeheim in Vienna, and given to the world in his work, *Der Waffenschmiede*, etc. Dr. Cornelius Gurlitt has also thrown much light on the masters of Saxony in his booklet entitled *Deutsche Turnier*, etc., of the sixteenth century. We owe much to these savants for their arduous labours in rescuing the names, and much besides, of so many of these great artists from an undeserved oblivion; and also by the identification of their work in providing valuable and reliable material for fixing the dates of armour within comparatively narrow limits.

Scale armour is but very rarely found in the fifteenth century.

Monograms are not often seen on armour of English make, but they were common in Germany towards the end of the fifteenth century, when armour was occasionally inscribed with the year. The comparatively few instances of dated armour are intensely valuable, as we have then no inferences or doubtful ancestral legends, but the actual year of make. Examples of both fifteenth and sixteenth centuries occur at Nuremberg and Berlin. There is an idea generally prevailing that the stature of the men of the middle ages was shorter than nowadays. After the comparison of many suits, both at home and abroad, it is certain that this is not the case, but the average development of the calf of the leg is greater now. An ordinary-sized leg of to-day would not fit into the average cuisse and jamb of the sixteenth century, but it must be remembered that a very large proportion of the suits preserved, made in the fifteenth and sixteenth centuries, were for Italy, South Germany, France, and Spain. The build and stature of these peoples were slighter than that of the Englishman. The wearers spent such a large proportion of their time on horseback, that the calves of their legs were naturally like those of the “horsey” man of to-day.

From early in the sixteenth century the changes were greatly matters of detail, the differences in suits being principally those of form. The shell or tile-

formed tuilles, after having been in use for nearly a century, gave place to the more comprehensive tassets of overlapping plates. Épaulières developed into pauldrons, which gradually increased in size, covering both shoulders and upper-arm, and at length extended over each breast, and then diminished again in size. Pikeguards were introduced to protect the neck from pike thrusts, and there are instances of these plates as early as the middle of the century. Sometimes they are double on each shoulder—see the brass at Qui, Cambridgeshire. In cases where a pikeguard appears on one shoulder only, a close examination will generally reveal holes for the fixing of its fellow. At the beginning of the sixteenth century, or a few years later, the so-called “Maximilian” armour superseded that termed “Gothic,” when a large proportion of this armour (the “Maximilian”) was fluted everywhere except the jambs. It had pauldrons, with pikeguards, and great “bear-paw” or “cow-mouth” shaped sollerets. This style became *à la mode*, in imitation of the prevailing fashion in dress, which was then largely puffed and slashed. It must be understood, however, that fluted suits were in a majority of the armour made, but not to the exclusion of plain armour. The cuirass is shorter than in the later Gothic form; it is more globose, with the top cut straight, and the breastplate is usually in one piece. The headpiece is the armet and burgonet. Sliding rivets (Almayne) gave increased elasticity to armour of this period. As may be seen from some notes in *Archæologia*, vol. li., p. 168, written by Viscount Dillon, P.S.A., the term “Almayne rivets” was sometimes applied to complete harnesses; for an order sent to Florence by Henry VIII., in 1512, runs: “The 2000 complete harness, called Almayne ryvettes, were to be alway a salet, a gorget, a breastplate, a backplate, and a pair of splints (tassets) for every complete harness at 16s. the set.” There is a sixteenth century specimen of an armourer’s pincers, with claw and hammer head for riveting armour, in the Rotunda collection at Woolwich. It was soon found that arms of attack would not glance so well off fluted suits, and smooth armour was again generally reverted to. Blackened armour was not uncommon at this time; and a black, white, or coloured tunic of stuff was often worn over bright. The first instance of black armour that we have met with is mentioned by Froissart, under the year 1359.¹⁸ While in “Gothic” armour the taste of the period found expression in beauty of outline, already in the fifteenth century it had become fashionable to have armour engraved and otherwise ornamented. Perhaps the only brass that is to be seen in Spain represents a beautiful specimen of inlaid armour; the figure is of Don Parafan, Duke of Alcola, who died in 1571. The pikeguard has ceased, sollerets are the shape of the foot, and he wears a

morion. The morion and cabasset were late sixteenth and seventeenth century helmets, while armets and burgonets were greatly worn early in the sixteenth. Late in the fifteenth and during the sixteenth centuries there was a description of armour called “penny-plate.” It consisted of round pieces of steel riveted on to leather. There is a specimen of this kind of armour at the Rotunda, Woolwich.

By the end of the fifteenth century heavy tilting-suits had attained their greatest strength, and as the sixteenth century advanced so did ornamentation. Under the Emperor Maximilian skirts or petticoats of plate began to be worn—another illustration of the influence exercised on armour by the prevailing fashion in dress, in fact the form was reproduced in the surcoat before 1470; and indeed the application of taces during the fourteenth and early in the fifteenth centuries, before the introduction of tuilles, was also something in the same fashion. These skirts were called bases or lamboys. There is an example in the Tower of London, and another on the Hertford tomb (1568). Another example is in the author’s collection, of which a detailed description and drawing is given later in these pages ([Fig. 25](#)). These lamboys were specially designed for fighting on foot, but there is often an arrangement by which a portion is detachable in order to enable the wearer to sit on horseback. There is a style of armour the Germans call “Pfeifenharnisch,” which has embossed pipings in high relief like puffs. Such a harness was made by Hans Seusenhofer for Prince Charles, later the Emperor Charles V. Visors of this period were often wrought in the form of a grotesque face. There is more than one example at Vienna, and indeed they were far from uncommon; the author possesses a couple. Bards had become highly decorated, and with the housings were sometimes designed in close imitation of the dress fabrics of the period. Such a suit of bards on a charger, on which is mounted a rider in a piped suit of the “Maximilian” type, may be seen in the Kungl. Lifrustkamar, in Stockholm. An illustration is given in [Fig. 3](#).



Fig. 3.—Mounted Suit with Bards, in the Kungl. Lifruskammars Collection, Stockholm.

Towards the end of this century (the sixteenth) defensive armour had reached its highest point of development. Tassets gradually became lowered to cover the knees in a series of lobster-shell plates, as on a brass of Queen Elizabeth's reign, that of Sir William Harper, in St. Paul's Church, Bedford. Examples of these elongated cuisses occur, however, much earlier. Jambes and sollerets were at length laid aside in favour of jackboots, and plate armour fell gradually into disuse, mainly owing to the new tactics rendered necessary by the general use of firearms, and the growing desirability of lightly-armed squadrons and companies; indeed, before the accession of Elizabeth the use of armour in campaigning had ceased to be a *sine qua non*, and, all regulations notwithstanding, a constantly increasing proportion of campaigners, especially among the infantry, insisted on discarding it. It became at length more used for purposes of display rather than for actual service, and hence armour became more and more decorative. There is a scarcity of plate armour of the fourteenth century, and but little remains of the fifteenth. This is not surprising, as the quantity made in those days was strictly limited; but what does seem strange is the scarcity of armour of the sixteenth century, and especially of the first half, over which time such immense quantities were in use. One explanation of this may be found in *Archæologia*, vol. li., p. 222, when Viscount Dillon gives examples of great quantities of armour having been converted, during Elizabeth's reign, into "targets" and "jacks" for the navy.

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Now that the armour period has been roughly covered, the evolution of each important piece will be followed to its decadence, when hand-to-hand fighting was rarer, and strategy in masses more developed, as the proud knight had at length become of minor importance as against organised infantry, which was now "the strength of the battle," and when the use of various weapons of attack, especially the harquebus, became general. Tactics in warfare were at a very low ebb during the fourteenth century, and the military scandals of that time were many. Agincourt is an example of confusion among the French ranks that had many parallels at the time; but with the advent of the fifteenth century, much systematic improvement was effected. It was not before the reign of Elizabeth that any large body of troops could advance in close column without breaking its formation. Armies in the sixteenth century no longer consisted of mere feudal and communal levies, but were organised into companies and regiments, the battalion becoming the recognised unit for the infantry in the reign of James I. Systematic tactics were introduced, and the

proper proportions of horse, foot, and artillery in the field determined. The effective use of gunpowder in battle, and its influence on armour and tactics, was very gradual, but during the sixteenth century it progressively compassed great changes in both. Boys in the fifteenth and sixteenth centuries were taught the use and practice of arms at an early age. An interesting group of boys' harnesses, of various sizes and periods, may be seen at the Dresden Museum. Numerous dints on the armour, some of them heavy, show that very hard knocks had been exchanged.

The mode indicated of treating the subject will be clearer than any attempt made at elaborate contemporary classification as a whole. Representative suits, especially from local and foreign collections, will now be taken more or less in detail, thus showing the combinations of the various periods they represent, leaving separate chapters for tilting suits, extra tilting pieces, and the tournament generally, besides enriched armour and a slight sketch of prominent armour-smiths, and some of the most important collections of arms and armour.

A large proportion of the armour used in England continued to be imported from Italy and Germany. Henry VIII. bought and received in presents, harnesses, both for foot-fighting and horseback, from these countries; indeed, the trade in armour and arms formed a not inconsiderable item in the importations of the Hanseatic Bund already mentioned, and the bulk of the armour in private collections of fifteenth and sixteenth century make is of German or Italian origin. Not only was armour imported, but foreign smiths and artificers, principally of German nationality, known as Almayne armourers, were introduced. Milan armourers were working at Greenwich in 1514.¹⁹ Exportation from England was not allowed without royal licence.

Although the matchless Beauchamp effigy ([Fig. 16](#)) was the work of an Englishman, it is probable that most of the fine suits in English collections, with the least possible pretensions to any historic connection with this country, were principally of Italian or German make, up to the meeting of Henry VIII. with the Emperor Maximilian; but a good deal of English armour was turned out later in Henry's reign, and in that of Elizabeth, by the "Almayne" smiths, already referred to, brought over from Germany and Italy. The Armourers' Album at South Kensington, with drawings of twenty-nine harnesses, throws much light on the armour of the earlier Elizabethan period, and some of the suits mentioned therein have been identified. It is certain, however, that the influence exercised by the imported German and Italian smiths on armour of

English make was of comparatively short duration, for suits made by armour-smiths in this country after the early portion of Elizabeth's reign were characterised by a vast inferiority in design, execution, and material to those turned out by their German and Italian confrères. With the exception of the fine specimens in the collection at the Tower of London, it is in Germany where most of the Gothic and Maximilian suits have been preserved, and a few are still to be met with in Italy and Spain. It is a great pity that the armour possessed by the nation should be scattered over so many places, instead of being concentrated in one grand national collection. Could this be arranged, we would possess an armoury worthy of the empire. The Wallace armour is a great accession to our store, but this collection still remains unpacked. The almost constant warfare, both in Germany and Italy, during the middle ages naturally made the manufacture of armour more of a speciality in these countries than in England, and the effect of the Italian "renaissance" was especially seen in profuse and artistic ornamentation, which at length came to be more regarded even than strength itself—it was, in fact, a fine art. Much of the armour was covered with embossed figures, engraved, chased, and damascened with gold. The work of the Augsburg, Nuremberg, and Innsbruck armourers was really, if not quite, equal, both in design and workmanship, to that of Italy; and many historic suits until recently classed as Italian have been since proved to be of German workmanship.

The counties of Northumberland, Cumberland, and Durham are not rich in armour, especially in that of the sixteenth century, and the only Gothic suit is, we believe, one in the author's possession, and there is no perfect harness of the "Maximilian" type in the district. As many as possible of what may be termed north-country examples will be given in these pages.

Military experts of the sixteenth and even seventeenth centuries differ widely in their estimation of the value of steel armour in battle, and many of them strove valiantly against its growing partial abandonment. James I. is said to have made the remark that body-armour was a double protection; for it secured the wearer from being injured, and also prevented him from injuring others! It became impossible to forge armour, for man and horse, proof against the improved musketry fire; and little by little the old chivalry of battle had to give way against overwhelming odds. The full effect of the movement was, however, much retarded by various causes. The earlier firearms were clumsy, dilatory, heavy to carry, and ineffective in practice; besides new supports, formations and tactics took time to organise and develop before firearms could reap the full benefit of their superiority, which they eventually achieved with

the musket, in conjunction with “covers” of halbardiers, and especially pikemen, before these footmen’s weapons were superseded by the bayonet. These causes, and the increasing demand for lighter and more easily manœuvred troops, and newer tactics demanding greater mobility and longer marches, brought about the downfall of the man-at-arms, who was effective only on the level; and with his disuse plate-armour had ceased to be generally worn.

PART VI.

**A SLIGHT SKETCH OF SOME OF THE
MORE IMPORTANT COLLECTIONS
ABROAD.**

The Königliche Zeughaus at Berlin.

THIS museum is rich in staff weapons and firearms, and is rapidly accumulating a very fine collection of armour, which has been greatly enriched by the purchase of the remarkably fine series of suits and weapons formerly belonging to Prince Carl of Prussia. The present emperor takes a great interest in the place, and has himself added several suits of armour.

The Königliche Historische Museum at Dresden.

This is perhaps the best collection for the student to visit, and is intensely valuable by reason of the strictly historic character of most of the specimens. The only weak spot is in the absence of any complete “Gothic” harness, but there are some fine pieces on exhibition. Next to suits with the date inscribed, those that are known to have been worn by historic personages provide valuable means of comparison for the student, and define the features and details presented within narrow limits as to time. The collection was, in a manner, begun by dukes Georg and Heinrich of Saxony from 1471 to 1541, and continued under the Kurfürsts. The first inventory was ordered by Kurfürst August, 1526–86, and then comprised twenty-eight mounted tournament suits for “rennen,” with their accessories and reinforcing pieces, as well as thirty-

four tournament suits for “stechen.” Under the section in this volume headed “Tournaments” will be found explanations regarding the differences between “rennen” and “stechen.” The next inventory taken, 1576–84, exhibits the addition of a number of enriched suits, and between this time and 1611 many more were added. A large number of these historic suits stand, so to speak, almost *in situ*. In 1893 many suits and weapons were secured by purchase from the collection of Richard Zschille, and the gathering together of suits and arrangement of the foot-tournament hall accomplished. The collection thus forms a historic series of armament most unique and instructive, and at the same time most decisive in its influence on the many questions of form and opinion that have so agitated the minds of many writers on the subject. The collection of weapons of the “renaissance” and later may be described as unique in its beauty and arrangement. This section was founded in 1730, and contains an immense number of the choicest specimens, including many weapons for the chase. The collection of tools used during the sixteenth century for armour-making is most instructive and comprehensive. The catalogue by the curator, Direktor Max von Ehrenthal, is an educational book of the first order.

The Armeria Real, Madrid.

This collection has most in common with that at Vienna; and if not actually founded by the Emperor Charles V., it contains a good deal of his armour, and many weapons used by him. It was Philip II. who ordered the arrangement of the collection then existing, and his successors continually added to it; and when one considers how it has suffered from the robberies of Napoleon, and the neglect consequent upon the unsettled state of Spain for so many years, it is a matter of surprise that it has survived in its present fine condition. The collection comprises a number of most beautiful examples of armour, especially of the reigns of Charles V. and the Philips II. and III. A harness made by Koloman Helmschmied of Augsburg, for the emperor, is very notable. It bears the armourer’s mark, in conjunction with the guild monogram of the city; the suit has tuilles. There are many mounted suits, all remarkable specimens of the armourer’s art; and with the bards of one of them is a chanfrein with ram’s horns. Suits with lamboys are finely represented; as also is enriched armour. The collection of helmets, swords, shields, daggers, and separate pieces of armour and arms generally, represents almost all schools and

varieties. The “Catálogo” prepared by Count Valencia is very fully and splendidly illustrated.

Die Waffensammlung des Kaiserl. Hauses at Vienna.

This collection includes that of Ambras, and the range of examples, especially armour, is even more complete and comprehensive than that at Madrid. It is rich in the most important of all schools, viz., the “Gothic”; and the general arrangement of the examples leaves little to be desired. With Custos Wendelin Boeheim at its head, it has provided the great educational agency in Europe in the determination of both arms and armour of the different periods covered in this book.

The Musée d’Armures at Brussels.

This collection has been placed in the Porte de Hal tower, an old fortress built in 1381, and all there is remaining of the old fortifications of the city. The museum is not in possession of a complete “Gothic” suit, but “Maximilian” fluted armour is worthily represented; and a later suit, with a tournament shield, is very notable. Armour of late sixteenth and early seventeenth century is there in quantity, and the collection of arms and cannon is very important. The catalogue, compiled by the accomplished antiquary, the late Hermann Van Duyse, leaves little to be desired.

The Historische Vaabensamling at Copenhagen.

This collection is placed in the old historic Tøjhus, built in the reign of Christian IV. It is practically an arsenal. The collection of arms is arranged under the reign of each king, this giving obvious chronological data. A harness, with a tournament shield, reminds one strongly of the work of Peter von Speyer; the leg armour is missing in this case. Another suit in this collection is mentioned in our text.

The Armeria Reale at Turin.

This collection is especially rich in weapons of the sixteenth century, and is one of the most important in Europe.

The Germanisches Museum at Nuremberg.

This is a worthy national collection, and one of the most important and educational in Europe, by reason of the great range and excellence of the specimens both of arms and armour. Gothic armour is well represented. Examples are mentioned in our text.

The National Museum at Munich.

This collection is large, excellent, and varied, containing many important and historic examples of arms and armour. It possesses three Gothic harnesses, and each period is fully represented. Examples occur in our text.

Kungliga Lifrust Kammaren, Stockholm.

This collection contains some very fine specimens, most of them historic. One of the suits of armour is mentioned in our text ([Fig. 3](#)). A fine set of drawings, with an interesting and very correct text, has been given to the world by the curator, C. A. Ossbahr.

The Musée d'Artillerie at Paris.

Many specimens in this museum have been alluded to in our text. The collection has been exposed to frequent casualties, but it is worthy of France. It is, however, regrettable that so many of the excellent examples are incomplete. This collection deals more especially with the sixteenth century, and is very rich in guns and artillery.

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There are important collections of arms and armour at Erbach, Sigmaringen, St. Petersburg, Graz, Emden, Antwerp, and many other cities of Europe.

PART VII.

THE TOURNAMENT.

THE word is derived from the French “tournoyer,” to wheel round, and the name in old French was “tournoiement.” Tournaments were first instituted as training schools for the practice of arms, and were later tempered by the rules of chivalry. Jousts or justs of peace (*hastiludia pacifica*) were single combats, or a succession of such, for a prize or trial of skill; while the tourney was troop against troop. The term “passage of arms” is often used somewhat generally; but, strictly interpreted, it was a combat where several knights on each side were engaged, some fighting on foot, others on horseback. The sword was often blunt and pointless, being of whalebone covered with leather and silvered over. When actual swords were used no thrusting was permitted, but striking only. The length of the lance proper was usually about fourteen feet, the shaft being of ash; but there were several varieties of the weapon for the different “courses”; and in very early times, like the lance for battle, it was both thinner and shorter. An ordinance of the thirteenth century provides that the lance should be blunted, but this having been systematically evaded, another ordinance of the century following required the lance-head to be in the form of a coronal; and this law was for a time strictly enforced. There are examples in the Tower of London, and specimens exist in most museums abroad, notably at Dresden. The courses to be run were generally three in number. “Joustes à outrance” were to the death. Tournaments had their birth in Germany, in which country warlike games, probably inherited from the Romans, prevailed as early as the ninth century; indeed, there was an important “passage of arms” at Strasburg in the year 842.²⁰ They continued very popular after the breaking up of the Franco-Germanic empire, and formed the pastime of the higher class up to the Thirty Years’ War. These early warlike games, in spite

of all precautions, were often attended with great loss of life, and as many as sixty combatants have been put *hors de combat* at one “passage of arms.” They were always popular in France, and held there on a large scale; indeed, it is claimed that the “tournoiement,” properly so called, had its birth in that country, where it is said to have been instituted by Geoffrey de Preuilli, who died in 1066; and these warlike games were very much in vogue during the reign of Philip Augustus. The armour and weapons for the tournament at this time were the same as those used for battle, and continued so until after the reign of Edward III.; but the lighter form of lance was common in France long after it had been discarded in the other countries mentioned, and the French shaft was made of sycamore or fir. It was not before the beginning of the twelfth century that jousting or fighting with lance in rest became common; in fact, until then the lances in use were unsuitable for that purpose. Much information regarding the armament of combatants, the usages to be observed, and the regulations as to heralds, pursuivants-at-arms, esquires, and varlets, besides many interesting details, is contained in the *Statutum Armorum ad Torniamenta*, written towards the end of the thirteenth century. New and more stringent rules had become necessary, because of the frequency of the “joust of peace” degenerating into one “à outrance.” This evil had become so great that the Pope forbade the games in England, and King Edward III. repeatedly issued fiats against them, and so also did his successor; still the Crown frequently issued licences for tournaments being held. An excellent description of the arms and armour employed at a later age may be found in the Tourney Book of King René d’Anjou (*Tournois du Roi René*), illuminated by himself, with a most minute statement of the rules, ceremonial, and courses; and in it is a graphic account of the combat between the Dukes of Brittany and Bourbon. A miniature in this book exhibits a knight entering the lists with great ceremony. The first regular tournament in England occurs in the reign of Stephen, and another was held very early in that of Henry II., but its consequences were of such a nature as to induce that monarch, at the pressing instance of the priesthood, to forbid these games. So great, however, was their popularity that they continued to be held in spite of the king’s fiat, though it was not before the reign of his heroic son that they became common, and were then kept in strict bounds by royal ordinances. Henry III. charges his subjects that “they offend not by tourneying,” and, as already mentioned, even as late as 1299 edicts were issued against the

games. There were only five authorised centres for lists in England, and four of these were south of the Trent. Tournaments in the northern counties required a special licence. Earls competing were obliged to pay twenty marks to the king, barons ten marks, and knight-bannerets and bachelors two to four marks, according to estate. The plan of the earliest lists was circular with palisades, but the form was afterwards changed to square rather longer than broad, and the latest were often made oblong. They varied very much in size, and were ornamented with tapestry and heraldic devices. Permanent lists were often enclosed by a ditch or moat. Roofed-in wooden erections, sometimes with sloping galleries for the spectators, were usually placed at the sides of the lists, and were often highly decorated. The marshals of the lists, heralds and pursuivants-at-arms, were stationed within the enclosure to take note of the various incidents taking place among the combatants, and it was the duty of the first-named to see that the rules of chivalry were strictly observed. Varlets were in attendance to assist the esquires in looking after their masters, especially when unhorsed. Trumpets announced the entry of each competitor, who was followed by his esquires into the lists. Each knight usually bore on his person some token from his lady-love, which was disposed on his helmet, lance, or shield. A prize was bestowed after a tournament, and presented with great pomp and ceremony. The arms and armament of the vanquished fell as spoil to the victors, unless ransomed by a payment in money. This was, however, only the case in jousts of courtesy, not in combats “à outrance.”

During the fourteenth and fifteenth centuries an immense amount of artistic skill was freely lavished on armour for the lists, as well as on that for purposes of parade. It was common to hold a “passage of arms” for three days; two of them for contending on horseback, and the third on foot. Lances were used on the first day, swords and maces on the second, and pole-axes on the third. Those open to all comers were termed “joutes plenières.” Pluvinel, who wrote at the close of the reign of James I., says: “There ought to be at each end of the lists a little scaffold, the height of the stirrup, on which two or three persons can stand, viz., the knight, the armourer to arm him and his assistant, and hence he mounts his steed.” Froissart, writing towards the end of the fourteenth century, gives a graphic account of the tournament in his day. Judicial combats were common throughout the century, and usually took place within the nearest lists. Trial “by ordeal,” or the judgment of God, was a strange outcome of the

Christian faith as practised during the “dark ages” of our era. It implied, of course, a strictly personal God, who specially interested himself in the doings of every one, and a simple, child-like faith that the Omnipotent would order victory to the just cause and protect the innocent from injustice. The “ordeal” was by fire, hot iron, boiling water, and by the sword. It was suppressed towards the end of the twelfth century, and was followed by that of single combat, “God shewing the right.” This method was in full accord with the chivalrous spirit of the times. Old persons, women, and minors were represented by “champion.” The combat might continue from noon to sundown, and if it lasted as long the innocence of the accused was established and proclaimed. This form of combat was only applied in the cases of crimes punishable with death, and only when merely circumstantial evidence was available. A figure of a judicial combat occurs in the *Conquêtes de Charlemagne*, a manuscript of the fifteenth century in the National Library at Paris. The combatants wore chain-mail, with genouillières and coudières, the period represented being late thirteenth or early fourteenth century. An angel superintends the duel.²¹

The custom of “judicial combats” fell into disuse in the fifteenth century.—We must confess to a lively partiality for the history of Sir Walter Scott, in spite of his facile imagination and palpable inaccuracies, and think the graphic picture of “The Gentle and Joyous Passage of Arms” at Ashby-de-la-Zouche, with “La Royne de la Beauté et des Amours,” gives as delightful an account of a tournament in the times of Richard Cœur-de-Lion as need be wished for. The gallant knights are distinguished by their belts and gilded spurs.

“The knights are dust
And their good swords are rust,
Their souls are with the saints, we trust.”

In the specification for arms and armour for the tournament of Windsor Park (1278) we see of what each suit consisted, viz., “one coat of fence, one surcoat, one pair of ailettes, two crests (one for the horse), one shield (heraldically ensigned), one helm of leather (gilded or silvered), and one sword made of whalebone.” The cost of each armament varied in price from

about ten to thirty shillings. The shields were of wood, costing fivepence each. The total cost of the combined thirty-eight armaments was about £80. Chaucer refers to tournaments in the following lines:—

“The heralds left their pricking up and down,
Now ringen trumpets loud and clarion.
There is no more to say, but east and west,
In go the speares sadly in the rest,
In goth the sharp spur into the side,
There see men who can just, and who can ride;
There shiver shaftes upon shieldes thick,
He feeleth through the heart-spone the prick;
Up springen speares, twenty feet in height,
Out go the swordes to the silver bright,
The helms they to-hewn and to-shred:
Out burst the blood with stern streames red.”

The leading “courses” of the tourney are fully described later in the paragraph devoted to German methods, which, though there were many more varieties, were practically those of England, where there was also the round-table game, etc. Matthew Paris mentions a “round table game” held at the Abbey of Wallenden in 1252; and Earl Roger de Mortimer held one at his castle of Kenilworth in 1280, and Edward III. another at Windsor in 1344. This form of tournament seems to have been very popular in England during the thirteenth and fourteenth centuries; but there is no clear definition of its peculiarities given by any of the few chroniclers who mention the subject. The idea of the knights sitting round the table seems to have been an assertion of the principle of equality so as to avoid questions of precedence—one full of difficulty in all ages.

Tilting was practised during the fourteenth century very much as in the century following. A joust of about 1330 is figured on “The Codex Balduin Treverenses,”—the horses bear housings, and the knights mantles. The armament for jousting and battle began in this century to show some difference from that of earlier times. The games continued in unabated vigour throughout the middle ages and the “renaissance,” and until the

general use of firearms rendered such exercises no longer of much practical value.

The necessary limits of this work will not admit of any detailed description of the many and curious rules, usages, and limitations which were absolutely necessary for carrying on these dangerous games without great and unnecessary bloodshed and the loss of many valuable lives, but much can be seen in a set of regulations prevailing under Henry VIII. in the tournament roll preserved in the Heralds' College. Students of the subject will do well to read an able paper in the *Archæological Journal*, vol. lv., No. 219, entitled "Tilting in Tudor Times," written by Viscount Dillon in 1898; and a most excellent and comprehensive account of the German "turnier," and weapons used, exists in Herr Wendelin Boeheim's work, *Handbuch der Waffenkunde*. This is a veritable text-book.

Tournaments and tilting generally were, however, rendered less dangerous than might have been expected by the addition of reinforcing armour, which pieces were screwed on over the more vulnerable places, on armour made for ordinary fighting purposes, and for some courses of the tournament, mainly on the left side, which received most of the blows; indeed, these extra pieces constituted a double defence of iron for the head, chest, and left shoulder. This was obviously rendered necessary when one considers the terrible impact of the lance in full career with the breastplate or helmet. These extra tilting pieces made their appearance in the reign of Edward IV., but they were known in Germany several decades earlier. It was early when suits of armour were made differently for battle and for tournaments, as William Lord Bergavenny bequeathed to his son "the best sword and harness for justs of peace and that which belong to war."

Late in the fifteenth century there were complete tilting harnesses of such immense weight that a knight once unhorsed lay on the ground absolutely helpless, and often could not rise without the assistance of his varlets. His movements when on horseback were very restricted. These suits were of such resisting power as to give practical immunity to the wearers so far as wounds were concerned, but they were far too heavy to be used in the *mêlée*. A tilting harness with the Nuremberg mark, in the splendid collection at that city, is of immense weight and strength, and the example is specially valuable, as the date 1498 is inscribed on the cuirass. The knight could barely move in the saddle, and was able only to guide his horse and aim his

lance. Armour made specially for the tilt-yard will be described later in these pages, and illustrations given.

There is an account of a tournament held in the reign of Henry VIII., in a tournament roll preserved in the Herald's College. The challenged (*Les Venantz*) were nine in number. The armour worn was of the heavy tilting class, with lamboys; and the horses were fully barbed, with housings. It would appear from the barrier between which the knights ride that this was the "Italian course," known in Germany as the "Welsche Gesteck." This barrier was first of cloth hung on a rope, but afterwards of wood; and then the great knee-guard came into use to protect the knee from being crushed against the barrier, the height of which was usually about five, or even six feet. The meeting between Henry and Francis on the Field of the Cloth of Gold, in 1520, was the occasion of at least one tournament. The king himself was one of the challengers. One of the drawings shows him as breaking a lance with his opponent. It is certainly desirable at this point to give somewhat full particulars of the leading modes of jousting as practised in Germany at the end of the fifteenth, and during the sixteenth century, as it was here where these games were most frequently practised, and the German archives fortunately yield us very full particulars, which throw much light on the subject generally.

The Emperor Maximilian and our Henry VIII. were great patrons of the tournament, often taking part in it, and so were all the German princes of the sixteenth century. We find very full particulars of Maximilian tournaments, as held during the emperor's reign, in the *Turnierbuch des Kaisers Maximilian I.*, a synopsis of which has been written by Quirin von Leitner. This "Triumph of Maximilian," dictated by the emperor in 1512, affords much information on this subject; and in it many of the forms of tourney are represented, with the various weapons and armour used in the different courses. The *Turnierbuch of the Emperor Maximilian I.* would have been both incomplete and inconclusive without the masterly drawings by Hans Burgkmair, painter and engraver, of Augsburg. This artist seems to have been closely associated with the great master Lorenz Kolman, surnamed "Helmschmied," and doubtless did designing and engraving work for him. Courses of rather a later period are described in *Hans Schwenkh's Wappenmeistersbuch*, written in Munich in 1554; besides which there are several "tournament books" of the German courts giving not only general

descriptions of the games, with the rules and regulations practised, but also full accounts of particular encounters concerning which we have the harnesses fought in standing before us for reference to-day. There are also many original prints preserved giving particular examples of these games. Furthermore, Dr. Cornelius Gurlitt has given an excellent *résumé* of tournaments from the middle of the sixteenth century up to the Thirty Years' War, derived greatly from the archives at Dresden. Herr Wendelin Boeheim, the curator of the imperial collection of Vienna, gives many details in his great work, *Handbuch der Waffenkunde*. The author has had the advantage of many personal hints concerning the German forms of tournament from Max von Ehrenthal, the accomplished curator of the Dresden collection, and he owes much information and several of the illustrations given under this heading to this gentleman's kindness and liberality. Dr. von Ubisch, the director of the collection at the Zeughaus, Berlin, has also assisted him greatly, especially concerning ordinary fighting suits and other matters.

Tournaments of the sixteenth century were mostly for diversion and practice, and it was very rarely that any great injury was sustained. It will be seen from the descriptions here given that it was mainly a question of concussion, in the splintering of lances, or being rolled on the ground, the hardness of which was greatly modified by a liberal covering of tanning refuse. The stunning effects from the strokes of the sword and mace, as felt on the inside of the thick defences used, must have been very trying, and one fails to understand how so comparatively little damage to life and limb was experienced in the riders being hurled from their steeds, encased in their heavy panoply of more than two hundred pounds in weight; and what makes this the more extraordinary is that the rider was helped on to his horse again after a fall and ran again, and this sometimes happened several times: but judging from the records preserved, and there are many, the casualties in the tilt-yard of the fifteenth and sixteenth centuries were little, if at all more numerous and serious than those in the hunting or football fields of to-day; or in the duels that were common so recently at German universities; or for the matter of that in the accidents arising from the use of the cycle. This comparative immunity from serious injury in the tilt-yard was partly accounted for by the assistance rendered by the varlets in helping the horse to keep his feet, and the rider his seat after impact, and also in assisting in breaking the fall of the rider.

SHARFRENNEN.

This form is characterised by heavy lances “sharp,” as the name for the course implies. The main object was “unhorsing,” and the saddle was unprovided with front and rear supports; it was, in fact, quite unlike the ordinary war-horse saddle—indeed, more resembling the English saddle of to-day. The object of this was that there should be nothing to impede the rider’s fall. The lances used in this course were not expected to break or splinter, though they did so sometimes. On the moment of impact each combatant dropped his lance to avoid injury to the arm from splintering, and this was the case in the other courses also. The consequence of a true impact was, as a rule, that at least one rider was unhorsed; but sometimes both riders fell, and occasionally both horses as well, so that all four combatants, for the horses may be said to have fought also, bit the dust. In cases where a rider was able to keep his saddle for a moment after impact and swaying in the endeavour to retain his seat, his varlets rushed forward to support him. Sometimes in case of lances slightly deflecting, or missing altogether, one and even both horses have been known to fall forward. There was a “rennen” between the Emperor Maximilian and Duke John of Saxony at Innsbruck in 1498.

The tournaments held at the imperial and princely courts were strictly games, the hosts often personally challenging their guests to a trial of skill. Much depended naturally on the training of the horses, which were sometimes ridden blindfold. The legs and feet of the competitors were without armour, except the “diechlinge,” so that the rider could sit firmly supported on the saddle. The “diechlinge” served as a protection for the thigh and knee. Such a defence was necessary, owing to the risk of these limbs of the combatants colliding. In the Dresden Museum, in the “Turnierwaffen-Saal,” an interesting and very realistic representation of a German “Sharfrennen” may be seen, the combatants facing each other, fully armed, with lances in rest. The defences are double throughout, each harness weighing about two hundred pounds. The period is 1550–53, and most of the riders in the “Saal” have sat their horses since the year 1591. The body-armour is engraved and fluted, and the helmet is the sallad. The breastplate of the harness nearest the entrance to the hall bears the monogram of the armour-smith Sigmund Rockenberger of Wittenberg, the

other was made by Hans Rosenberger of Dresden. The grand-guard, volant-piece, and left shoulder-guard are of wood, strengthened with plate, and covered with leather. A curved plain shield is screwed on over the left shoulder, while an enormous vamplate, or shield with a bouche, guards the right, and through this the butt end of the lance rests.

The armour itself is of the heavy tilting kind, over which is a dress of stuff with bases, a sort of petticoat like the civil dress of the day. Stockings and slippers are worn, and there is no defence of plate over them excepting at the knee, over which is the great “diechlinge” already mentioned. The woollen stockings and slippers in these instances are restorations; but there is an actual tilting shoe of the period in one of the museum cases at Dresden. Spurs with long necks are used. The horses are barded and fully housed. Housings reaching nearly to the ground are usually highly and fancifully decorated, bearing the “arms” or “cognizance” of the rider, and are often ornamented with the figures of birds or animals. In the Royal Library at Dresden is a representation on parchment of a “Sharfrennen” between Kurfürst August of Saxony and Johann von Ratzenberg, and afterwards with Hans von Schönfeld, in 1545, at Minden. It was drawn by Heinrich Goding, the court painter in 1584. This combat was termed a “Gedritts,” signifying that the victor, in order to gain the prize, had after the first encounter still to dispose of a second antagonist—three were thus engaged, and hence the term. A copy of this interesting record follows in [Fig. 4](#). An example of the armour worn in this course is given in [Fig. 5](#). It was made for Kurfürst August, by Sigmund Rockenberger of Wittenberg in 1554. The form is graceful, and the ornamentation of a chaste character. The details are clearly marked, such as the screw for the volant-piece; the sharp-pointed, spearhead-like projection standing forward from the centre of the breastplate, a fashion that only endured for a few decades; the ponderous lance-rest, and heavy abdominal extra plate,—all being characteristic of a suit used for “rennen.” The elegant sallad differs materially from the earlier form, and is very shapely. Only persons of noble birth or those subsequently granted “arms” were permitted to take part in “rennen.”

DAS DEUTSCHE STECHEN (THE GERMAN GESTECH).

Herr Wendelin Boeheim, in an article in the *Zeitschrift für historische Waffenkunde*,²² says that the “old German Gestech” was far from having been introduced during the reign of the Emperor Maximilian I., as has often been supposed, but is of much earlier origin. This course depends much more on adroitness and skill than in the Italian joust, when the knights tilt with a barrier between them, and the rider gets no assistance from his charger, as the chanfrein is without ocularia, and sometimes its ears were stopped with wool. The lance, unlike that used in “rennen,” is tipped with a coronal.²³ The “Stechtarsche,” or small ribbed shield which is tied to the left shoulder by laces, affords grip to the coronal of the lance; and this is the point aimed at. The saddle used for this course has an upright front plate, but none behind, so that there was no impediment in “unhorsing.” Later, the front plate disappeared. The horse bears no bards beyond the chanfrein, but there is a cushion filled with straw fastened over the horse’s breast, as a protection against collision. There are several kinds of “Stechen,” but the rule in all is to have no leg armour, and this was in order to give the rider greater command over his seat: the lance hand bore no gauntlet. Quirin von Leitner gives a figure of the Emperor Maximilian I. armed for the German Gestech.



Fig. 4.—Sharfrennen at Minden, ran in 1545.



Fig. 5.—Suit at Dresden for Sharfrennen, date 1554.

Instead of the fifteenth century sallad, a “Stechhelm” was worn in the sixteenth; and it was shaped something like a bucket. Brassards were always used in this course, while jambs and sollerets were usually dispensed with.

Early examples of armour made for this course may be seen in two very fine suits on exhibition at Nuremberg. The ponderous lance-rest stands free by reason of the cuirass being flattened on the right side. The breastplate, which bears the date 1498 and the Nuremberg guild monogram, is in two pieces, one of which is a reinforcing plate for the extra protection of the lower body, and this is fastened on to the main plate by large screws with very big heads. The lance-rest is supplemented with a queue screwed on behind, and curving downwards to hold the butt end of the lance. The right arm has heavy brassards; while on the left the heavy vambrace and gauntlet are in one solid piece, and quite plain. There is an immense rondelle on the right side, with a bouche cut out of the lower part to make room for the lance. The older of these suits has a sallad, while the one that is rather later is provided with a “Stechhelm,” which is very heavy and in one solid piece with the mentonnière, and strongly fastened on to the breastplate by screws; while a permanent socket and screw attach it to the backplate. The helm is thus immovable when fixed; it is roomy, and permits the head to move about freely within. These suits are so heavy and ponderous that the combatants could do little more than hold their lances in position; and if unhorsed, lay like logs where they fell, being unable to rise without the assistance of their varlets. In arming, each piece required to be screwed on, one after the other. The later of these suits is taken for illustration in [Fig. 6](#), as it bears the more characteristic “Stechhelm.”

THE ITALIAN COURSE, OR “WELSCHES (ITALIAN) GESTECH,” ÜBER DIE PALLIA (OVER THE BARRIERS).

This course first appeared in Germany about 1510, but it doubtless originated in Italy, as its name implies, and the Italian name for barrier is “pallia.” It was fought with lances tipped with a coronal, the same as in “the German Gestech,” but the main difference between that course and the

others under discussion is the presence of a wooden barrier about five feet high, along which the two riders charge, with it between them. In this course the legs and feet were generally armoured, though there were exceptions to the rule. There are very full particulars in *Freydal*, the book in which the tournaments of the reign of the Emperor Maximilian I. are drawn and described; and this form of tournament is figured in the tournament roll of King Henry VIII. preserved in the Heralds' College. The knights in their career had to hold their lances on the left side of the horse's head. Originally the main intention was to unhorse: still the splintering of lances was of more frequent occurrence than in the before-named courses, as the saddle here was furnished with high front and rear supports, rendering it in fact "well" shaped, so that the riders sat much more firmly in their seats than on the "renn" saddles, especially those which were without supports. Soon after the middle of the sixteenth century a change took place in the armour for tournaments; and with this came a modification in the lances also, which became lighter, so that they mostly splintered on impact, and in such cases the riders were but rarely hurled from their saddles. For the later Italian course harnesses were worn, as in [Fig. 7](#).



Fig. 6.—Tilting Suit at Nuremberg, for the German Gesteck.



Fig. 7.—Tournament Suit for the Italian Course (Welsches Gestech).

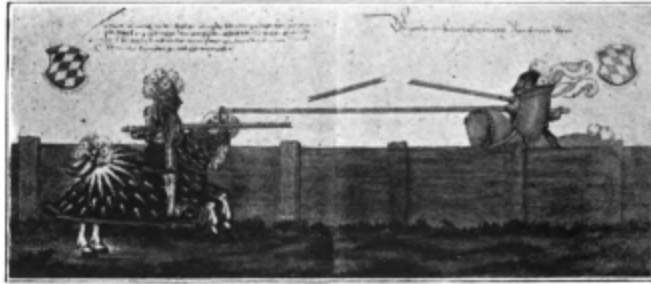


Fig. 8.—An Italian Course at Augsburg in 1510
(Welsches Gestech).

The helm for this course differs somewhat from that worn in the others in being provided with a little opening or window on the right side for fresh air. The cuirass is not flattened on that side, as in [Fig. 6](#). There are other differences, all of which may be seen on a suit in the Armeria Real de Madrid. In the old form of “Welsche Gesteck” the rider wore sometimes the armour used for the ordinary “Stechen” course. In Leitner’s *Freydal* an example is figured; and there is an actual harness, by Wolf von Speyer of Annaberg, in the Turnierwaffen-Saal at Dresden.

The illustration ([Fig. 8](#)) gives an excellent rendering of this course as it was run between Duke Wilhelm IV. of Bavaria and the Pfalzgraf Friedrich bei Rhein, at Augsburg, in 1510. It has been taken from Duke Wilhelm’s tournament book.

THE FREITURNIER, OR FREE COURSE.

This course received its name in contradistinction to the “Welsches Gesteck,” because it was run in the free field or lists, without any barrier between the combatants. In this respect it resembled the old German “Stechen,” and to a certain extent grew out of it. This form however does not occur, under the name, before the second half of the sixteenth century. The armour for the Freiturnier differs from that of the “Welsches Gesteck” (Italian Course) in the particulars that a grand-guard was screwed on to the left shoulder and chest, instead of the tournament shield used in the Italian course. To the left elbow was screwed a garde-de-bras of larger dimensions than that used for the Welsches Gesteck. Armour for the tournament was now usually so arranged that by the interchange of reinforcing plates the same suit could be made available for both these forms of tournament. The lance and horse furniture were exactly the same in both cases, and the body armour of the rider very similar, subject to the interchange of the reinforcing pieces already alluded to. The suit selected for illustrating the armour used for this course ([Fig. 9](#)) forms part of the remarkable collection at Dresden. It is a fine example in plain armour of about 1580. The breastplate, it will be observed, is the “peascod.”

THE FUSSTURNIER.

This is the foot-tournament which originated in the sixteenth century, and is very different from the courses on horseback. Full particulars can be seen in the *Akten des Dresdener Oberhof-marshallamtes*, anno. 1614. An extract (in translation) from this work by Dr. Cornelius Gurlitt runs as follows, viz.:—

“The one who shivers the greatest number of lances in the most adroit manner shall have the lance prize; and he who in five courses strikes the bravest and strongest with the sword shall have the sword prize.”

This extract furnishes a sufficient outline of the game. Like the “tourney,” it was troop against troop. Each combatant had to exchange three charges with the lance across a sort of barrier; and five strokes with the sword, all directed towards the head, not only with one but with every opponent on the opposing side; and prizes were awarded as set forth in the extract. No prize was awarded unless the lance splintered, nor was any given in cases where a combatant had stepped or been driven backwards in any way. Striking below the belt was forbidden, for no leg armour was worn. The locking gauntlet was expressly forbidden.



Fig. 9.—Armour for the Freiturnier at Dresden.

It is very interesting to find that a suit used in a “fussturnier” by Kurfürst Johann Georg I. of Saxony is now in the Dresden collection. It is by Anton Peffenhauser of Augsburg. The harness used was the ordinary fighting kind. The lance was held in both hands.

KOLBENTURNIER, OR BASTON COURSE.

This is a variety that first appeared early in the fifteenth century. It was a dual combat on horseback, and was not in vogue for more than a century. The weapon used was a “baston,” a short wooden polygonally cut mace, thickening towards the end. The helmet for this course was heavy and round, with a strong grated front. The head did not touch the helmet at all, for the “baston,” being made of very heavy wood, was a dangerous weapon for striking. An example of the saddle used in this course may be seen at the Nuremberg Museum. It is so constructed that the rider cannot well fall off.

There were a number of other courses, but the differences were only trifling, consisting mainly in humorous devices and fashions in costume. During the closing twenty years of the sixteenth century, and the first twenty in the following, the Hungarian tourney was much in vogue. This course obtained its name solely from the dress worn—the spurs used were very long.

Running at the ring can hardly be classed under the tournament category. It was called “Ringelrennen” in Germany, and was much in favour at the Saxon court from 1570 to the end of the seventeenth century. The lance used was shorter and much lighter than that for tournaments. There is a specimen at Dresden which is tipped with a cone, to hold the ring when hit, and there is naturally no vamplate.

REINFORCING PIECES FOR THE TILT-YARD.

These may be divided into two classes, viz., those extra pieces appertaining to purely tilting armour, made specially for the lists, and those used to augment the strength of ordinary fighting suits donned for the lists.

The former class comprises the grand-guard and volant-piece, often in one plate, but sometimes screwed together, the latter piece being provided with an ocularium on the right side only. These plates defend the breast and face. A small wooden shield, plated and covered with leather for the left shoulder, screwed or tied on. This piece is in some courses the objective of the lance. The heavy elbow-guard which protects the elbow, and half-way up and down the arm. The German tilting arm-guard and gauntlet was often in one piece from the shoulder. The right side is further protected by an enormous vamplate, which in the German form covered half the arm on both sides of the elbow. There is also a large knee-plate, the Germans call a "diechlinge," which is sometimes fastened to the saddle, the leg passing between it. This piece is more especially used in "Sharfrennen." Suits for "rennen" and "stechen" were usually made so that they could be worn by any man of anything like medium height, and it was quite common for one knight to borrow the harness of another.

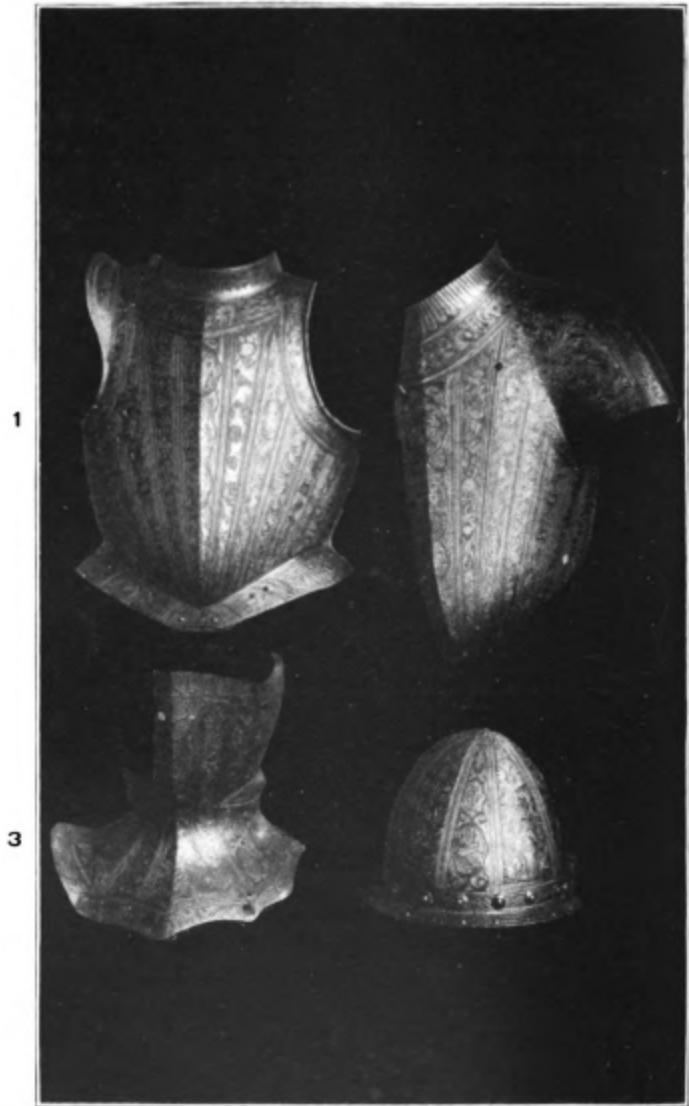


Fig. 10. Reinforcing Pieces.

No. 1 is the Breastplate for Tilting, and on it are the holes for the insertion of the screws of the Lance rest. It differs but little from the ordinary Fighting Breastplate with the suit.

No. 2 is an extra protection for the left breast and shoulder. This is the Grandguard.

No. 3 is the Volant-piece, a protection for breast, neck, and face. [24](#)

No. 4 is the Cabasset for the suit.

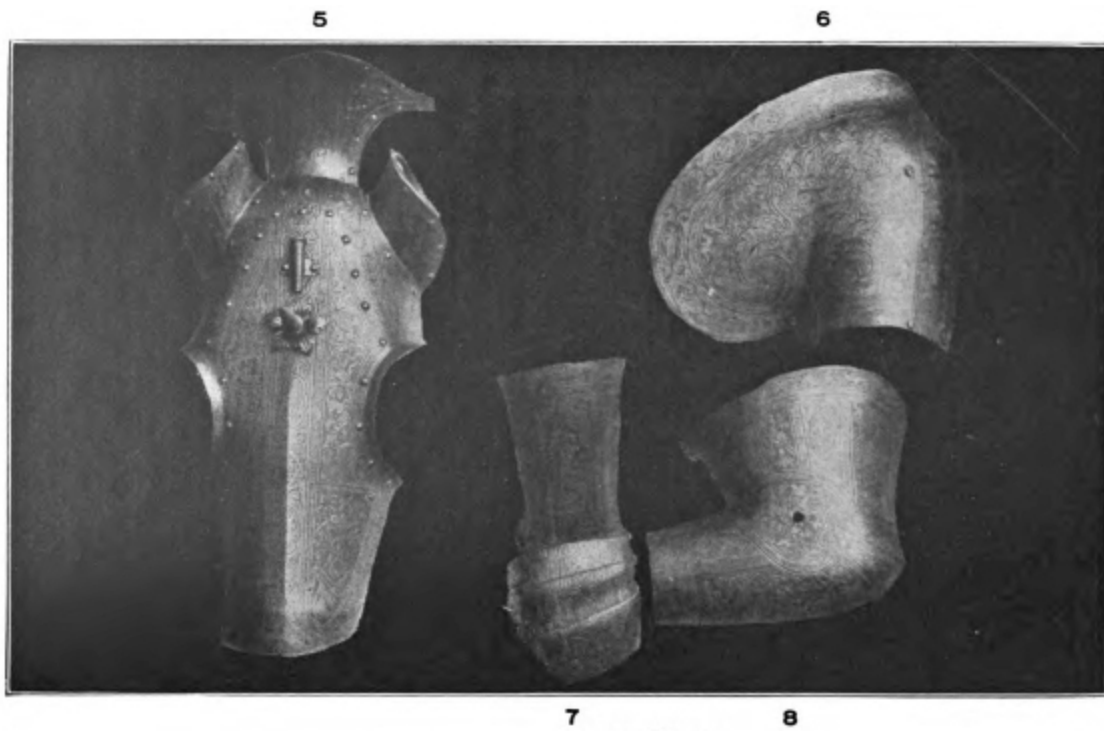


Fig. 11.—Reinforcing Pieces.

No. 5, the Chanfrein for the horse's head.

No. 6, the Extra Shoulder-guard.

No. 7, the Manifer or Tournament Gauntlet for the bridle hand.

No. 8, the Elbow-guard, or Pass-guard. [25](#)

As to the reinforcing pieces for screwing on to ordinary armour, drawings are given of a series of these plates, belonging to a splendid suit at Munich that was worn by the Prince-Bishop of Salzburg (Wolf Dietrich von Raitenau), illustrated in [Fig. 35](#). The pieces are numbered on the drawings for reference, one and upwards in Figs. 10 and 11.

A projection called the queue, screwed on to the back plate, supports the butt-end of the lance. The suit and all the pieces are richly inlaid with gold, with the Bishop's arms engraved on the breastplate. There is a suit very similar in form and details of the pieces in the Tøjhus, Copenhagen, but the ornamentation of that suit is much bolder, having the thistle for its theme throughout. It is of French make. As in the Alnwick suit ([Fig. 33](#)), the cuisses are in two parts, the upper being detachable, and the tassets bear evidence of missing detachable portions. An interesting feature of this suit is that the lance-rest is so adapted as to be capable of being either raised or lowered. Boeheim states that he has not seen any examples of these reinforcing pieces of an earlier date than about 1510. These pieces, with interchangeable plates, were very numerous in some cases where expense was no object. A suite made for the Archduke Ferdinand of Tyrol, by Jörg Seusenhofer, consisting of a field-harness and a suit for foot-fighting, had appertaining to the two suits as many as thirty-four interchangeable and reinforcing pieces. They were made in 1547, and are now at Vienna.

PART VIII.

DETAILS OF DEFENSIVE PLATE ARMOUR.

THE GREAT HELM.

THE real great crested helm, so often seen pillowing the head in effigies, dates from the last quarter of the thirteenth century, but it was rarely used except in tournaments after the fourteenth. This helm has been described in a previous section. It was replaced for fighting purposes by the visored bassinet, the movable aventail being added about the reign of Edward II. There is a perfect specimen of this helm at Berlin; it was found near Bubad, in Pomerania. An illustration has been given in [Fig. 2](#).

The great jousting helm of the fifteenth century was made wide, very strong, heavy, large, somewhat flat at the crown, and often in the lighter tilting helm had an aperture on the side for speaking. It was crested, and rested on the shoulders, being attached to the body armour by screws front and rear, and was so large that the head of the wearer did not touch it in any part; a cap was worn over the head. The attachment to the cuirass was a new departure. The top is flatter, and the ocularium, which is wider than in the older forms, can only be used for sighting by lowering the head. The plates meet sharply in front, producing a ridge, the higher end forming a beak-like projection. It fell a good deal into disuse during the reign of Henry VIII. There are two very fine tilting helms in the Rotunda collection, Woolwich, one of which was formerly in the triforium of Westminster Abbey, and weighs 18 lb.; the other, which was acquired from the "Brocas" collection, weighs 23 lb. A north-country example of the German "Stechhelm"

([Fig. 12](#)) is in the collection of W. D. Cruddas, Esq., M.P., of Haughton Castle, Northumberland. The kolbenturnier helm is a variety specially used for that course; the whole front is composed of transverse bars. These helms were firmly screwed on to the breastplate and therefore immovable, as may be seen on the tilting suit ([Fig. 6](#)).



Fig. 12.—Tilting Helm at Haughton
Castle, Northumberland.

THE BASSINET.

This helmet, the German “beckenhaube,” was round or conical, with a pointed apex. The large bassinet of the thirteenth and fourteenth centuries was very similar in all the countries of chivalry. It fitted close to the head, and was covered by the great helm in tilting. An example may be seen in Lincoln Cathedral. Before the visor appeared it was often fitted with a detachable nasal. As soon as the helm became visored, say in the first half of the fourteenth century (see an example in Alvechurch, Worcester), it assumed a great variety of form, and towards the end of the century often projected to a point like a beak. Other forms were concave, convex, and angular. Most of these may be seen in Stothard. There was also the small bassinet or cervelière, sometimes called cerebrerium. It was often worn under the hood, with a small quilted cap next the head. In the reign of Henry V. the bassinet became more like the sallad. The effigy of the Black Prince shows how the camail was attached to the bassinet by a silken lace through staples. There are some fine examples of the visored bassinet in the Johanneum at Dresden.

THE SALLAD.

Visored sallads, with a peak behind and slits for vision, appear in the reign of Henry VI. The form is a low obtuse oval ridged in the middle; it replaced the bassinet, but was never used as an under helmet. It was generally associated with armour of the second half of the fifteenth century, and used with the mentonnière, which, when fixed, afforded excellent protection for the face and throat. The distinguishing feature is the peaked collar behind, which rests between the shoulders, and the helmet was occasionally, in the earlier forms, provided with a hinged nose-guard. It was worn at an angle, so that the ocularium came in the direct line of vision, and had often a movable visor. It measured in extreme cases as much as nineteen inches from back to front. An example of the time of the Roses hangs in St Mary’s Hall, Coventry, and there is another in the Priory Church

at Hexham. The earliest representation of this form of helmet in England, that the author knows of, may be seen at Castle Donnington, Leicestershire, on a brass of Sir Robert Staunton, who died in 1458. This sort of helm is in several varieties, and a simple form was in use among the rank and file, especially by archers. There are several of these helmets in the “Rhodes” collection at the Rotunda, Woolwich, and actual specimens of typical Italian and German forms are to be found in most of the German collections of armour; there are examples in the Tower. Illustrations of sallads are given in [Fig. 13](#).

ARMET AND CLOSE HELMET.

This is the most perfect form of helmet and the most familiar, so much so indeed as to render any description almost unnecessary. It may be said to have been evolved from the sallad and mentonnière, in the sense that the bavier took the place of the latter; but instead of being slipped on over the head like the bassinet and sallad, it was constructed to hinge over it, and strictly followed the outline of the head and neck. Its form is globular, with a guard for the back of the neck, and in front round the chin is the bevor or bavier. The space between this piece and the rim of the crown-piece is filled in by a movable visor, which is pierced with narrow openings for vision and air. It thus consists of at least three pieces—the skull-piece, the visor, and the bevor; the visor is usually in one piece. It is beaked, and exhibits a series of ridges with air slits in the indentations. The crown-piece is usually combed. During the second quarter of the sixteenth century the visor was made in two plates, the upper closing inside the lower—the upper plate could be lowered at pleasure, without moving the one below. The Seusenhofer armet in the Tower is a masterpiece of the kind, being composed of six pieces, working one within the other. English armets date from the last decade of the fifteenth century, perhaps a little later. They were to be met with in Germany as early as the middle of that century. It is impossible to make much distinction between the armet and close helmet, which latter was the improved armet of the sixteenth century. A camail was sometimes used with the earliest form of armet. Illustrations of this head-piece may be seen on several of the suits given in this volume.



Fig. 13.—Sallads and an early Burgonet.

BURGONET.

This is a helmet of the sixteenth century of Burgundian origin, as its name implies, with a hollow rim at the bottom, which fitted over the projecting edge of the gorget. It was made in close imitation of the head, and in either three or four parts. This helmet was designed to meet a defect in the armet, for there was a weak place, where the casque came in contact with the body armour. This arrangement permitted the head to move freely to the right or left without leaving the neck unguarded. There is a handsome specimen of the first half of the sixteenth century at the Rotunda, Woolwich, weighing nearly eight pounds, with a fluted crown-piece, and round the neck a wreath of roses is engraved. There are holes in the crown for the wreath and mantling. There are some important beaked varieties at Dresden and Berlin. The more modern burgonet has neck-guard and oreillettes or ear-flaps of steel. An illustration of an early burgonet is given in [Fig. 13](#).

MORION, CABASSET, AND CASQUE.

The morion first appeared in England in the reign of Henry VI., and was introduced into Europe by the Spaniards, who got the design from the Moors, as the word implies. It is an oval helmet, and has a high comb-like crest and almost semicircular brim, peaked at both ends. The cabasset is a helmet similar in character to the morion, and generally peaked. Both varieties were worn for foot fighting, and are often lighter than earlier helmets, and usually richly engraved. The Baron de Cosson²⁶ says that “the cabasset first appears in an ‘ordonnance’ of Francis I., who orders that men-at-arms wear the armet, light horse the sallad, and ‘les arquebusiers seulement le cabasset pour viser mieux et avoir la tête plus délivre.’ The cabasset did not impede the aim, and was therefore the proper headpiece of the musketeer.” Casques are open helmets like the others, and of classical design. There are illustrations of a cabasset in [Fig. 11](#), and of bassinets, morions, etc., in [Fig. 49](#).

GORGET AND MENTONNIÈRE, OR BAVIER (BEVOR).

The mentonnière was used specially with the sallad, and the chin-piece fulfils the same purpose with that helmet as the bavier does with the armet; it fastens on to the breastplate by a staple and cusped catch, or goes partly under that piece. The upper portion, to cover the mouth and chin, is of laminated plates, which move up and down at pleasure, but always from below. In conjunction with the sallad, it has this advantage over the visored bassinet of allowing a free supply of air, and only required to be closed just before an onset. This piece is generally omitted in effigies, for obvious reasons; but there is an example on a brass already referred to at Qui, Cambridgeshire, of a date near the middle of the fifteenth century. The actual piece is, of course, to be seen on almost any suit of the period. There is a specimen at the Royal Artillery Institute. The portion going over the chest is, of course, a sort of gorget; but the gorget proper is the piece for the neck, going all round towards the shoulders and back, closing with sliding rivets. This piece followed the mentonnière, and was certainly not common before the beginning of the sixteenth century; but there are much earlier examples, for instance, a gorget with a turned-down collar at the throat is attributed to Albrecht Achilles of Brandenburg, 1414–86. It is a piece closely connected with “Maximilian” armour, and prevailed up to the decadence and after. We find an early instance of the plate gorget on a brass of the D’Eresby family in Spilsby Church, Lincolnshire, representing armour of a date very late in the fourteenth century—this covers a gorget of chain-mail. A brass of Sir John Fitzwaryn in Wantage Church, Berkshire, shows the plate gorget pure and simple. The date of this monument is 1414. Towards the end of the sixteenth century it was far from uncommon to find the gorget joined on to the shoulder-pieces.

THE CUIRASS.

The cuirass consists of breastplate and backplate, which pieces are usually fastened together by straps and buckles, but screws are sometimes used, especially for tournament armour. It was probably introduced into England in the reign of Henry V., and the form is an excellent guide as to

date. The word, or rather its prototype “quirettæ,” occurs in a “Roll of Purchases” (1278) preserved in the Tower of London. The armour for the breast was considered next in importance to that for the head, and inventories of the fifteenth century frequently refer to “pairs of plates, large, globose,” which sufficiently indicate the period. Breastplates were used by the Franks in the eighth century, and probably by the Norsemen about the same time; that of the fourteenth century was without the salient ridge in front called the tapul. The Rev. T. N. Roberts, vicar of Cornforth, county Durham, to whom the author is indebted for several hints, reminds him that it is difficult to say whether it is correct to speak of the fourteenth century breastplate as a cuirass or not. In effigies, brasses, and illuminations this part of the armour is always concealed by the jupon. When the jupon disappeared (temp. Henry V.) the breastplate is revealed always in two pieces; afterwards (temp. Edward IV.) in only one piece, as a true cuirass. On a monument in Ash Church, Kent (dating about 1335), where the lacing of the surcoat at the side permits the body defences to be seen, “rectangular plates like tiles riveted into a flexible garment” are discernible. The only remains of an actual cuirass of fourteenth century date were found at the castle of Tannenberg. The figure of St. George in the Cathedral square at Prague has a flexible garment covered with very small rectangular plates like tiles, and over this a breastplate—not a complete cuirass. All this leads one to suppose that fourteenth century breastplates were not cuirasses so much as additional plates of various shapes over the hauberk, the skirts of which appear below the jupon on effigies, etc., of the fourteenth century. Still, it must be remembered that an effigy of the preceding century in the Temple Church exhibits both front and back plates. The standard of mail is a feature of the close of the fourteenth and beginning of the fifteenth centuries. It was designed to protect the weak place between the gorget and top of the cuirass—it grew, in fact, out of the camail. Almost all effigies of the period exhibit these pieces. The tapul first appeared in the fifteenth century; this ridge after being discontinued reappears later, when it often swelled out to a hump, either over or below the navel. This, indeed, was a decided feature of the second half of the sixteenth century, when the cuirass had often one overlapping plate under the arm. Occasionally it was provided with transverse bars, forming a cross. The Gothic type is very beautiful, and is usually in two or three plates, the second rising to a point in the middle of the breast, and the third running nearly parallel with it and

converging to a point below it. At the top of the breast is a socket for attachment to the mentonnière by a cusp-headed bolt. There are, however, exceptions to this, as shown in examples at the Dresden Museum, where the top of the breastplate projects in a piping. In one of these cases an open helmet had been worn, and the suit probably used by the leader of a company. A suit of which an illustration is given in [Fig. 18](#), shows how the mentonnière goes under the cuirass. The same would also be the case in [Fig. 19](#), but here the mentonnière is missing. The English form of the fifteenth century is usually in two plates, as in the Redmarshal and Downes effigies.²⁷ The first examples occur before the middle of the century.

The lance-rest is on the right breast, and on the left are screw holes for a tilting shoulder-guard when this piece is used, or for a grand-guard. The Maximilian form, which followed the Gothic, is sometimes in one piece with the taces—that is to say, riveted with them—and is more globular in character. In the sixteenth century the cuirass is lower and flatter, and cut straight at the top, and frequently had the tapul already mentioned. In the middle of the century it tends to lengthen somewhat, and is provided with a ridge along the top and round the arm-holes for turning a stroke, and has often, as already mentioned, a single lamination round the arm-holes. A feature of the breastplate about 1560 is the hump or projection over the navel; while usually a little later we have the “peascod” form, where the projection is found lower down. The “peascod” is obviously copied from the doublet of the period, but whence the idea of the middle hump sprang we cannot say. The cuirass made specially for tilting is fully described under the heading of “tilting suits.” In the seventeenth century the breastplate becomes very flat and very short.

ÉPAULIÈRES AND PAULDRONS.

It is not easy to follow the development of épaulières in the earlier stages, as the shoulders on monumental effigies are usually draped by the surcoat, but the principle of laminated or overlapping plates, so early applied to sollerets, was not long in being extended to the upper arm and shoulder, where special mobility for striking and parrying was so needful—indeed, we have instances of articulated épaulières late in the first quarter of

the fourteenth century. These pieces at their highest development were admirably adapted for giving great freedom to the arm. Plates over the shoulders, as distinctive from ailettes, first appeared in England late in the thirteenth or very early in the fourteenth century, but they were merely rondelles or discs. Articulations, as already mentioned, came a little later, and rondelles protect the shoulder-pit and inner arm. A brass of a knight of the Cuttes family in Arkesdon Church, Essex (1440), is a good example of what may be termed the development of épaulières into pauldrons. Pikeguards, generally applied to “Maximilian” armour, are to be found occasionally much earlier—see example in Southerly Church (1479).²⁸ The Beauchamp latten figure at Warwick (1454) shows these pieces. Viscount Dillon mentions an example as early as 1424. Suits are often seen with only one of these projections, but it will always be noticed on examination that there are screw holes in the other pauldrons for its fellow. They are guards against pike thrusts, and are occasionally found double on each shoulder. These shoulder-guards are usually known by English writers as pass-guards, but Viscount Dillon considers this to be a mistake, as he thinks the real pass-guard to have been an extra tilting-piece. The absence of these pieces is far from always implying that they have been omitted, for in many cases a close examination will reveal holes on the shoulders, showing that they have been originally present. Pauldrons were usually attached to the cuirass by straps and buckles, and consist of plates in successive lames over the shoulders and upper arm. Sometimes the attachment is by a pin, as in [Fig. 22](#). In armour of the second half of the fifteenth century the upper plate scarcely reached beyond the shoulder, while in “Maximilian” and later armour they came well over the chest, assuming a resting wing-like form before and behind. They were sometimes very large and uneven in size, that for the right arm being the smaller, for using the lance. There are many instances late in the sixteenth century where gorget and pauldrons are joined together in one piece, and then elbow-gauntlets are used. This is the case in armour called “allecret.” In the second half of the sixteenth century pauldrons were often smaller and wingless—indeed, more like the older épaulières, and then rondelles reappear for the protection of the weak place, “defaut de la cuirasse.”

PALETTES, RONDELLES, OR DISCS,

were plates attached to the armour, variously applied for the shoulders or any weak places, later specially to defend the armpits, where there was a vulnerable place called “vif de l’harnois,” and later, “defaut de la cuirasse,” and leave the arms free to parry or strike. These pieces assume various forms, and were not invariably in pairs; in cases where they differ, that over the right armpit is the smaller—an instance of this may be seen on a brass in Harpham Church, Yorkshire (1420). In this instance the left rondelle is round, while the other is scroll-shaped. There is a portion of a “Gothic” suit at Dresden with an oblong rondelle on the right side, while a projection on the épaulière, to a certain extent, protects the left armpit. They appear very early, and may be seen freely and beautifully applied on a figure in Alvechurch, Worcestershire, of the earlier half of the fourteenth century. They vary very much in size, and in armour of the next century were very handsome, being ridged throughout with scalloped flutings, and often charged with a heraldic rose, and sometimes spiked in the centre. They became very large in tilting suits, little short of a foot in diameter. The earliest application of these discs was to the elbow-guard. Rondelles for the armpits reappear freely in the second half of the sixteenth century, as shown in examples at Dresden and Berlin. They are frequently on the right side only.

REREbrace, COUDIÈRES, AND VAMBRACE (FRENCH BRASSARD, ITAL. BRACCIALE).

These pieces are the armguards—the rerebrace for the upper arm, the vambrace for the lower; they first appear in plate in the second quarter of the fourteenth century, and became general a quarter of a century later. Coudières for the elbows first appeared in the thirteenth century in the disc form, about the same time as genouillères for the knees; and these pieces exhibit one of the earliest applications of plate to body armour. Both may be seen on an effigy of William Longespee the younger (1233) in Salisbury Cathedral. Coudières are elementary in the early stages, with rondelle, then cup-formed and laminated both above and below the elbow, with shell-like

side expansions to protect the inner bend of the arm, and later going all round the elbow joint. This was the completed form, but all these improvements did not come at once. The De Bohun effigy exhibits the second-mentioned form. The outer guard assumes many forms, fan-shaped, bivalve, escalloped, etc., and is sometimes preposterously large. The rerebrace and vambrace do not appear in England before the fourteenth century. The effigy of the Black Prince at Canterbury exhibits these pieces. The armour for the arm, that is the three pieces dealt with, is termed brassards or brassarts. The garde-de-bras, an additional protection for the left arm for tilting, attachable to the elbow plate by a screw, was introduced in the fifteenth century.

GAUNTLETS.

The earliest form after chain-mail was of cuir-bouilli, both plain and fortified with scale work, and such largely prevailed in the thirteenth century, and even later. An example occurs on the tomb of Sir Richard de Burlingthorpe, of about 1310. The earliest form of plate gauntlets occurs in the middle of the fourteenth century, and shows articulated fingers—see an example on a brass of Thomas Cheyne, Esq. (1368), at Drayton Beauchamp, Bucks; after which mitten gauntlets of laminated plates, with a separate thumb-guard and peaked cuffs, prevailed. Late in the fourteenth century an attempt is made to copy the finger nails. An example occurs on the monument of Sir Robert de Grey, at Rotherfield Greys, Oxfordshire. Late in the fifteenth century, the earlier form, with articulated fingers, was reverted to. Gadlings, or knuckle and finger spikes, were in vogue throughout the century—a truly dangerous weapon of offence for the *mêlée*. Again, later we have the fingers covered with overlapping plates, very narrow and flexible. Another common form, though late, is the elbow gauntlet. There is a pair in the Castle of Newcastle-upon-Tyne, and there are others at Naworth Castle and in the author's collection. A locking gauntlet was invented in the latter part of the sixteenth century, the object of which was to prevent the weapon from being knocked out of the hand, to which it was fastened by a hook and staple. It is stated that this gauntlet was often barred in single combats, it certainly was in foot-tournaments. There is an

example of this contrivance in a suit in the Tower of London. Gauntlets were sometimes made of brass.

TACES, TUILLES, TASSETS, BRAYETTE, AND GARDE-DE-REINE OR RUMP GUARD.

Taces were the laminated plates at the bottom of the cuirass, and to these the tuilles or upper thigh guards were attached by straps and buckles. It was common to wear mail below the taces, often with scalloped edges, but the lower portion was often the bottom of a shirt of mail still worn beneath the cuirass. The mail skirt appears so late as 1578 on an effigy at Whitchurch, Denbigh. Taces usually consisted of three, and sometimes of five, and even of eight lames, as noticeable in the brass of Sir John Lysle (died 1407), whose armament is entirely of plate; but early examples are in one piece, and indeed late examples also. An early example, with taces only, is to be found on the brass of Sir John Drayton, but part of the lower portion is missing. Laminated taces first appear late in the fourteenth century; the brass of Nicholas Hawberk (died 1406), at Cobham, is an example. The introduction of "Almayne" rivets (sliding) gave great elasticity to the armour. Tuilles are peculiar to armour dating from the second quarter of the fifteenth century; the earlier form is short and square, but later it becomes pointed and an scalloped shell or tile-like plate in one piece, extended down so as to cover the top of the cuisse, and was attached to the taces by straps and buckles as a guard against an underthrust of the sword. There is an early example on the brass of John Leventhorpe, in Sawbridgeworth Church, Hertfordshire (1433). This, like all tuilles of its time, was small and attached by straps or hinges to the lowest rim of the taces—indeed, it differed but little in shape from the plate to which it was attached. It lingered long in England, as shown in the Stanley and Lementhorp brasses in Westminster Abbey and Great St. Helen's Church, 1505 and 1510 respectively; and there is a very late example on a suit of armour of the time of Philip II. of Spain, but this may perhaps more properly be looked upon as a solid tasset, the suit having been used for tilting. The Beauchamp effigy shows four tuilles, two large and two small. Tassets followed on these pieces, though they were for a time contemporaneous. They were

practically the same piece as the *tuille* in laminated plates, but were generally attached directly to the bottom rim of the cuirass, *taces* being then usually dispensed with, unless in one plate, forming the connecting link. It was not uncommon to find them in two parts during the second half of the sixteenth century, as shown in the Alnwick example ([Fig. 33](#)), and there are also cases where they are in one solid piece, as shown on a gilded suit in Windsor Castle, and in the other example referred to. *Tassets* gradually increased in length as time went on until they reached the knees, forming then the *cuisse* itself of laminated plates. This was the last stage before the introduction of the jackboot. The *brayette* or *cod-piece* is a hollow cap-like projecting plate for fixing on to the bottom of the cuirass for the protection of the fore-body. Fig. 14 represents this piece in chain-mail. We are not aware of the existence of another specimen in chain-mail. The fortunate possessor of this unique piece is Dr. Edgar von Ubisch of Berlin, and we are indebted to his kindness for the illustration. The *garde-de-reine* was a projecting piece attached to the rim of the backplate; it was of overlapping plates, and protected the rump and small of the back.

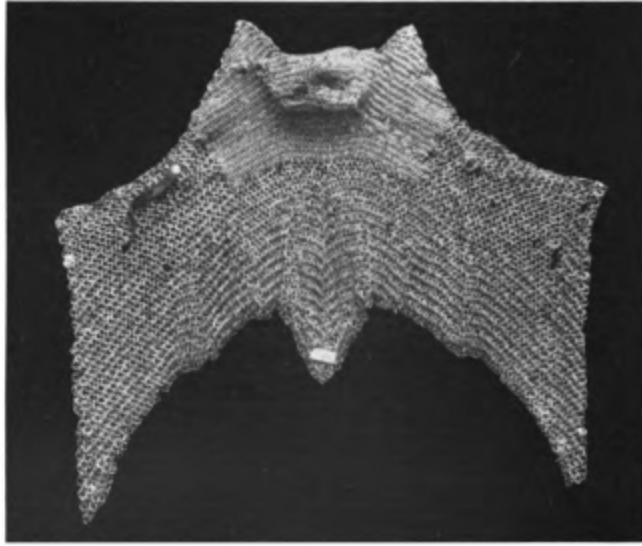


Fig. 14.—Brayette in Chain-mail, at Berlin.

CUISSE, GENOUILLIÈRE, AND JAMB.

Up to the Conquest there was probably no leg armour in England other than thongs, but there are early German examples. Chausses would naturally suggest themselves after Hastings, as William bore them; while Harold, who did not, was wounded in the leg. The term applied to the upper leg armour, or breeches of mail, was *chaussons*. Soon after the Conquest *cuir-bouilli* was largely used, and this was followed by stockings of mail and *sollerets* of the same, as may be seen on the seals of Richard I. Wace mentions iron chausses. Even up to the middle of the fourteenth century it continued common in England to wear these pieces in chain-mail with attachable *genouillières*. An example of this kind may be seen on the effigy of Robert de Vere (died 1221) in Hatfield Broad Oak Church.

The *cuisse* was the plate going round the front of the lower thigh, fastened by strap and buckle. It first appeared in France and England in the second quarter of the fourteenth century, and became general towards the close. In armour of the latter half of the fifteenth century it was often embellished by consecutive laminations at the top. In the second half of the sixteenth century it was sometimes in two detachable pieces, for foot fighting and horseback.

Genouillières (defences for the knee) were the first body pieces of plate, except perhaps the *plastron-de-fer* or breastplate, and possibly the *coudière* also. They were called *poleyns*, and first appear in the thirteenth century; an example, about 1250, is figured in Plate XXX. of Stothard. The side of the knee became further protected by *rondelles* later in the century; and from that time these appendages became more ornate and comprehensive. As soon as plate armour was completed, *genouillières* became articulated both above and below the knee. In armour of the second half of the fifteenth century they are specially beautiful, assuming a shell-like form, often bivalve and butterfly shape with scalloped edges and flutings. The *chausse*, or shin-piece, was used in chain-mail, indeed earlier still in fortified leather, and early in the fourteenth century it became plate and was termed *jamb*; first only in front attached by strap and buckle, and later going round the leg hinged, and fastened by sliding rivets. These

pieces were also called greaves. The inventory of Piers Gaveston (1313) catalogues “three pairs of hinged jambs.” These pieces were generally plain. Both they and sollerets disappeared with the advent of the jackboot.

SOLLERETS.

Sollerets are a better guide as to date of armour even than gauntlets, particularly after the fourteenth century, for reasons given under the head of the last-named. The earlier sollerets of overlapping plates were of extravagant length. This form followed the prevailing fashion in shoes, and hence the name “à la poulaine” from “souliers à la poulaine.” The long form was much modified during the last quarter of the fourteenth century and well into the fifteenth, but it became in vogue again later in the century with enormous tips, the length from toe to heel being up to twenty-four inches. The instep of chain-mail was not uncommon in the fourteenth century and even later. The sollerets of the Black Prince were of enormous length. The tips could, however, be disconnected at pleasure. The shorter form was styled “demi-poulaine” or “ogivale lancette.” A variety called “ogivale tiers-point” largely prevailed in the second half of the fifteenth century. When ridged and scalloped armour was replaced by “Maximilian,” sollerets were wide and short—in fact the shape of a bear’s paw or cow’s mouth, spreading out at the sides, and requiring very broad stirrups; but when fluted armour was discontinued the shape became gradually narrower, and after the middle of the century more like that of the foot; still there are very late instances of the “bear-paw” form. This variety was styled “bec-de-cane,” which differs, however, from the “tiers-point” of the fifteenth century. Sollerets disappeared altogether with the jamb, the jackboot taking their place.²⁹ These pieces in laminated plates are shown on the Daubernoun brass, and continue to occur on such monuments.

SHIELDS.

This subject is too vast for more than a mere outline in these pages. The kite-shaped, round, and triangular shield appears in the twelfth century. The

two first-named are long, and either bowed or flat. They were held over the breast by a strap going round the nape of the neck, called a “guige.” Shields of the thirteenth century were either small and “heater” shaped, or larger and rounded. Pavises were very large shields to be placed before the bowmen as a defence, and were provided with an inner prop to hold them upright on the ground. As to ordinary shields, most of the thirteenth century forms extended into the fourteenth, when the bouche, or hole cut in the right corner as a spear-rest, was introduced. They were pear-shaped, triangular, heart-shaped, circular, oval, curved, and sometimes nearly square. The round buckler was carried in the hand, while the larger shield was borne on the arm. The material was generally of wood or leather, or both combined—the latter often embossed. They were more or less fortified or bossed, and sometimes partly or wholly of iron. For tyros, basket-work was used. Shields generally bore a heraldic device, or other cognizance, and were frequently curved, bossed, and spiked. The usual shield of a knight of the fifteenth century had the bouche, was convex, and about two and a half feet long by about a third of that broad, and pointed at the bottom. In the sixteenth century ordinary shields were seldom used, but an immense amount of fine artistic work was lavished on the pageant shields of that period, an example of which is given in [Fig. 15](#). The tournament shield is described under the heading devoted to these games.



Fig. 15.—Pageant Shield, formerly in the collection of Prince Carl of Prussia.

CALTROP, OR CROW'S FOOT.

This, the Roman murex or tribulus, was a sharp point of iron standing upright, fashioned like a crow's foot. They were strewn broadcast on the ground, for the purpose of maiming horses in a charge of cavalry, or placed on a moat filled up with fascines, or on a breach to resist an attempt at escalade. Knightly spurs have been known to have been used for this purpose. The name is an abbreviation for cheval-trap. There are some specimens in the Rotunda, Woolwich, varying in height from 1.25 to 2.5 inches. [30](#)

SPURS.

These goads were used by the Romans, and the gilded spur was one of the badges of the knight of mediæval times. The earlier are of the "goad" type, and fastened by a single strap; they were probably first used singly, and were called "prick spurs." An example of the goad prick may be seen in the Daubernoun brass (1277). We get the rowel prick late in the thirteenth century. The D'Argentine brass (1382) furnishes an example of a spur of the fourteenth century. The number of points or pricks in specimens of the middle ages approximate the date. Early in the fourteenth century there are usually eight, but in the fifteenth as many as twelve points to the rowel, and spurs were long-necked. Later, the fashion in style and form was "legion." In heraldry the knightly spur was a "goad" up to 1320, and called a "pryck-spur," later the "rouelle-spur." The tournament spur of the sixteenth century was straight and long in the neck. In the case of a knight's degradation his spurs were hacked off by the king's master-cook. During the fourteenth century it was usual, when orders were given to men-at-arms to fight on foot, for their spurs to be taken off, so as not to impede their movements; and these were then often used as caltrops. This was notably the case at the battles of Courtray and Poitiers.

PART IX.

“GOTHIC” ARMOUR, 1440–1500; AND SOME ARMOUR-SMITHS OF THE PERIOD.

THE “Gothic”³¹ school, as it is termed, exhibits the highest embodiment of artistic beauty as applied to defensive armour; and it inaugurated a new epoch in warlike panoply. The armour-smith’s best efforts were directed not only to give increased protection to the limbs and make the armour light, flexible, and impenetrable; but the flutings and scalloped edges were designed to produce beauty of form and outline, as well as with a view to deflecting the weapon of attack from vital points; and the armour was equally mobile for fighting on foot or on horseback. We owe its initiation doubtless to Italy, in which country, together with Germany, it reached its highest pitch of excellence; but the style itself is really a reproduction of the mediæval Florentine dress. Gothic armour is greatly associated with the *sallad*, large *mentonnière*, *tuilles*, and *sollerets* “à la poulaine.” The *cuirass* is decorative; the earlier form being somewhat short with many *taces*, and the later with a longer breastplate and fewer *taces*, thus exhibiting the evolution from the still earlier fashion. It has been fully described under the heading devoted to this piece. There is an English example of this style of armour shown on a brass in St. Mary’s Church, Thame, Oxfordshire, about 1460; and another in the effigy already mentioned of Sir Richard Beauchamp, Earl of Warwick, in St. Mary’s Church, Warwick. There are only very few Gothic suits preserved in this country, our practical people having used up so many as old iron, just as they let the weather into our fine abbeys and churches by tearing off the roofing lead for the melting pot.

A few suits are attributed to particular ancestors in some of the castles of long-descended German families, but, in almost every case, with but

slender foundation in fact; the only specimens in England that may be termed historic are those in the Rotunda at Woolwich, and these are only fragments. Few of the “Gothic” suits in this country, if indeed in any other, are quite homogeneous, and many of them are more or less made up of odd pieces. This is the case with the “Gothic” armour at Parham, which is said to have come from the Church of St. Irene at Constantinople. Many of the details of this armour are of the most exquisite and obviously authentic character, while pieces, such as the sallads, apparently never went with the other armour. Reliable armour of this period is very scarce, and difficult to buy. Four thousand pounds was recently asked in London for a suit! Fashion was as absolute regarding armour as in dress; and with the advent of the “Maximilian” period the “Gothic” form was greatly laid aside, for it could not be adapted, and therefore became obsolete. This is the main cause why so few specimens have been preserved. A historic example in the collection at Sigmaringen Castle, the cradle of the Hohenzollerns, is described in detail, and an illustration given ([Fig. 17](#)). Another example may be seen on the brass of Sir Robert Staunton at Castle Donnington (died 1458), on which the épaulières extend over the armpits. This brass probably presents the earliest English instance of the sallad. The “Beauchamp” effigy in latten, a “species of fine brass metal,” affords a beautiful example of the earlier Gothic school. The suit from which the models were taken is probably the work of Tomaso da Missaglia of Milan. This effigy, and its probable origin, raises the question as to which country we are indebted for the “Gothic” form until comparatively recently freely attributed to Germany; but it is tolerably certain that it originated with the Missaglias. There is a further interesting point brought out by the effigy itself, which was the work of an Englishman, viz.: that the smith who could copy a suit so faithfully would probably be able to make real armour of a high character. We read in Blore’s *Monumental Remains* all about the contracts for this truly magnificent monument, where it is stated that Dugdale, the historian of Warwickshire, has very fortunately preserved a recapitulation of the agreement between the executors of the Earl and the artisans employed in its erection. This document is given *in extenso* in Blore’s work, and, as he says, it throws considerable light, and affords some extremely important information, on the construction of ancient monuments in general. The original was found among the muniments of the bailiff and burgesses of Warwick, and bears the date June 13, 32 Henry VI. The Earl died in 1439,

so that the contract for the monument was given out in 1454. Various subsidiary agreements of an early date are included in the main contract. The names of the contractors were John Essex, marbler; William Austin, founder; and Thomas Stevyns, coppersmith. The clause of the contract regarding the effigy runs as follows, viz.:—

“The said Will. Austin, xi. Feb., 28 Henry VI., doth covenant to cast and make an image of a man armed of fine latten, garnished with certain ornaments, viz., with sword and dagger; with a garter; with a helm and crest under his head, and at his feet a bear musled, and a griffon, perfectly made of the finest latten, according to patterns, all which to be brought to Warwick, and layd on the tombe, at the perill of the said Austin: the said executors paying for the image, perfectly made and laid, and all the ornaments in good order, besides the cost of the said workmen to Warwick, and working there to lay the image, and besides the cost of the carriages, all which are to be born by the said executors, in totall xl li.”

A further clause refers to the agreement made with Bartholomew Lambespring, Dutchman, and goldsmith of London, 23 Maii, 27 Hen. VI., who “covenanteth to repaire, whone, and pullish, and to make perfect to the gilding, an image of latten of a man armed, that is in making, to lye over the tombe, and all the apparell that belongeth thereunto, as helme, crest, sword, &c., and beasts; the said executors paying therefore xiii li.” The accounts of one of the executors show that the monument took twenty-one years to erect and finish, and that the total cost was £2481 4s. 7½d. Mr. Blore continues: “The monument of Richard Beauchamp, Earl of Warwick, consists of an altar-tomb of grey marble, in the finest preservation. Within canopies admirably wrought, are whole-length sculptures of fourteen of the immediate relatives of the deceased, executed in latten, which was a species of fine brass metal, and richly gilt; these figures are disposed five on each side, and two at either end of the tomb. Underneath every figure, in starred quatrefoils, is a shield with armorial bearings enamelled on brass, and between the larger canopies, alternately, a smaller, containing an angel executed in similar metal with the portraiture of the mourners, and holding in one hand a scroll, on which is engraven in Gothic letter,

“Sit deo laus et gloria, defunctis misericordia.”

The image, excepting the hands and head, is in complete armour, with the garter encircling the left leg. The head rests upon a helm, surmounted by the family crest, and at the feet are a bear muzzled and a griffin, badges of the ancient house of Warwick. The armour may be considered as real, from the extreme care and exactness that have been bestowed on it by the original artist. Mr. Charles Stothard had the figure turned over, and found that the armour for the back was as carefully and minutely finished as that on the front. The suit exhibits the cuirass as shorter than we find it in later "Gothic," while the taces are correspondingly more extended, and consist of five lames. The breastplate has a gracefully curved groove on either side, and a catch for the mentonnière on the breast. The mentonnière is usually omitted in effigies for obvious reasons. A remarkable feature of the effigy is that there are four tuilles; the larger two do not converge so abruptly to a point as they usually do rather later, but the smaller ones are more sharply pointed. The coudières are of the beautiful butterfly type, and very large; while the sollerets are far from being extravagantly tipped. The most unusual feature of this effigy is the early presence of pikeguards. The Earl died in 1439, so the figure could not well be copied from any armour he left behind him, for the general aspect of the suit would fix the date about 1450–60, which would correspond with the date of the contract for the tomb. As already stated, the figure was probably fashioned after models supplied by Tomaso da Missaglia, and seems to represent his later work. This impression is strengthened by the following comparisons with two harnesses at Vienna, viz.: a suit by this master, made for the Pfalzgraf Friedrich am Rhein, about the middle of the century, exhibits points of contact with the Warwick figure, especially in respect to the number of the taces; while another by Antonio da Missaglia, made about thirty years later for the Count of Gajazzo (died 1487), shows a relatively longer breastplate and fewer taces. The latter suit bears pikeguards, which the earlier does not. One may perhaps deduce from these examples that the Beauchamp effigy represents the later work of Tomaso. The illustration given in [Fig. 16](#) represents the effigy in an upright position. It is a reproduction of that given in Blore.



Fig. 16.—Effigy of Richard Beauchamp, Earl of Warwick, in St. Mary's Church, Warwick.

Tomaso and Antonio da Missaglia, the illustrious sire and son, were the great Milan armour-smiths of from the end of the first quarter to the close of the fifteenth century. To the first it seems certain that we are indebted for the form called “Gothic,” which was, however, merely a graceful improvement on the fashion immediately preceding it. There was nothing very abrupt in the transition, as was the case in the radical change from “Gothic” to “Maximilian.” The work of Tomaso is conspicuous for purity of style and nobility of form, and, from an artistic point of view, it has no rivals. Armour of his period was generally plain, but the more pronounced passion for decoration of the time found expression in the work of his son; an example of whose skill may be seen in the case of a helmet in the Tower of London, and at Vienna is the superb Gothic suit with pikeguards, already referred to, made for the Count of Gajazzo. Tomaso was, it is believed, the first master to use armourer’s marks. His monogram is the letter “M,” surmounted by a crown. The Negrolis, who worked after the Missaglias, seem to have been of the same family, and, as Boeheim points out, the name “Missaglia,” like that of Ferrara, seems to have originated as a “place” designation. Examples of the work of the Negrolis may be seen both at Vienna and Madrid. Their work represents the full swing of the “renaissance.”

Milan, where the Missaglias worked, is not the only town in Italy where there is a Via degli Armorari and a Via degli Spadari, showing that there were then separate guilds for armour- and sword-making in that country.

SIGMARINGEN SUIT.

This beautiful “Gothic” suit, by Lorenz Kolman of Augsburg ([Fig. 17](#)), is said to have belonged to one of the Counts of Hohenzollern-Eitel. Demmin refers to it as being erroneously ascribed to Eitel Frederick I. of the thirteenth century. The mistake is obvious, as there were no Counts of Hohenzollern-Eitel then! There were two Eitel Fredericks in the fifteenth century. On consulting the Stammbaum at Hohenzollern it appears that

Eitel Frederick I. reigned 1426–1439.

Jost Nicolaus I.	”	1439–1488.
Eitel Frederick II.	”	1488–1512.

And the character of the armour conforms closely to the early portion of the reign of the last-named. There was no later “Eitel Frederick.” A later suit, made for this Eitel Frederick about 1510, is now at Vienna. It is “Maximilian” and partly fluted, and very possibly by the same master; for we see by the Berne example, referred to somewhat later in these pages, that Lorenz Kolman turned out Maximilian armour after that fashion had superseded the “Gothic.”



Fig. 17.—Gothic Suit at Sigmaringen.

The sallad ([Fig. 17](#)) is very heavy, and of the usual German form. There are traces of a leather lining, and besides the ocularium there are two small holes above the forehead. The mentonnière is fastened to the breastplate by a cusped clasp; the neck and chin-piece can be raised or lowered at pleasure, and there is a spring catch for the purpose. The cuirass is most elegant in shape, and being much longer than that on the Beauchamp effigy brings out clearly its later date. It consists of three plates, the two lower slightly overlapping, leaving decorative margins, and they converge to points along the tapul at the breastbone and below. The lower plates are riveted, and add both strength and elasticity to the piece. There are holes on the right breast for fixing a lance-rest, and on the left are two holes for fastening on a grand-guard for tilting. The taces consist of three lames, and to these the tuilles are attached by straps and buckles. The tuilles are very graceful, with angular flutings, and terminate in a point. The cuisses are decorative, while the genouillières are small, with bivalve guards. The épaulières and rerebraces are laminated, the coudières pointed, and held in their places by straps. The rondelles are unfortunately missing. The gauntlets are articulated, with sharp gadlings over the knuckles and first finger joints. The garde-de-reine consists of three lames. The sollerets are “à la poulaine” in an extreme form, but the tips can be disconnected at pleasure for foot fighting, like those on the effigy of the Black Prince. The lower part of the body is protected by a skirt of mail.

The Sigmaringen harness exhibits many points of contact with a beautiful suit recently acquired by Prince Ernst of Windisch-Graetz, which is a glorious specimen of the armourer’s art at his very best. The tuilles of this example are not pointed, as is the case on the Sigmaringen suit.

A “Gothic” suit from the collection of Prince Carl of Prussia now in the Zeughaus at Berlin, of which an illustration is given in [Fig. 18](#), is very beautiful. The finely modelled breastplate has a fluted rim across the upper chest, a feature that is uncommon in Gothic armour, but of which other examples are given under the heading “The Cuirass.” The rondelles are ornamented with curved radiating flutings, in the matrix of which a projecting spike is fixed. The coudières are sharply pointed at the elbow, while the tuilles are large, with a shape not unlike that of the larger pair on the “Beauchamp” effigy, though bevelled and pointed. The sollerets are “à la poulaine,” in the extreme form.

The remarkable armour-smith family of Kolman of Augsburg occupied a similar position in Germany, during the reign of the Emperor Maximilian I., to that held by Antonio da Missaglia in Italy. Lorenz, surnamed Helmschmied, is perhaps best known to-day for the beautiful “Gothic” harness, made about 1490 for the Emperor Max, which now adorns the collection at Vienna. It resembles the Sigmaringen example somewhat closely, the points of difference lying mainly in the form of the sallad, the shape of the tuilles (square-cut at the bottom in the Vienna example), and the Vienna harness has an extra plate on the breastplate. That Lorenz Kolman was employed during the later portion of his career as Court armour-smith to Maximilian in making fluted armour, as he had been engaged earlier in turning out “Gothic,” is shown by an early and interesting example of fluted “Maximilian” armour at Berne. This harness compares closely with that represented in [Fig. 24](#), though the Swiss example shows pikeguards, which the other one does not. The early character of both suits is shown by the swelling out of the breastplate over the abdomen. The figure sits on horseback, and the horse is fully barded contemporaneously with the figure. The saddle has the deep seat of the “renaissance.” Lorenz Kolman died in 1515. The armourer’s mark of this family is a helmet surmounted by a cross. The mark of Hans Grünewalt of Nuremberg has not been fully determined, so that his work cannot be identified with absolute certainty; but a breastplate that belonged to Philip the Fair, and a shield at Vienna, have been attributed to him. They are of exquisite workmanship, and the mark on these specimens is a stag on a shield, which clearly refers to the “greenwood.” He was the great rival of Tomaso da Missaglia, and died in 1503.



Fig. 18.—Gothic Suit at Berlin.

The Gothic armour at the Rotunda, Woolwich, is especially valuable from its direct association with the Knights of Rhodes. It is fragmentary in character, consisting mainly of isolated plates and portions of plates. There are several sallads of the Italian type, a beautiful breastplate in two plates, a backplate, some gardes-de-reine, a cuisse with a small butterfly genouillière guard, rounded flutings radiating from the centre of the upper thigh, several broken gauntlets, besides other fragments, and a tilting helm of the end of the fifteenth century, on which fifteen of the staples remain; the helm is perforated on one side only.

The last Gothic suit given is one in the author's collection, and an illustration of it is here given ([Fig. 19](#)).

GOTHIC SUIT IN THE AUTHOR'S COLLECTION, 1460–1500.

This suit, like so many of its period, is incomplete. The armet with it, when acquired, never belonged to the suit, and there is no mentonnière. The sallad, shown on the figure, was made recently to give the general effect of the period. The suit is otherwise complete, and of fine material, proportions, and workmanship. The steel of this period is of excellent quality. The details, with a few exceptions, somewhat closely resemble those of the Sigmaringen suit. There are rondelles at the armpits on this suit which are ornamented with radiations, and these, together with the elbow-guards, are beautifully ridged and bevelled. The tuilles are larger and squarer than those on the Sigmaringen suit, and the sollerets not so long in the tips. The cuirass is in two plates, with a rim across the chest, as shown on the Berlin suit ([Fig. 18](#))—the mentonnière therefore went partly below the cuirass. The general details greatly resemble those of a suit at Vienna, attributed to Sigismund of Tyrol, which is also an incomplete suit. As the gauntlets of this suit are distinctly typical, it may be well perhaps to go somewhat into detail concerning them. They are of fine workmanship and material, as well as light and graceful. The surface of the steel is very hard. The cuff is sharply pointed, and deep flutings run in parallel lines towards the extremity; while similar perpendicular flutings join the lowest of these

lines. Three supple articulations lend flexibility to the gauntlet, and connect the knuckle-plate with the cuff. The last-mentioned plate and four finger plates all work in slots, and are beaten into ridges for fitting over the knuckles and fingers. The thumb-guard is also articulated. An illustration is given in [Fig. 19](#).

Transitional Gothic, where laminated tassets replace tuilles and merge into the next stage in various ways, is also very beautiful. In both varieties you have lovely scalloped and fluted rondelles, often charged with a heraldic rose. A fine example of this description may be seen in the National Museum at Munich, and an illustration is given of it (Frontispiece), because of the beautiful details. The rondelles are especially fine, and the mentonnière and breastplate, which latter is in two plates, are clearly shown.



Fig. 19.—Gothic Suit in the Author's Collection.

PART X.

MAXIMILIAN ARMOUR, 1500–1540.

THE strong military tone lent to this period by the bent and character of the three great monarchs who then ruled the destinies of Europe, had great influence on armour, civil dress, art, and display generally. The tendency, as in architecture, was towards redundancy of detail, and the abandonment of simpler and more truly artistic forms for something more ornate. This tendency found expression more in the details and ornamentation of armour than in the intrinsic beauty of the form itself. The third estate emerged more and more from its long vassalage, bringing trade and opulence in its train; besides a corresponding diminution in the power and prestige of extreme feudalism. The imagination was cultivated, as it had not been before, and luxury, with the means of gratifying it, had correspondingly increased; indeed, the society of the time had already passed the threshold of the “renaissance”—one of those periods of revival, in long course of incubation, suddenly bursting into life. Harnesses were more solid and altogether less mobile than in the “Gothic” form.

The “Ehrenpforte” of Maximilian, supposed to have been decorated from the designs of Albert Dürer, gives a vivid representation of the meeting between Henry VIII. and Maximilian. This work, and much literature with illuminations, filled in details of the times which are invaluable to us now. These monarchs revelled in pomp and parade which found expression greatly in the tilt-yard; and the influence exercised on the arms and armour of the period was immense. Now the man-at-arms was completely encased in plate. Immensely heavy “Gothic” suits of armour already began to be laid aside in tournaments in favour of harnesses made for battle, supplemented with reinforcing pieces.

Armour then underwent a great change about the end of the fifteenth century, during the reign of the Emperor Maximilian (died 1519), when fluted armour (*armatura spigolata*) came into fashion. The change was radical and abrupt, being obviously suggested by the civil dress of the period. The transition was so sharp as to convey the idea that the change was by order. The beautiful Gothic lines, ridgings, and indented outlines disappear, and the form becomes stiffer and less elegant in every way. The breastplate is shorter and more globular, and fringed at the top by a projecting piping; the more graceful épaulières change into pauldrons, often of unequal size, and the pretty rondelles become unnecessary for the time being; but they were resumed at a later period. Coudières and genouillères are smaller, while tuilles are replaced by tassets of laminated plates. Sollers became very broad and clumsy, in absolute contrast to the “soulers à la poulaine.” It seems in every way probable that this style of armour, though like the “Gothic,” so closely associated with Germany, may have had its origin in Italy; for the Germans in contemporary writings call it “Milanese.” Henry VIII. ordered many suits at Florence. The helmet, the armet, and a little later the burgonet, are nearly as much associated with “Maximilian” armour as the sallad is with the “Gothic”; and the gorget proper replaces the mentonnière, or in other words, the bavard of the armet took the place of the neck and chin-piece.³² Another prominent feature is the general use of pikeguards, which stand out at the head of the pauldrons to protect the neck of the wearer from pike thrusts. There are some fine suits of this armour in the Tower of London, presented by the Emperor Maximilian to our Harry the Eighth. An illustration is given in [Fig. 20](#) of a typical suit in the collection of Prince Carl of Prussia, now in the Zeughaus, Berlin. The details are as follow, and bear out the general descriptions of the class already given in these notes:—The suit is fluted throughout, except the jambs, which are nearly always plain. The helmet is the armet, and this example sufficiently indicates the date of the armour; both form and workmanship are good. Instead of the large “Gothic” mentonnière, there is a gorget and the bavard. The pauldrons, which are uneven in size, are surmounted by pikeguards; the left pauldron is the larger. These pieces consist of front and back plates, an innovation of the sixteenth century. The cuirass is shorter than the later Gothic form, more globular, and cut straight at the top with a rope-like rim. The backplate terminates in a garde-de-reine of three lames. Gauntlets are of the mitten type, with narrower lames than in

the form immediately preceding, and there is a twisted ridge across the knuckles. The coudières are sharply rounded over the elbow joint with bivalve guards. The taces are in four lames, and the tassets buckled on; there is the usual arrangement in the centre for the insertion of the brayette or cod-piece, which is missing. The armet-collar is laminated behind. The sollerets are of the “bear-paw” form.



Fig. 20.—Fluted Maximilian Suit at Berlin.

There is a remarkably fine suit of Maximilian armour in the Königl. Bayer Armée Museum at Munich. It is not, however, quite such a characteristic example as the ones already given, inasmuch as the pauldrons, besides not being winged, are without pikeguards. The armpits are protected by spiked rondelles. In all other respects this suit is identical with the one preceding.

A suit at the National Museum, Munich, of which a drawing is given in [Fig. 21](#), is more shapely than the one preceding, and differs in some rather essential particulars. The armet has a very projecting and grated visor. The pauldrons are more comprehensive; the cuirass more globose. The mitten gauntlets with fluted cuffs are very beautiful, and the finger plates are wonderfully flexible. This is rather an early form of the “Maximilian” gauntlet, and would date the suit between 1505 and 1510.

Armour was often worn at this period with helmets of a grotesque character. A drawing is given in Fig. 22 of a suit at Nuremberg, badly set up, with an armet of this character. The armour is fluted. There are some of these grotesque helmets, of the same period, at Vienna, and the author has a couple of a later time in his own collection.

Although armour of the Maximilian period is usually fluted this is by no means always the case, and a smooth suit of that school in the author’s collection is now described, and a drawing of it follows in [Fig. 23](#), which somewhat incongruously exhibits the knight as holding a flamberge, which is a footman’s weapon.



Fig. 21.—Fluted Maximilian Suit at Munich.



Fig. 22.—Fluted Maximilian Suit, with Grotesque Helmet.



Fig. 23.—Plain Maximilian Suit, in the Author's Collection.

Though not fluted, this suit belongs to the style and period of fluted armour. It is of noble form and fine workmanship. The armet is graceful in outline, with a twisted comb, and there are twin perforations on each side of the crown-piece. The visor exhibits the series of ridges so characteristic of the period, and there is a projecting peg on the right side to work it, and a spring catch on the same side to close it, while a similar catch connects the bavair with the crown-piece. The collar terminates in a grooved rim, which is articulated behind. The gorget is strengthened by an extra inner plate in the centre, riveted on to the outer; and a lamination towards each shoulder lends elasticity to the piece. The cuirass differs radically from the Gothic form. It is globular without a tapul ridge, and is shorter in the waist. The "movement" below the breastplate is a combination of taces and tassets. The former consist of three lames over the abdomen joined on to the rim of the cuirass; and the latter are in five lames, being riveted on to the lowest rim of the former. The breastplate is cut short at the top, along which runs a thick twisted projecting rim, and just below this are two small perforations in the centre. This rim is continued round the armpits on the outside edge of a laminar plate attached to the breastplate. A lance-rest is on the right side. The brassards are apparently of a somewhat later date than the rest of the suit, the pauldrons being exactly the same in form as those on a suit, of German origin, made for King Philip II. of Spain about 1540. The gauntlets are of the mitten type, and finely wrought. The knuckle-piece has a twisted ridge, and a smaller piping decorates the edge of the cuff and the last plate over the fingers. The cuffs are hinged, and clasp with a hole and peg. The cuisses have one lamination at the top, on which is a narrow twisted rim, and below it a very thick twisted ridge. The genouillières are small and "butterfly," while the sollerets are bear-paw, thickly ridged over the toes, and very handsome. This suit presents many points of contact with a harness made by Koloman Kolman for Count Andreas von Sonnenberg, about 1506. There is another fine unfluted suit of about this period in the Tower collection, said to have been made for Henry VIII. The visor of the armet is grated, and the tapulled breastplate is rendered more mobile by two laminated plates at the bottom. The taces and tassets are riveted together, the former consisting of four lames, and the latter of seven. The pauldrons are a pair, and there is only a pikeguard on the left side, but whether the other shoulder was holed or not for a fellow, as is generally the case when only one, the author does not remember. Viscount Dillon states that the suit

is composed of 235 interlocking pieces, and weighs about 93 pounds. It was specially made for foot fighting.

We will close the “Maximilian” examples pure and simple by briefly referring to a fine fluted suit on horseback formerly in the collection at the Chateau De Heeswijk, near Bois-le-Duc. This suit ([Fig. 24](#)) is almost identical with that already referred to in the Königl. Bayer Armée Museum at Munich, and the figure carries a tournament lance, with the coronal. The bards are contemporaneous with the armed figure, and the same theme of repoussé ornamentation runs throughout the entire armament.



Fig. 24.—Mounted Maximilian Suit, with Bards.

PART XI.

ARMOUR WITH LAMBOYS OR BASES.

As already mentioned, a very distinctive feature of this period, which lasted only four, or at the very most six decades, is the skirt of mail called “lamboys,” or in the language of the day, “bases,” which resembles a full gathered or plain petticoat, or kilt of laminated hoops, held together with “Almayne” rivets. A drawing is given of this kind of armour from an example in the author’s collection ([Fig. 25](#)), which is said to have come from an old castle in the Tyrol into the family from whom he obtained it. The suit could only be traced back some seventy or eighty years. Armour with long skirts was current during the reign of Henry VI., but this description differed from the “bases” of the reign of Henry VIII. in the plates being flexible in a vertical direction; capable, as Viscount Dillon says in *Archæologia*, vol. li., p. 258, of being lifted up like a Venetian blind. As shown by the fine suit with lamboys or bases, by Conrad Seusenhofer, in the Tower of London, which will be commented on somewhat later in these pages, it is obvious that this style of armour was to the fore during the later years of Maximilian’s reign, but it became more *de rigueur* in that of his successor. The general pose of the suit ([Fig. 25](#)) is excellent and characteristic. The armet is fluted and “Maximilian” in three pieces, and is a most perfect specimen and graceful in outline. There is a small comb on the crown-piece, and a plume-socket. The visor moves on rosettes of nine petals, and it projects sharply forward to a point, the front consisting of four deeply indented bevels, with two broad lights above them, and two smaller slits in each bevel. There is a spring-catch for closing the visor. The bevor is attachable to the crown-piece by a similar catch. The helmet has a collar of three lames, and weighs five pounds. It is almost identical in form with one catalogued No. 47 among the helmets exhibited at the rooms of the Royal Archæological Institute in July 1880. The date given is 1515–30. In all

probability the helmet on [Fig. 25](#) was made earlier than the date fixed upon for the suit, and perhaps was not worn with it. The cuirass has a tapul with a projection near the base, like the “peascod,” and this feature seems at first to be indicative of a rather later date than 1550–60. The same form is present, however, on a suit with lamboys in the Vienna collection, made by Mathaus Frauenpreis of Augsburg in 1550. This armour, like the one in the author’s collection, is for fighting on foot. The lamboys in [Fig. 25](#) consist of nine lames, the lowest much broader than the others, with a band studded with rivets for an inner lining, terminating with an ornamental string-like piping. These skirts are attached to the lower rim of the cuirass by sliding adjustable screws, and each lame is provided with a similar screw on both sides for attaching the back and front portions together. The back of the lamboys is the same as the front. These sliding rivets are believed to be the “Almayne” rivets so often referred to in inventories of the reign of Henry VIII. They are present also on the fine suit with lamboys in the Tower, made by Conrad Seusenhofer of Innsbruck for Henry VIII. The Tower suit is earlier than the one under discussion, has pikeguards, and the “base” has a brass border, which was doubtless once gilded or silvered. The pauldrons of the author’s suit are very large, and of equal size both back and front, while the rerebrace is freely laminated. The coudières are cup-formed, and go nearly round the elbow joints. The heart-shaped guards, the tops of the pauldrons, and bottom of the rerebrace are enriched by a small piping. The gauntlets are “mitten,” quite complete, and of fine workmanship. The cuffs have their upper edges adorned with a similar piping to that on the other pieces, and the same design is repeated at the base of the last finger plate. Over the knuckles is a bold twisted piping, and the laminated plates over the back of the hand number five above the ridge, while those below are the same in number. The gauntlet is of the type prevailing about 1535–40. The cuisses and jambs have a ridge running down to the sollerets, while the genouillières are ornamented with a double bevel in the centre. The knee-guard is oval, and bevelled in the centre. The sollerets are small, and of the “bec-de-cane” type.



Fig. 25.—Suit with Lamboys, in the Author's Collection.

PART XII.

SOME ARMOUR-SMITHS OF THE FIRST HALF OF THE SIXTEENTH CENTURY.

THE armour-smiths who stand out prominently during this period are the Negrolis of Milan, who have already been referred to, the later Kolmans of Augsburg, and the Seusenhofers of Innsbruck. An example by Koloman Kolman, son of Lorenz, may be seen at the Armeria Real, Madrid (Catálogo No. A65), in a harness made for Charles V. Tuilles are here replaced by tassets, and the figure has a “stechtarche” or tournament shield at the shoulder. Another example, in a noble unfluted suit, is at Vienna. It was made for Count Andreas von Sonnenberg about 1506, and has been already referred to. Desiderius, son of Koloman, also turned out work of the highest character. A specimen of his handiwork is in the Madrid collection. The Kolman’s mark is an armet surmounted by a cross, with the Augsburg armour-smiths’ guild badge. Very little is known concerning the work of Hans Seusenhofer of Innsbruck, beyond the curious “piped” harness at Vienna, made for the Emperor Charles V. when a youth. We have an example of that of his brother Conrad in the exquisite mounted suit, with lamboys, in the Tower of London, made by order of the Emperor Maximilian I., and presented by him to Henry VIII. The date is 1514, and it is chastely engraved with the cognizances of the king, and of his consort Katharine of Arragon. The general theme of the ornamentation throughout is the legend of St. George. The suit is referred to by Viscount Dillon in the *Archæologia*, vol. li. The armourer’s mark is on the helmet, and the suit has been originally silvered over. Jörg Seusenhofer, son of Hans, worthily closes the line; specimens of his work are at the Musée d’Artillerie, Paris, and there is a splendidly enriched harness at Vienna made for the Archduke Ferdinand of Tyrol, about 1547. The collection at the Königl. Zeughaus,

Berlin, is in possession of a fine example by this master in a suit made for Francis I. of France. It is engraved and gilded in the French style, evidently in compliment to the King, or by his orders. The breastplate exhibits an early instance of the "peascod." The leg armour and sollerets are palpable "restorations." Other examples by this master are given under the heading "Enriched Armour." The engraving for this suit was done by Hans Perckhamer. Another celebrated armour-smith, who worked under the Emperor Maximilian II., is M. Frauenpreis of Augsburg, of whose work an admirable specimen with lamboys, which has been already referred to, exists at Vienna.

PART XIII.

DEFENSIVE ARMOUR, 1540–1620, AND TO THE END.

DEFENSIVE armour experienced another change a little before the middle of the sixteenth century, viz., in the casting aside of fluted armour, for the reasons already stated, and the resumption of plain steel. Suits became generally lighter, and the form of the breastplate changed, with a hump over the stomach or the abdomen. During the second half of the century the cuisse and tasset tend to combine in a series of laminated plates to the knee, and sollerets were smaller and more the shape of the foot; indeed, greaves and sollerets began to be replaced by leather boots. This period was specially remarkable for profuse and artistic ornamentation. Armour was engraved by hand and manipulated with aquafortis, as well as embossed and damascened with gold, in a manner that has never been surpassed in any work of the kind whatever. There is a very fine suit of the period, 1550–60, at the Königliche Zeughaus, Berlin, made by the elder von Speyer; and though the armour is enriched, it has been described in this section by way of showing a typical harness of the period in its order. It was undoubtedly made by Peter von Speyer in 1560 for the Kurfürst Joachim II. of Brandenburg, and is thus historic. The letters P. V. S., with the year, appear several times on the armour, and the Brandenburg arms decorate the breast. The helmet is the burgonet, the cuirass is shorter than the fashion immediately preceding, while the rim of the breastplate stands out sharply beyond the tassets. The breastplate projects a little below the centre, and the shoulder-pieces and general pose, with the before-mentioned features, are all characteristic of the year of make. The ornamentation in repoussé work is very fine. This suit has been fully and ably described by Dr. Edgar von Ubisch in the *Hohenzollern Jahrbuch* of 1899. (See illustration, [Fig. 26.](#))

Descriptions in detail and illustrations are given of various suits of the second half of the sixteenth century. During this half century (sixteenth) defensive armour may be said in some respects to have reached its highest point of excellence; but towards its close unmistakable signs of decadence began to appear, and cap-à-pie suits fell gradually into disuse. This was caused by the inability of the armour to resist the then more penetrating firearms, or perhaps even still more, because the newer tactics demanded lighter cavalry and fighting more in masses, and less from individual efforts hand-to-hand. A style of demi-armour, called the “Allecret,” largely prevailed during the second half of the sixteenth century. The name is a corruption of “allekraft” (all strength). The peculiarities of this fashion will be shown in an example from the author’s collection ([Fig. 27](#)), which will be fully described later in these pages. This half-armour was often worn by light horsemen, household troops, and leaders of companies. It is very common to find, especially in family collections, some particular suit or suits ascribed to a great ancestor, but this is nearly always romance. It is an uncommon advantage to find a harness dated with the year, as some few are. There is a suit of this kind in the National Museum, Munich, with the date 1597 inscribed, and others at Nuremberg and Berlin. The more that is seen of armour, 1560–1600, the greater is the difficulty in many cases of fixing any approximate date, or arriving at any standard for suits covered by the period. Many suits were restored again and again, and this naturally gives rise to great perplexity. With this period closes the pre-eminence in the field of the knightly order, as such.



Fig. 26.—Suit by Peter von Speyer of Annaberg,
dated 1560.

The decline of armour may be said to have already commenced contemporaneously with the period of its greatest elaboration, in the sense that half-armour began to be freely worn early in the second half of the sixteenth century, indeed, a figure of a Swiss halbardier, given in Holbein's "Costumes Suisses," of the first half of the century wears merely a light sallad, with cuirass and taces; and the rank and file of pikemen, billmen, and harquebusiers generally bore a similar equipment. Even the "Allecret" description, which is half-armour, was greatly used by the leaders of companies and mercenaries generally; while what might constitute a battalion or combined body of troops was often under the command of a captain belonging to the knightly order, still armed cap-à-pie. The fact is, that full armour could not be constantly worn during a long campaign as then conducted without injury to health, and its use became more and more restricted to the knightly order and men-at-arms, who were not generally exposed to the same hardships as the common soldier. The man-at-arms of the sixteenth century became the pistolier and cuirassier of the seventeenth, and then wore half-armour. The example of demi-armour ([Fig. 27](#)), sometimes called "Allecret," dates from late in Queen Elizabeth's reign; but a demi-harness, with other details, was worn much earlier, and notably by the German Landsknecht and the Swiss. The main features of this suit are that there are elbow gauntlets, a fashion adopted from the Asiatics; and that the gorget and épaulières are riveted together. The specimen under discussion is probably of English make. A shirt of mail was possibly worn beneath it, but this defence was generally dispensed with by the end of the sixteenth century. The "Triumph of Maximilian" shows leaders of footmen wearing half-armour. Black and white demi-armour was very common at this time, and an interesting example of this description is given in [Fig. 28](#). Its general characteristics are as follows:—The burgonet is open, and the gorget, which is riveted to the épaulières, has two laminations at the neck, around the highest of which is a corded rim. The breastplate is short, with a projection over the navel. The taces are riveted to the tassets, which descend to the knee. There are no brassards, but short elbow gauntlets protect the hands and lower arms. The figure has jackboots, and is of early seventeenth century date.

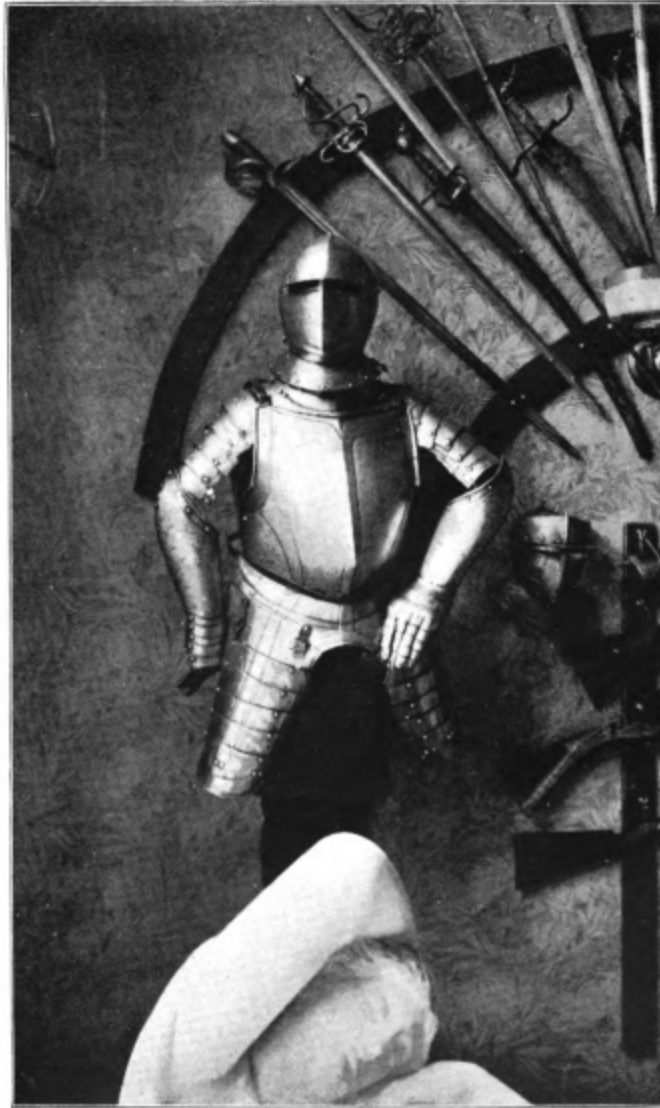


Fig. 27.—Plain Demi-suit, in the Author's Collection.



Fig. 28.—Black and White Demi-Suit, in the Author's Collection.



Fig. 29.—Late Suit at Munich, 1590–1620.

Cap-à-pie harnesses tended to become lighter as time wore on, and in the last quarter of the century the tasset and cuisse became combined in a series of light overlapping plates, directly attached to the cuirass and riveted on to the genouillières; which in their turn become attachable to the jambs by an adjustable screw. A representation is given of a late suit of armour of this description in [Fig. 29](#), where the helmet is the collared burgonet, which is characteristic of the end of the sixteenth and early in the seventeenth century. The cuirass has three horizontal laminations over the abdomen, while the upper leg and thigh armour is the combination already referred to. The elbow gauntlets of the suit are very characteristic of the period. The harness dates from very late in the reign of Queen Elizabeth, or possibly even later.

Many writers lay far too much stress on the use of firearms as the main cause of the gradual disuse of armour. Coming easy to hand, it was eagerly adopted by many writers on the subject, but like most generalisations it is misleading. That it was a potent factor in this direction is certain, but it was only one of the many causes which have been already touched upon in these pages. The general demand for cap-à-pie armour languished from the end of the sixteenth century forward, and with it vanished the taste and skill of making and decorating it; for we have very little more of the exquisite work of the “renaissance,” the vigour and force of which had spent itself. Here and there a fine suit is met with, usually made for royalty, but always lacking finish in the details; the majority are sadly inferior in material, workmanship, and decoration—indeed, the character of the work is coarse in every particular, and became more so as time moved on. The change in armour during the first half of the seventeenth century was very great. The breastplate became flat and very short, and open helms were much worn. The representation ([Fig. 30](#)) of a very early seventeenth century suit is from the armoury at Brancepeth Castle, Durham. This suit probably dates from very early in the seventeenth century. The helmet has an umbril over the eyes. Immediately under the peak is the ocularium of two very broad slits—the visor is grated. The suit is freely studded over with rather large-headed rivets, the gorget is pointed, cuirass short with lance-rest, but no garde-de-reine. To a broad rim at the bottom, tassets, consisting of nine lames, are attached by straps and buckles. The coudières are sharply pointed at the elbow. The most remarkable and distinctive feature in connection with this

suit is the protection given to the inner arm by a series of small and very mobile laminated plates, attached to the rerebrace and vambrace by rivets; another example with a similar arrangement may be seen in the Tower. Cuisse and jamb have a high ridge running down the centre in front, the genouillières having a thicker projection, bevelled at the sides, in a line with the ridge on the other two pieces.

Plate armour fell into discredit during the seventeenth century and gradually disappeared, the pikeman being the last of the foot soldiers to use it. The cuirass was the last piece generally worn, and this in time gave place, except in the case of the cuirassiers, to the buff coat and jerkin.

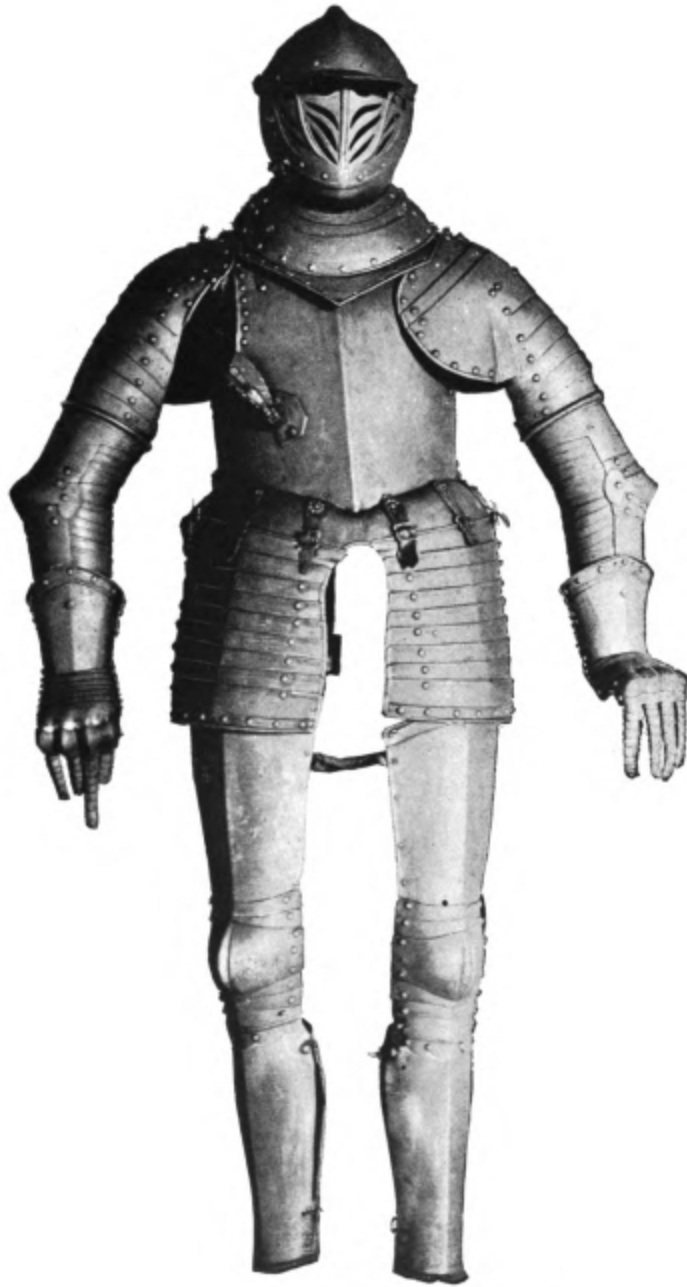


Fig. 30.—Late Suit at Brancepeth Castle.

Among the great armour-smiths who worked from 1540 to the end of the century may be mentioned Kunz Lochner of Nuremberg, who was perhaps the greatest artist in steel of the German “renaissance.” A suit made for Duke Johann Wilhelm of Weimar about 1560–65, at Dresden, is very typical of his time. The comb of the burgonet is high, the neck-piece consists of three lames; the breastplate is short and “peascod”; while the cuisses exhibit an early instance of coming to the knees. This suit is referred to under the heading “Enriched Armour.” Anton Peffenhauser of Augsburg began somewhat later, and worked up to the end of the century. A notable example of this master may be seen at Madrid, in an enriched harness made for Don Sebastian of Portugal in 1576 ([Fig. 39](#)); and there are others at Dresden. Sigmund Rockenberger of Wittenberg; the von Speyers of Annaberg in Saxony, and the two Wilhelms von Worms, and Heinrich Knopf of Nuremberg; Giovanni Battista Serabagio, and Lucio Piccinino of Milan, were all great artists of their time; and examples of their work may be seen at Vienna, Dresden, and Berlin. Mention of the work of Jakob Topf of Innsbruck first appears about 1575, and an attempt has been made to identify this armour-smith with the “Jacobe” of the South Kensington Album, but with very slender foundation in point of evidence, as it seems to us. Some further sifting of the matter would be interesting.

PART XIV.

ENRICHED ARMOUR.

THIS class of armour was more for parade purposes than for actual service in the field, and it was much used in the lists. Most suits of the kind were provided with a set of reinforcing pieces for jousts and tourneying. These pieces have already been fully described under "The Tournament" heading, and illustrated in Figs. 10 and 11. The amount of artistic skill of the very highest order that was lavished on the ornamentation of armour in the later "middle ages," and especially during the "renaissance," was a remarkable feature of the times, and artists of the greatest repute found constant and lucrative employment in designing for this purpose. Suits were finely and delicately chased, engraved, russeted, and enriched with gold, embossed, damascened, appliqued, and decorated with repoussé work.



Fig. 31.—Suits by Jörg Seusenhofer of Innsbruck.

Italy and Germany were the workshops for the finest specimens, and Milan, Brescia, Nuremberg, Augsburg, Innsbruck, Venice, Florence, besides other places, vied with one another in the production of specimens of consummate skill and elegance. French examples were coarser and less artistic in every way, while there was but little of knightly armour made in England, and that little, excepting for a very brief period, was of a vastly inferior description. The number of artists and craftsmen, in widely different branches of art and manufacture, who were employed to design, turn out, and finish a suit of armour, or a weapon for war or for the chase, was simply legion; and, of course, in the case of enriched suits, or arms, still more were brought into requisition. There is the designer, modeller, steel, silver and gold smiths, carvers, enamellers, inlayers, engravers, repoussé or workers in hammered work, damasceners, polishers, and hosts of other craftsmen, each contributing his quota of industry and skill to one complete whole. Artists of the very highest celebrity, such as Donatello,^{[33](#)} Michael Angelo, Albrecht Dürer, Leonardo da Vinci, Benvenuto Cellini, and Hans Holbein, had no higher ideal than in designing for this kind of work, and some of them were engaged in engraving also. It is well known that many armour-smiths employed other artists for designing and ornamentation, while others, like Kunz Lochner of Nuremberg, did their own embellishing as well as the smith's work. An illustration is given in [Fig. 31](#) of two very fine suits by Jörg Seusenhofer of Innsbruck. They are both tastefully engraved, and appear to be of a somewhat earlier make than the archducal suit by the same master, referred to in a previous chapter, and differ from it, as well as from each other, in some rather important features, especially in the form of the cuirasses and tassets. Only one of the three has pikeguards. These suits were made about 1540.



Fig. 32.—Cuirass and Tassets, at Dresden.

There is a chastely enriched harness in the Kriegswaffen-Saal at Dresden attributed to Wilhelm von Worms. A drawing is given in [Fig. 32](#) of the cuirass and tassets. On the left side of the breastplate is engraved a figure of a knight kneeling before the crucified Christ on the cross. The top of the breastplate is tastefully ornamented with a shield, with foliations on either side. This example is specially valuable, as it bears the date of make —1539.

An example in black and white may be seen at Berlin, the bright spaces being engraved. The breastplate is adorned with an engraved figure of Christ on the cross, and the gorget bears the legend: SOLVS SPES MEA CHRISTVS. A rondelle protects the right armpit. The left pauldron is a restoration. The suit dates about 1570.

There is a remarkable harness at Berlin, dating from about the middle of the sixteenth century. The cuirass, taces, and tassets are banded with an ornamentation of chevrons, which are bright and black alternately. Each row is defined with lines of brass, probably originally gilded. The cuisses are bright on the upper portions, which are enriched alternately with piping and small overlapping plates like shillings; the lower portions are black, and so also are the jambs. The sollerets are small and “bear-paw,” the extremities adorned with alternate bright and black flutings; the pauldrons are treated in the same manner. The rerebraces are ornamented with thick, circular coils to resemble puffs; there are no coudières, but the joint is rendered mobile by eleven narrow lames. There is a boy’s harness of similar make at Vienna, by Hans Seusenhofer, dating about 1511. This suit is obviously a copy of the civil dress of the time.

SUIT AT ALNWICK CASTLE.

This is a very chaste and elegant Italian suit ([Fig. 33](#)), dating from the last quarter of the sixteenth century. It is ornamented in the banded Italian style; the ground of repoussé work, with its rich minute foliations in low relief, is gilded, while the rest of the steel remains bright. The general style of the ornamentation is alternate chevrons of bright steel and minute repoussé work. The decorative work on the pauldrons and genouillières is,

however, much bolder in character than on the rest of the armour. A very similar style of ornamentation may be seen on a tilting suit given in Skelton, vol. i., Plate VIII., and dated by him 1543. The Alnwick harness is freely studded with brass-headed rivets which have been gilded.

The helmet is in four pieces, and highly characteristic of the Italian school of the period.

The gorget is comparatively modern, but conveys the idea that it was copied from the original piece owing to dilapidation, and but for the ornamentation it would pass even with close observers when the suit is set up.



Fig. 33.—Suit at Alnwick Castle,
Northumberland.

The pauldrons are very beautiful, and laminated at the shoulders and upper arm. The rerebrace and vambrace are finely formed and ornamented; the former is laminated.

The coudières are pointed at the elbows, with side-guards which continue round the arms.

The gauntlets are articulated, with thumb-plates, and a salient ridge runs across the knuckles. One of them, like the gorget, is of a more recent date than the main portion of the suit.

The cuirass is specially long and handsome. A broad piping borders the top and arm-holes. A tapul runs down the centre, projecting in a hump towards the middle. On the right side is a lance-rest, and on the left are holes for affixing a grand-guard. The lower portion of the cuirass consists of three narrow laminated plates, running almost horizontally, and fastened together by brass-headed rivets, which were originally gilded. The tassets are riveted to the bottom rim of the cuirass. These pieces consist of ten lames, with gilded rivets. A special feature is that the tassets can be shortened or lengthened at pleasure, the last four lames being detachable—clearly an arrangement for fighting on foot or on horseback. Other examples of this kind have already been given. The upper section is complete in itself with an ornamental rim, as is the lower one. This is a contrivance often met with in the second half of the sixteenth century. The attachment is accomplished by a screw catch and sliding rivet.

The backplate, which terminates in a garde-de-reine, has a piped border round the top and shoulders, and there are two lames at the bottom.

The cuisse, like the tasset, is in two sections, with similar means of attachment. The genouillières are attachable to the jambs by catch and sliding rivets. The knee-guards are small. The jambs are banded down the centre, in a line with the genouillières and cuisses. The sollerets are the variety styled “bec-de-cane,” being almost the shape of the foot. Both jambs and sollerets must be classed with the gorget and one gauntlet as restorations; they are all most beautifully done. Some details will be clearly seen in [Fig. 34](#).



Fig. 34.—Some details of the Suit at Alnwick Castle.

The harness already referred to as having been worn by the Prince-Bishop of Salzburg about the year 1600, and illustrated in [Fig. 35](#), is a beautiful suit by the celebrated Milan armour-smith, Lucio Piccinino. It is profusely inlaid with gold, and the ornamentation is most elegant. The sumptuous and elaborate decoration, which is in the banded Italian style in repoussé or hammered work, with arabesqued foliations, is interspersed with medallions encircling male and female figures. The helmet and suit throughout is closely in touch with the elegant Italian school of the end of the sixteenth century, which, however, already erred on the side of redundancy in ornamentation. The close of all great periods culminates with this great fault, sharply marking the beginning of the end; the waning vigour of the theme eked out by a profusion of detail. The prince-bishop's arms are engraved on the cuirass, and the historic character of the suit invests it with special interest and importance. The series of reinforcing plates belonging to it may be referred to in Figs. 10 and 11. Lucio Piccinino's style marked the last stage before the decline of art. He came of a family of artists; his father was the celebrated sword-smith, Antonio Piccinino. Other examples of Lucio's handiwork may be seen in a richly decorated helmet and shield at Vienna.



Fig. 35.—Suit by Lucio Piccinino, of Milan.

SUIT AT NAWORTH CASTLE.

This suit is very rich and handsome, being freely engraved and inlaid with gold—the gilding has, however, greatly worn off. The ornamentation is somewhat rude, both in character and in execution, and vastly inferior to either Italian or German work. The cuirass is ornamented with a “George” badge on either side, indicating a knight of the Garter, the execution of which is good. The genouillières are attachable to the jambs by reversible catches, which pass through the plate—they are the same catches as shown on the Osuna harness. There is a tapul and garde-de-reine. The sollerets are square-toed, but very narrow, not “bear-paw” like the “Maximilian.” The Earl of Carlisle suggests the possibility that the harness may have belonged to the last Lord Dacre, who died in 1566. This would, of course, point to an even earlier date of make, but this seems incompatible with the general aspect of the suit, which would appear to date from late in the reign of Queen Elizabeth.



Fig. 36.—Repoussé Armour at Berlin.

SUIT FROM THE COLLECTION OF PRINCE CARL OF PRUSSIA, NOW IN THE ZEUGHAUS AT BERLIN.

This beautiful suit ([Fig. 36](#)), decorated in repoussé work in very high relief, dates about the close of the sixteenth century, and the ornamentation is instructive as well as artistic in the classical battle-scene and details it depicts. The mitten gauntlet, with expanded cuff, is very fine. The ridge over the knuckles is bold, and smaller ridges continue to the finger tips.

THE OSUNA SUIT.

This is highly characteristic of the period it represents. The armour is freely ornamented in repoussé or hammered work, and bears traces of gilding. The suit was probably made in Italy, is very handsome, and has seen much service. Being well authenticated, it has a special interest. The suit belonged to Don Pedro Fellez de Giron, Duke of Osuna and Infantado, Knight of the Black Eagle Order, etc., Viceroy of Sicily about 1600, and later of Naples (about 1610). It was saved from the fire at the old De Giron family seat in Belgium—the castle of Beauraing, in the Province of Namur, not far from Dinant. The place was burnt on the 3rd December, 1890.

DETAILS.

The whole suit ([Fig. 37](#)) is freely ornamented with arabesqued foliations on a ground of fine vertical lines, banded in the Italian style, interspersed with human heads, some of them grotesque, and enclosed in medallions; and a series of armed figures, which would richly repay a close examination. The helmet is a remarkable piece of workmanship, and forged in a single piece; it weighs seven pounds. It is an Italian casque of a most graceful and classic form. The repoussé ornamentation on it is banded like the rest of the armour. The comb is very high, and fluted all over the crest.

There are remains of a leather lining inside, fastened all round with gilded rivets. The plume socket has two holes for adjustment, and there is another hole in the comb for firmly securing the plume of feathers. The oreillettes are provided with six holes on one side, and three on the other, for hearing; and have each a round projecting eye, with fluted edges, presumably an attachment for keeping the flaps up when not required, or for fastening them across the throat. Both peaks are of overlapping plates, with fluted borders. A very similar helmet, formerly in the possession of Baron de Cosson, was ascribed by him to 1530–40. He writes concerning it: “Many rich suits had one of these light open helmets as well as a close helmet,” a fact proved by existing examples at Madrid and elsewhere. We have already quoted an example in the description of the suit of the Prince-Bishop of Salzburg, which has a close helmet and a cabasset. The cuirass has a tapul, with a projection near the bottom; this particular form was termed the “peascod” in England. Both these pieces are bordered round the chest and arms with a thick ridged piping. This piping was a contrivance to stop a stroke from penetrating beneath the gorget. The tassets consist of six lames, and are attached to the tace, which is in one piece, by straps and buckles; all the rivets have gilded heads. The lower body is protected by chain-mail. The left pauldron is the larger; both have free laminations at the shoulder and upper arm. The coudières are cup-formed over the elbows, and go round the arm. The gauntlets have highly-rounded articulations for the fingers, with a separate thumb plate. Both leg armour and sollerets are freely decorated in “banded” ornamentations, with enclosed medallions, besides gilded rivets. A sharp ridge runs down the front of the cuisse, genouillière, and jamb. The genouillières are fastened round the back of the knee by straps, and on to the jambs by a reversible turning pin on the latter, passing through a hole in the former; and a turn of the screw secures the attachment. Jambs, which are hinged, and sollerets are riveted together, with lames above the ankle. The sollerets are “bear-paw.” All these pieces are held together by gilded rivets. The suit was probably made in the third quarter of the sixteenth century, or possibly as late as the fourth quarter, though the shape of the sollerets would point to a somewhat earlier period. [Fig. 38](#) exhibits some details of the suit. The stand on which the armour is hung is very old, and has probably stood in the armoury of the castle of Beauraing for centuries; and the face is very possibly a portrait of the Duke d’Osuna.



Fig. 37.—Suit of the Duc D'Osuna.



Fig. 38.—Some details of the Osuna Suit.

The beautifully embossed harness at Vienna, made for the Archduke Ferdinand of Tyrol, about 1560, is the work of the Milanese master, Battista Serabaglio. The casque is of classic form.

An embossed suit ([Fig. 39](#)), made by Anton Peffenhauser of Augsburg, about 1570, for Don Sebastian of Portugal, is in the Armeria Real at Madrid (Catálogo, page 94, No. A290); it is a notable example of the time.

The collection in the hall set apart for enriched armour at Dresden is especially valuable in exhibiting a remarkable series of fourteen historic suits, blazing with ornamentation, and covering a period of from something like the second quarter of the sixteenth to the end of the first quarter in the seventeenth century. All these suits are royal specimens of their school. The earliest is the harness of Kurfürst Moritz of Saxony, 1521–53. The rider sits on horseback in his field-harness, which is freely decorated with gold arabesques on blue bands. The Kurfürst bore this armour on his entry into the conquered city of Magdeburg in 1551. The bards are enriched in the same manner as the armour borne by the Kurfürst. Another suit is that of Duke (afterwards Kurfürst) August, 1526–86. It is fluted and richly ornamented, bearing the Saxon arms inlaid. This harness was the gift of the Archduke Ferdinand of Tyrol, and is probably the work of Jörg Seusenhofer of Innsbruck. The figure holds a field-marshal's baton in the right hand. The legend, "Semper suave," is inlaid on the bards. Another suit of this Duke's is a specimen of blackened harness with white bands; a description much worn in campaigning in the second half of the sixteenth century and later, because it was easily kept clean in all weathers. It is a fine piece of work, and is inscribed with the date 1546. The Duke bore this suit at the battle of Mühlberg in the year following. A harness of Duke Johann Wilhelm of Weimar, bearing the mark of Kunz Lochner of Nuremberg, date about 1560. A suit for man and horse of Kurfürst Christian I., 1560–91. Tournament reinforcing pieces stand by it—a tilting helm, grand-guard, garde-de-bras, etc. The harness for man and horse of Kurfürst Christian II. (1583–1611), a masterpiece of the armour-smith's art, is by Heinrich Knopf of Nuremberg, and cost £1,750. The ornamentation consists of arabesques on a gold ground with enclosed medallions. A rapier by Andreis Munsten of Solingen is with the suit. There is a second suit that belonged to this prince—the ground is a dull green, with chasings. This harness, according to an inventory of 1606, was bought at Augsburg in 1602—it bears no mark. The

latest harness of the series is that of Kurfürst Johann Georg I., and the date is 1622; it is the work of Hieronymus Ringler of Augsburg, and though very richly decorated exhibits unmistakable signs of the decline of art.



Fig. 39.—Suit by Anton Peffenhauser, at Madrid.

This remarkable series is as valuable from an educational as from an æsthetic point of view; still, though the differences in points of detail, over the various periods, stand before you, it must not be forgotten that fashions were far from being contemporaneous over northern and central Europe, and that new departures of fashion in armour, as in dress, took long to travel and get generally assimilated—far longer in the sixteenth century than to-day,—hence one or two salient features cannot always date a suit, even within a couple of decades. There is a fine series of plain gilded suits at Dresden, which were worn with boots.

To give a completer series of examples of late sixteenth and early seventeenth century forms and fashions would make this work far too voluminous. Examples of pikemen's later suits, etc., would make the chain more complete, but the varieties are so very numerous that it would be impossible reasonably to cover them without largely extending the size and scope of the work. Practically the illustrations close with the end of the sixteenth century; after which the general use of armour, from causes already referred to, rapidly declined. The interest in the later forms is comparatively far less to the student or collector, whether looked at from an artistic or historic point of view, than the grand period which has been imperfectly covered here.

SECTION II.

**THE WEAPONS AND ENGINES
OF WAR.**

PART XV.

INTRODUCTORY AND GENERAL.

DION Cassius refers to the armament of the Caledonians as being a buckler, dagger, and lance; while Tacitus says that the Britons used large blunt swords and small bucklers.

Excepting for a few specimens found in peat mosses and burial mounds, we are indebted to monkish chronicles for all our knowledge regarding the weapons of the “dark ages” of our era, together with a few glimpses and suggestions obtained from the “Sagas” handed down, partly *vivâ voce*, from generation to generation. There are many errors in the best classifications of arms, and many weapons in museums and private collections scheduled as belonging to the “iron age” are really of mediæval origin; still, this state of things has vastly improved of late years, and some of the newer museum catalogues leave but little to be desired, having been compiled by men who have made a close study of the subject, and who have had the advantages of ample opportunities for comparison in their surroundings.

Procopius, the secretary of Belisarius, gives some account of the arms of the Franks of the sixth century, whose weapons were the sword, the axe or francisca, and the spear. The ordinary battering-ram and the testudo, which was a movable shed containing a ram, were in use in this century, as well as a machine for boring walls.

The sources of information available from the seventh to the end of the tenth century are very scanty as far as Britain and the Germanic peoples are concerned; but more has been preserved relating to the Franks, a race also of Germanic origin, whose country, more than any other during the “dark ages,” seems to have been imbued with the continuity of Roman methods

and traditions. This was indeed a barbarous nation, with the corrupt remnants of the Roman Empire grafted on to it; and the Frankish kingdom only became consolidated some time after the introduction of Christianity, which provided a much needed common platform in the teachings and example of the monastic orders. The monks wrote and preserved the manuscripts, without which the “dark ages” of our era would have left but little trace behind them.

Double axes and the lance or javelin appear in the seventh century, and indeed up to the age of chivalry the weapons of the ruling class of the more civilised nations of Europe continued to be the axe, the lance, and above all, the sword; while those of the yeomanry or peasantry were the bow, the sling, and the fustibal or staff sling. The axes differed in shape and length, some blades curving like a halbard, of which it is evidently the prototype, while others were long and narrow. The form of the lance or javelin varied greatly, and some were barbed. Two kinds of swords prevailed—the true sword and a shorter weapon. The true sword was worn by leaders only; it was flat, double-edged and sharp, two and a half to three feet in length, with an obtusely pointed blade. The shorter sword was in general use, also the battle-axe and a dagger.

The Anglo-Saxon thane carried a sword, then solely a horseman’s weapon; while the footman was armed with a spear, an axe, a shield, and a dagger. The Anglo-Saxon spear was long in the blade, and the pole-axe narrow bladed and single edged.

Among the valuable Anglo-Saxon records we have, the Ælfric MS., which is profusely illuminated, and contains a good deal of information about swords, mentions the tri-lobed hilt; but the richest mine of contemporary history, for delineation of the weapons of the eleventh century, is undoubtedly the Bayeux tapestry. The arms given in that invaluable record are the lance, the sword, the mace, the axe, and the bow. This bow is shorter than the weapon known as the English longbow, which was not used in battle much before the reign of Edward I. Some of the Anglo-Saxons appear with javelins.

The weapons used by the Normans at Hastings still retained traces of their Scandinavian origin. Their army was rich in cavalry and archers, while their Anglo-Saxon adversaries were but ill-provided in these respects.

The sword was used in conjunction with the dagger as early as the reign of Edward I. As the great advantages of the use of infantry became more apparent, the yeomanry began to play a much more important part in the warlike combinations of the age; while even the peasantry had now become indispensable in all campaigning on a large scale. It was mostly, however, the freedman who went to the wars, while the serf remained at home to till the soil. This it was which brought the bow and other footman's weapons so much to the fore. Bills and scythe-knives³⁴ appear to have been in use early in the eleventh century, indeed probably long before, as this was the class of weapons most easily extemporised from the implements of husbandry. The goedendag, the weapon of the guilds and boors of Flanders, and later of the lower orders in France, is by some considered to have been a ploughshare mounted on a pole or staff; but this is a question which will be dealt with in the more detailed descriptions of the various weapons covered by these notes. The flail also, with its military adaptations, contributed its quota at a very early period towards the armament of the masses; and the English longbow was the arbiter of victory in many a stricken field, and was the main factor in breaking down the inordinate power and oppression of the English, or perhaps more properly speaking, of the Norman barons. English archers carried stakes pointed at both ends as part of their equipment. When driven into the ground with their points towards the enemy they formed an efficient stockade against a charge of horsemen, as the horses impaled themselves upon them. The mace and its kindred weapons, with their common prehistoric ancestor the club, and the long line of the more rudimentary axes, from the remotest times, all played their part in the wars of the earlier "middle ages."

The weapons of the fourteenth century differed but little in form from those of the thirteenth, and it was not before the fifteenth century that organised infantry became an indispensable contingent of the "establishment" of every army in the field; by which time halbards, pikes, partizans, and their kindred weapons were all in use. These weapons, with the glaive, voulge, holywater-sprinklers, and morning-star, continued more or less in vogue until the beginning of the eighteenth century. It is frequently affirmed that gunpowder was known to the Chinese before the Christian era began, and the embrasures in the Great Wall, erected 200 B.C., are often cited as proof that artillery of some sort or other was used in China

at a very early date. However this may be, it is certain that there must be an extraordinary wealth of facts and suggestions lying buried deep under the soil of that “old world empire” and Japan. In this age, so hungry for new developments, it will probably not be many years before some enthusiastic antiquary begins to look more closely into the possibilities of this virgin soil by digging investigation.

The honour of the invention of gunpowder is claimed, however, by several of the European nations. It is often stated to have been a fortuitous discovery in 1320 by Bartholdus Schwartz, a monk of Friburg; but there is a recipe for its production as far back as the ninth century of our era, the component parts then being six parts of sulphur to two each of saltpetre and charcoal,³⁵ but this acted by fusing and not by detonation, and was probably a form of Greek fire. The properties of gunpowder were thus more or less known long before its application as a motive force for projectiles. This did not take place, however, before the fourteenth century. It is often stated that gunpowder was not made in England before the reign of Queen Elizabeth. Henry VIII. bought gunpowder largely in Spain, but as he also purchased saltpetre and sulphur it seems certain that gunpowder was made in England during his reign. There are records at this time of payments for gunpowder to people with English names; and Carlo Capello, the Venetian, writes in 1532 that Henry made powder in the Tower then. Its adoption for application to projectile warfare gradually revolutionised both the armament and tactics of the middle ages and of the “renaissance,” especially in the direction of gradually discrediting the use of the bow in all its forms. The introduction of the epoch-making bombard and hand-gun changed the face of history.

Weapons may be divided into two classes, those made for the rank and file being plain and coarse; while an immense amount of artistic skill, frequently of the very highest order, was lavished during the later middle ages and the “renaissance” on the decoration of swords, daggers, crossbows, and staff weapons generally, as well as on armour of proof, for leaders and the higher classes. The hilts of both swords and daggers were richly chased and decorated in high relief with mouldings and even statuettes, while the blades were often inlaid as well as engraved. Even artists like Holbein and Albert Dürer exercised their utmost skill in

designing for such work. A beautiful example is given in [Fig. 40](#) of a sword that belonged to the Archduke Ferdinand of Tyrol.



Fig. 40.—Enriched Sword, second half Sixteenth Century.

The pageant weapons of a prince's guard, though formed like those used in actual warfare, were especially rich in this respect; and the stocks of crossbows, which afforded great scope for ornamentation, were not only beautifully inlaid with bleached stag's horn, ivory, and mother-of-pearl, but often adorned with mythological, historic, or biblical legends, carried out with rare elegance and finish. The great German smiths—Hans, Jörg, and Conrad Seusenhofer, Brockberger, Lorenz Kolman, Conrad Lochner, Swartz, Jörg Endorfer, Klemens Horn, Peter Munich, Wilhelm Wirsberg, etc., etc.; and the Italians—Antonio and Tomaso da Missaglia, Filippo Ciro, Giacomo and Francesco Nigroli, Ghinelli, Spacino, Antonio and Lucio Piccinino, and many others, vied with each other in the production of consummate creations of workmanship and art, some of them in armour of proof, others in offensive weapons, and many in both; and if the palm of excellence may perchance be awarded to the latter nation for originality and delicacy in design and finish, surely the Germans were but little if anything behind their confrères beyond the Alps in all these respects. The swords of Bordeaux and Poitiers were now far behind those of Toledo in renown, and the great Spanish masters, such as Antonio Ruiz, 1520; Juan de Almai, 1550; Francisco Ruiz, 1617; Tomas de Ayala, 1605; Sebastian Hernandez, 1637; and hosts of others rendered their cities and country illustrious by the excellence and beauty of their workmanship. Still, strangely enough, quantities of Solingen blades were imported into Spain during these centuries; for it will be noticed that the majority of rapiers picked up by collectors in that country have these German blades. The marks used by these smiths and many others may be found in the *Catálogo de la Armeria de Madrid*, and in a work by the learned curator of the Imperial collection at Vienna, entitled, *Meister der Waffenschmiedekunst von xiv. bis xvii. Jahrhundert*, and in the excellent catalogue of the Königl. Historische Museum at Dresden, compiled by Herr Max von Ehrenthal, the accomplished curator.

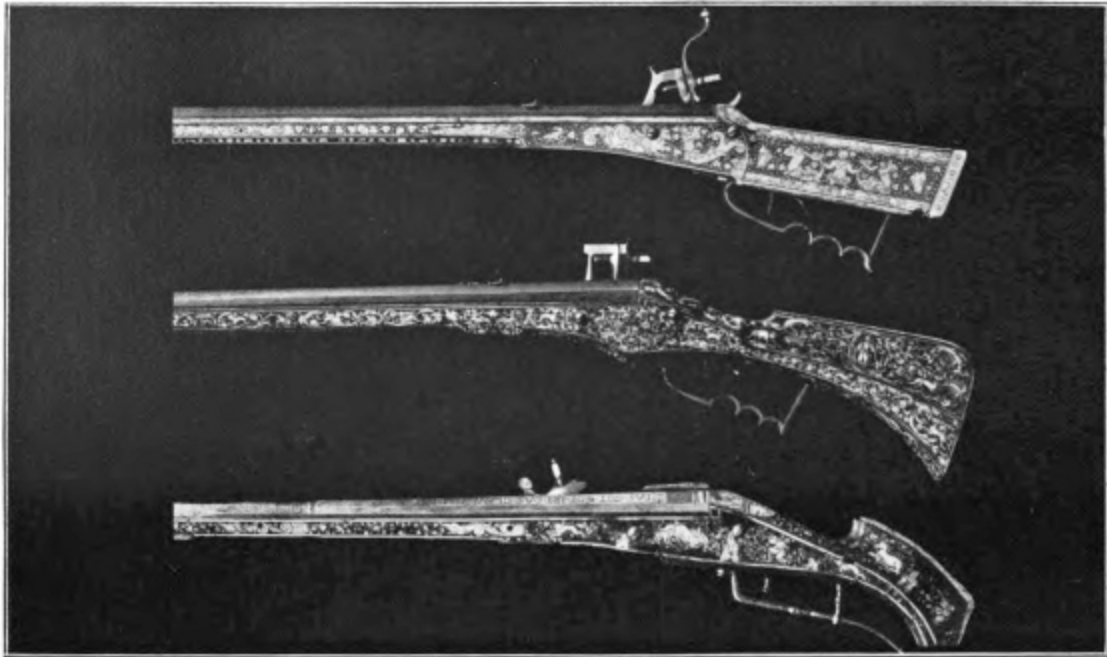


Fig. 41.—Hand-guns, Renaissance Work.

During the “renaissance” the gunsmith and his coadjutors lavished all manner of ornamentation on pageant hand-guns and their accessories. Barrels were chastely engraved, and stocks inlaid with bleached stag horn, silver, gold, steel, brass, stained wood, and mother-of-pearl; but these highly decorated weapons were not so much for real campaigning as for the use of body-guards, palace troops, and purposes of display generally, and especially for the hunting-field. [Fig. 41](#) represents three of these enriched weapons, inlaid with bleached stag’s horn. They are late sixteenth or early seventeenth century work.

The weapon of the Harquebusier and Musketeer was much plainer; and the matchlock was preferred to the wheel-lock by reason of its greater rapidity of discharge. There were, however, corps, especially cavalry, armed with wheel-lock weapons. The use of the longbow, which had for so many centuries played a predominant part in the combinations of English campaigning, had gradually languished with the greater mobility and precision of firearms; and the bayonet was soon destined to add new lustre to the British name. An order in Council of 26th October, 1595, ordains that the bows of the trained bands were to be handed into store, and calivers and muskets issued in their stead. In the year 1638 the stock of bows and arrows was omitted altogether from inventories of arms, thus showing that the weapon had become obsolete.

PART XVI.

THE SWORD.

THE sword has always been the most universal of weapons among almost all nations and ages. It is alike the symbol of honour and the vindicator of justice; though often, alas, the instrument of oppression. The history of the sword is almost that of humanity itself, and supernatural attributes have often been ascribed to it. There is something about an ancient sword that appeals to the dullest imagination—it is so suggestive of historic memories, both in heroism and treason. It is typical of the force behind the law; but the living sword of our forefathers is now but a memory. It would be fascinating to follow its forms, traditions, and ramifications from the “stone age,” and from Menes to Julius Cæsar and Charlemagne—in fact, something like such an enterprise was begun by Sir Richard F. Burton. His book is indeed “A Romance of the Sword,” but the priceless stores of information he has collaborated, and his fine florid imagination, help us but little in the present quest: sad it is that his researches stop at such an early stage.

The sword, and its diminutive of which it is doubtless an extension, forms a distinct class of arms, in contradistinction to the numerous family of hacking, clubbing, and staff weapons generally. It is difficult to draw any very arbitrary line between the sword and the dagger—the hilt is often the same in form, but some swords are short and some daggers long. Perhaps the best definition of difference is that the dagger is roughly under two feet in length, and was used rather as an auxiliary to the sword, for thrusting only; besides being more capable of concealment, and more efficient at close quarters than the larger weapon. Writers differ in their method of imagining the position of a sword for descriptive purposes—that is to say,

whether it be held downwards or upwards. It will here be regarded as being held in the right hand, point uppermost.

Bronze swords were deficient in hardness, so that they could not be adequately tempered; they were narrow and leaf-shaped, and this was the characteristic form everywhere. That recorded on Assyrian monuments is straight, narrow, and like the Greek, more for thrusting than cutting. The Roman type was longer, though still not of much use for parrying; and the leaf form became less accentuated.

The true sword had its birth early in the "iron age," which arbitrary period, though usually classified to close with the fifth century, might reasonably be prolonged to the dawning of the middle ages. It is during this interval that we have but little accurate information, still it may be taken generally that the weapon became both longer and broader after the fall of the Roman empire, when it was straight, double-edged, and of varying length.

The sword of Chilperic of Soissons (died 584) was found in his tomb at Tournay in 1653, and is now in the Louvre. The weapon has short straight quillons, and the pommel is also cruciform; it bears strong evidences of Oriental influence. Procopius describes the Frankish sword of his day as a short, straight, broad-bladed, and double-edged weapon, somewhat obtusely pointed, and usually about thirty to thirty-two inches long, just about the standard length of the modern small sword; while Agathias, his successor as a chronicler, records it as just the length of a man's thigh. To judge from the few specimens on record, it has both a cross-guard and pommel, but was by no means uniform either in form or size. Its extremity was rather rounded. A sword found in a grave on Chessel Down, in the Isle of Wight, answers very closely to that of the Franks, as described by Procopius.

The Scandinavian sword of the fifth, sixth, and seventh centuries was long, straight, and double-edged; while the Anglo-Saxon weapon of the eleventh century was about three feet long, cruciform, and rounded at the end. No one under the rank of thane was allowed to carry a sword, which accounts for so few specimens having been found.

The earlier Anglo-Saxon sword is, as far as can be ascertained, without cross-guard, but it has a small pommel. A MS. in the British Museum of the

tenth century gives an illustration of a sword of this kind, which is only two feet long.

We read in “Sagas” that the swords of heroes were often endowed with names or titles, such as the “Hrunting” of Beowulf, the “Excalibur” of Arthur, the “Tizona” of the Cid.

The component parts of the sword are of course the blade and the hilt. The tang is a piece of wrought iron welded into the shoulder of the blade, and inserted in the grip or handle, at the bottom of which is the pommel. The pieces or guards which pass across between the hilt and the blade are the quillons. Proving the blade was accomplished in various ways: an early method was by a heavy blow on a block of iron, first the flat, then the edge, and lastly the back; then bending the blade flatwise. The operation concluded by driving the point through a thin iron plate, which was called the “Toledo” test. A machine for testing swords was invented in England towards the end of the last century by Matthew Bolton, in which the blade was forced into a curve, reducing from 36 inches to 29 inches.

The Frankish sword of the eighth and ninth centuries is cruciform, with a pommel, which is itself sometimes surmounted with a cross. This may be seen in the Codex Aureus of St. Gall. The weapon of this period is, however, far from being uniform in shape, length, or breadth. The knightly weapon of the Bayeux tapestry is cruciform with a long, straight, two-edged blade, coming somewhat abruptly to a point, and a ridge running up the centre. The hilts are heavy and strong, with pommels. A Norman sword on the tapestry shows the pommel to curve on the grip. There is an actual specimen of this period in the Museum of Artillery, Paris. The blade of the footman’s weapon is much narrower than that of the knight. The sword of William Rufus is shown on a miniature in the Canterbury bible. The point is obtuse, the blade widens towards the quillons, the ends of which curve upwards, while the grip is short, and the pommel round.

There is not much change in the twelfth century, when swords vary a good deal in form; as also does the shape of the pommel. A specimen of the reign of the Emperor Frederic Barbarossa is in the museum at Dresden.

The cultellus or coustel is a short sword or long dagger. The weapon is mentioned in a statute of William of Scotland, 1165–1214. From this time

forward we have in military brasses and effigies figures of the knightly sword brought before us as it actually was.

The sword of the thirteenth century is more distinctly pointed, and has the cross-guard either straight or curving more or less towards the blade; the grip is rather short, and the weapon is usually about two feet six inches to over three feet long, and there is a large heavy pommel of various shapes. A good example may be seen on the Daubernoun brass. Some of the German swords of the century, actual specimens of which may be seen at Dresden, are, however, very much longer. The short handle could be rigidly gripped, so that the entire force came more from the arm and shoulder.

The sword blades of Damascus, India, and Persia were equal, if not superior, in temper, finish, and decoration to any made by the sword-smiths of Europe, but the Eastern smiths devoted much more care to the edge than to the point. In the main, they were curved blades. There is a good deal of romance in old Japan about the sword, and some very remarkable weapons have been turned out by their craftsmen. There were numerous distinct varieties of Asiatic swords and daggers; but to give even the merest outline of these would make the present notes far too long. Single-handed swords of Europe consisted of curved weapons like the scimitar or falchion, the dusack, cutlass and sabre, and those with a straight, double-edged blade.

The scimitar is of Persian origin, and was introduced into Europe during the first crusade; it did not, however, come very much into vogue before the middle of the fifteenth century. Like most swords of Asiatic origin, it is specially devised for cutting; and its curved blade, and the setting of the hilt, in relation to it, is well adapted for the delivery of a highly penetrating stroke. This weapon, the blade of which is short and single-edged, has probably its prototype in the "Acinace" of the Romans, a representation of which may be seen on that instructive monument of contemporary history, the column of Trajan. Possibly the Romans themselves derived it, like so much besides, from an Eastern source. The falchion, or fauchon, which is a smaller type of scimitar, appears in England early in the thirteenth century, and is mentioned in the fourteenth century romance of *Richard Cœur de Lion*, "broad fawchons and fawchons kene." It is in two varieties—a broad blade widening towards the point, with a concave back and sharp edge; and the other with a straight back. The curious tenure falchion of the Conyers is an example of the latter kind. This

weapon is figured in *Archæologia Æliana*, vol. xv.; and is also referred to in Blount's *Antient Tenures*. Sir Edward W. Blackett, Bart., in a communication to the Society of Antiquaries of Newcastle-upon-Tyne,³⁶ says that this weapon measures two feet eleven inches in length; on one side of the pommel are three lions, the arms of England, with remains of red enamel in the ground; and on the other an eagle with outspread wings, which Mr. Longstaffe considered to relate to Richard, King of the Romans, brother of Henry III. This statement would point to its being a weapon of the thirteenth century, which it undoubtedly is. The tenure is given in the inquest of Sir John Conyers in 1396. The Baron de Cosson mentions two examples somewhat similar, one in the Musée de Cluny, Paris; the other in the Brera at Milan. He compares the Conyers falchion with one given on the drawing from the Painted Chamber, Westminster, attributed by Mr. John Hewitt to the second half of the thirteenth century. The forms are certainly almost identical. The Conyers weapon has a nearly round pommel, with the quillons slightly curved towards the point at the extremities. The Paris falchion has a very large circular pommel, with the quillons on a sharp curve in the same direction. The guard of the Milan specimen is straight and the pommel a large oval, with small square side projections. The blades of all three falchions are similar in form, the Milan example being the largest. Drawings of the three falchions may be seen in the *Proceedings* (vol. v., p. 42) of the Society of Antiquaries of Newcastle-on-Tyne. *The True Tragedy of Richard of Yorke* (1595) says: "With purple fawchon painted to the hilts." Another local tenure sword, mentioned in Blount's *Antient Tenures*, is that under which the Umfravilles held their lordship of Redesdale in Northumberland. An instance of the application of the "tenure" principle in a humbler form and modern date, occurs in an agreement with the swordsmiths of Shotley Bridge, County of Durham, concerning rent for houses occupied by them. The rent is supplemented by an annual sword of their own make.

The sabre, which is a near relative of the scimitar, is of two kinds, both straight and curved; the latter form was in vogue as early as the thirteenth and fourteenth centuries, and of course later.

An interesting example of the curved form, which is attributed to Charlemagne (771–814), is preserved in the Treasury at Vienna. The form betrays its direct Eastern origin, and the tradition is too vague to base any

inferences on it. The sword is about thirty inches long, by over three-quarters of an inch broad, and would appear to date about the fourteenth century.

The sword of the fourteenth century continues cruciform, with the quillons either straight or curving towards the blade. The shape of the pommel varies greatly, being trefoiled, conical, circular, etc., and sometimes it is also charged with a cross. It was not uncommon for a ring to be fixed to the pommel for attachment to a chain connecting it with a mamillière. Examples of this kind may be seen on an effigy in the church at Ebersberg, temp. 1371; another at Borfe, in the Tyrol; and one is given by Hewitt in his *Ancient Armour*, vol. ii., Plate XV. The sword is fastened at the left side by a broad straight belt, called a “bawdric.”

Blades of this century, though far from uniform, become generally more ornate and longer than in the century preceding, sometimes attaining the length of four feet, and there are even longer examples.

Sword sheaths were usually of leather. The knight's sword-belt was greatly embellished in this century by quaterfoils, jewels, and enriched pendants.

The grip of the sword proper rather lengthens in the fifteenth century, and the tendency of the pommel is to become lighter, and is oftenest round or pear-shaped; there is still the plain cross-guard. The straight double-edged blade is long, and sometimes grooved. The pas d'ane guard is found in this century, though rarely. This guard projects over the base of the blade, its object being to protect the back of the hand, which it did but inadequately. It has often been assumed to have made its appearance first in the sixteenth century, but this is not the case, as a picture of the early part of the fifteenth century in a church at Mondoneda shows swords with this guard.³⁷ It forms, however, as a rule, an excellent guide as to date, and its presence would, under ordinary circumstances, indicate a weapon of the sixteenth century. There are some fine swords of this century (the fifteenth) in the Munich Museum, in excellent preservation, some with the original sheaths.

The knuckle-bow, called the finger-guard by some writers, is comparatively rare towards the end of the fifteenth century, but becomes common in the following. Mr. John Hewitt, in one of his contributions

towards the *History of Mediæval Weapons*, mentions an instance as early as the reign of Charles the Bold of Burgundy. It was long before this guard became united to the pommel. It clearly developed from the counter-curved quillons, one of which seems to have reached the pommel by stages. In Holbein's "Costumes Suisses" is a figure of a Swiss halbardier of the first half of the sixteenth century with a sword, the knuckle-bow of which unites the quillons and pommel.

The executioner's sword is broad in the blade. A German example in the author's collection is 39 inches in length. The pommel is circular, very heavy and flat, and engraved with an eagle; the quillons solid and plain, curving slightly towards the blade, which has a groove running up the centre. The blade is two and a half inches broad, and is inscribed with a cross, cross-bones, and a crown. Quillons are, of course, unnecessary on these weapons, and are unusual except in the case of German examples.

The sword used in the foot tournament was heavier and shorter than that for war.

The two-handed sword was introduced late in the fourteenth or early in the fifteenth century, and became a favourite weapon in the sixteenth, after which it was greatly superseded by the rapier. This long and very heavy two-handed weapon is a footman's sword, and was much used by the hardy mountaineers of Switzerland in battle, while the less robust Germans and Burgundians applied it more in the defence of fortified places. It was introduced into England early in the sixteenth century, when it was a favourite weapon of Henry VIII., and continued much prized there up to its close, when the rapier came into vogue. The handle is very long for both hands to grasp the hilt. The total length of the sword is up to five feet eight inches, and even more. This sword is the true *espadon*. Two-handed swords were usually worn without scabbard, but had a piece of leather permanently fixed on the blade above the quillons; they were rarely met with after the close of the sixteenth century. A variety with a wavy blade is called "flamberge." An example from the Meyrick collection is in the author's possession, and shown somewhat incongruously in [Fig. 23](#). This being a footman's weapon, ought not to be in the hands of a man-at-arms. Great strength of arms and supple wrists were necessary for cutting with these weapons; the point was rarely used. The true claymore is a two-handed sword. Some fine examples of two-handed swords and flamberges are given

in [Fig. 42](#). The thumb-ring appears in the fifteenth century, possibly a little earlier, and it was common in the sixteenth.

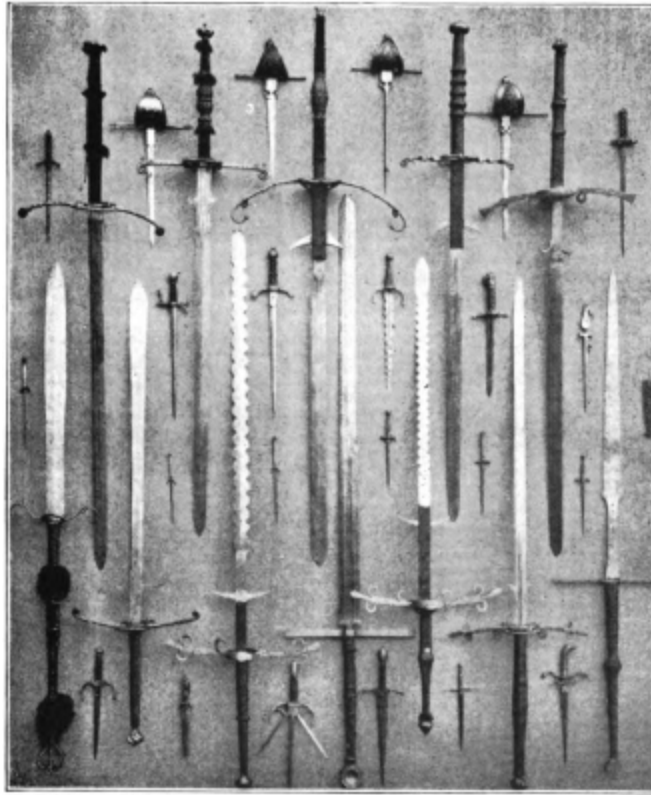


Fig. 42.—Two-handed Swords, Flamberges, and Daggers.

The anelace was a very common weapon of the fifteenth century. It is a short, broad sword or dagger, tapering to a point. The blade is usually about twenty inches long, by four broad, and double-edged. The weapon, called in Italy the cinquedea, is of Verona origin, and was styled oxenzunge by the Germans, and braquamart or épée de passot by the French. It is a very similar weapon to that carried by the ancient Greeks and Romans on the left side, called the parazonium, a late specimen of which was found at Sesto-Calende, and is now at Milan.

The dusack is a sword of the fifteenth and sixteenth centuries, with a blade like that of the curved sabre, while the hilt consists either of a hole in the rounded base of the blade for the hand to grip, or is a rounded continuation of the blade at the shoulder, forming a circular hole. The length is about 39 inches. The swordsman wore an iron or leather gauntlet reaching to the elbow.

Swords tended to become more ornate as the fifteenth century advanced, and towards the end and early in the sixteenth both pommels and quillons varied greatly in form and in size, the former being round, square, cusped, truncated, crescent-shaped, etc., while the latter tended both downwards and upwards, sometimes counter-curved, and curled at the extremities, but this feature became more pronounced later. The play of sword and buckler is very ancient, and was displaced in England by the rapier and dagger in the second half of the sixteenth century. The sword was of medium size and double-edged, while the buckler was about fourteen inches in diameter.

The usual form of the sword up to the middle of the sixteenth century is still cruciform, with or without the pas d'ane guard, a broad two-edged blade about three feet and a half long, and a large and frequently circular pommel; the quillons straight or slightly bent towards the blade, which tends to become narrower and lighter. There are, however, many examples of a greater elaboration of guards at an earlier period, when the guard formed like the letter S was not uncommon. An example of a sword by Ambrosius Gemlich, about 1530, is given in [Fig. 44](#). There is a calendar on the blade. The simple cross-guard disappears with the commencement of the second half of the century, and the pas d'ane guard becomes common. The sword-hand now becomes adequately guarded, and you get the counter-

guard, which later becomes amplified into one or more branches for encircling the back of the hand, while the quillons more generally assume curved forms and eventually merge into the knuckle-bow or finger-guard; and it was during the second half of the sixteenth century that the rapier hilt became completely developed. It was no longer the rule to wear the steel gauntlet; such guards had therefore become more necessary, and they were gradually evolved by reason of new developments in fencing strokes. Swordsmanship had now reached the point when the weapon, besides being for attack, was used more in a defensive sense. The term "shield" is applied to the flat piece of steel sometimes found at the base of the hilt, while the "shell" refers to a semicircular hilt. The growth of what are but inadequately described as counter-guards consists in a more or less complex system of perpendicular and horizontally curved and interlacing bars and hoops gradually evolving the S guard, cross and side ring, cross and finger loop, cross finger loop and half ring at the side, double branches, etc., which crystallised, so to speak, in certain classes of swords into the basket-hilt and the shell or cup. The practice and progress of the art of fencing had induced upward cuts and other movements that necessitated additional protection for the hand and wrist.

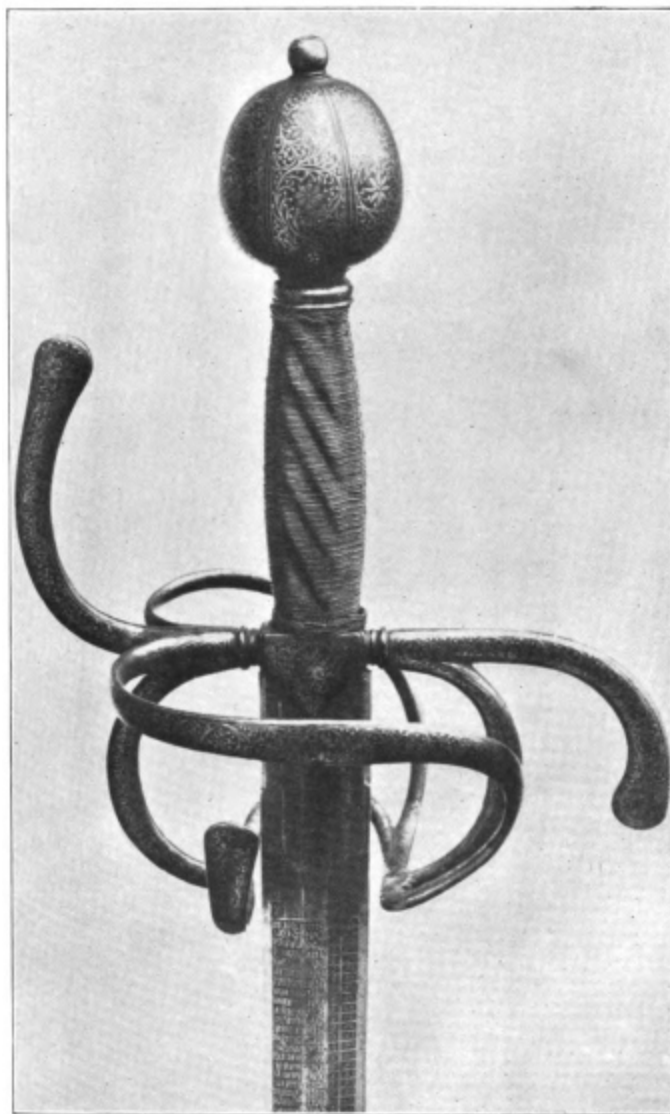


Fig. 44.—Sword of the Emperor Charles V., about 1530.

The lansquenette appears in the sixteenth century. It is a weapon about two and a half feet long, by two inches broad. The blade is broad and double-edged, and the grip thick and surmounted by a pommel. There is usually a counter-guard of two rings.



GERMAN.



SPANISH.



ITALIAN.

Fig. 45.—Rapier.

The mediæval estoc is a long, narrow stabbing sword of French origin. It was often used in tournaments, and is sometimes two-handed like the real claymore; it is a horseman's weapon.

The English broadsword appears in the reign of Edward VI.; both it and the cutlass are somewhat heavy and unwieldy.

Fencing is a purely European invention, and the time had now arrived when it had become more of a fine art, though still in its early stage; and this cause, more than anything else, brought about the general use of the rapier and small sword. The rapier is a sword with a great variety of guards, or with the basket hilt, either solid or perforated, and straight or curved quillons; it was introduced into England by Philip II., but appeared in Spain in the complex form during the preceding reign. This weapon has sharp edges, is grooved, and sometimes strengthened by a sharp central ridge. It was used mostly for thrusting, but not to the complete exclusion of cutting. The two-edged rapier is a military sword, but not useful for the *mêlée*, being more suitable for single combat in any form. Duels were sometimes fought with the rapier alone, but oftener with the rapier and main-gauche, the latter held in the left hand. Why the main-gauche should be specially named as left-handed is impossible to understand. Another form was with the rapier and a cloak, the latter being held in the dagger-hand. Examples of German, Spanish, and Italian rapiers are given in [Fig. 45](#).

Sir Frederick Pollock, Bart., in his admirable monograph, *The Forms and History of the Sword*, quotes George Silver (1599), the father of English broadsword play, who speaks of "that mischievous and imperfect weapon (the rapier) which serves to kill our friends in peace, but cannot much hurt our foes in war."

The small sword came into general use towards the close of the seventeenth century, and it had almost entirely superseded the two-edged lengthier and heavier rapier when the eighteenth century was in its second quarter.

The duelling sword and rapier are often confounded with each other, but the former was used mostly for thrusting only, while the latter was more adapted for a cutting stroke, although still a weapon more for thrusting than cutting. The elaborate Spanish hilts were followed in the seventeenth

century by the shell guard for duelling, and a hilt much lighter than before for military purposes.

The swords made at Toledo have a reputation which still endures; and the well-known name of Ferrara is derived from a Venetian family of the sixteenth century. The Ferrara blades are broad and of splendid temper, but the name was used by many smiths as a sort of “standard” mark. Andrea Ferrara or Ferara was established in business, in partnership with his brother Giovan Donato, at the town of Belluno, in the Venetian province of Friuli, in 1585. The *Trattato Militare*, published at Venice in 1583, mentions the brothers as the celebrated sword-makers of that day. Ferara blades, inscribed with the name, were, however, in existence much earlier than this; but whether all or part of these were made in Spain, where there are several towns of the name, is far from clear. The question, then, as to which city or country gave its name to the great master is not yet absolutely determined.

Andrea was probably born between 1550 and 1560,³⁸ and his master, Giovanni Battista, some of whose blades were marked “Zandona,” was called the “Barcelonian,” which circumstance might suggest the possibility that the brothers were emigrants from Spain; but it is much more probable that they came of an Italian family which had been domiciled in Italy for generations, as there are blades of a considerably earlier date than the “Andrea” span, bearing the names of Cosmo and Piero Ferara, both of which Christian names are undoubtedly Italian. A tradition exists in Scotland that Andrea Ferara, or Ferrara, came there as a fugitive from justice, and made swords there in great numbers, but there is no evidence whatever of this being the case. There are swords bearing the brand “Andrewea Ferrara” with a St. Andrew’s Cross, which clearly discloses their Scottish origin, or at all events is suggestive of their having been made in or for Scotland. Indeed, almost all Scottish blades bearing the name of Ferara, with variations, are of seventeenth century make, some even later. We know that it was a common practice of many of the German smiths during the “renaissance” to inscribe their blades with the names of Italian makers; and while Ferara blades are to be met with all over Europe, strangely enough very few are to be found in Italy. The practice of using the marks of celebrated sword-smiths by others less renowned cannot be looked upon as a deliberate forgery, unless perhaps in the earlier instances, when marks were taken possession of by one town or country from another,

proceeding, doubtless, from the importation of craftsmen; but even in such cases it was not uncommon for the maker to give his own name or mark in conjunction with such as those of Ferara, the running wolf, etc. Marks like the bishop's head, moor's head, Sahagun, Ayala, Piccinino,³⁹ were often used by others, though probably rarely in the sense of piracy. This is shown by the annexation of the Wolf of Passau by the Solingen makers, and that of Ferara by the Scotch. Mere legends, like the domicilisation of Andrea Ferara in Scotland, or that of Jakob Topf in London, require some more direct evidence for serious attention, which is certainly not forthcoming in these cases, though the probability is greater in the case of the latter than in the former. Excellent rapier blades were also made at Seville, Valladolid, and Solingen. The Solingen blades are stouter and more suitable for military purposes than those forged in Spain; they bore the stamp of the running wolf, but the mark came originally from Passau. A Passau sword of an early date, with the wolf-mark inscribed on the blade, is in the museum at Dresden. The general aspect would indicate a date in the second half of the fourteenth century. The wolf-mark of the Passau sword-smiths was borrowed from the city arms, which consist of "Or, a wolf-figure, statant gardant." Later, and especially in the sixteenth century, this mark was adopted in other places, and especially by Solingen smiths. These blades were known as "foxes" in England, doubtless from the "wolf" inscription, which might well be taken as a representation of the fox. The term constantly crops up in Elizabethan literature. This mark, like that of "Ferrara," was freely used by sword-makers up to the end of the last century; indeed, this was the case near Newcastle, where swords forged on the banks of the Derwent, in the county of Durham, bore the mark. The smiths came originally from Bavaria, and brought the brand with them. There are still descendants of these people living in the neighbourhood; and there is a specimen of their handiwork in the Black Gate Museum, Newcastle-upon-Tyne.

The spadron is adapted for cutting and thrusting, but is lighter than the sabre.

Spanish swords enjoyed a very early celebrity, the Romans having adopted them after the Carthaginian War, for they were never able to forge weapons of equal temper. The best early Spanish swords were made at Bilbilis on the Jalon, and the poet Martial writes of the excellence of the

waters of that river for tempering them; indeed, it was universally believed that the fine temper depended on the virtues of a particular river. Probably the steel produced from fine Spanish ores, so free from deleterious ingredients like sulphur and phosphorus, had most to do with the super-excellence of the blades. These weapons are mentioned temp. Julius Cæsar, when the poet Gracio Falisco adds his testimony to their admirability.



Fig. 46.—Schiavona, in the Author's Collection.

The schiavona is a Venetian sword of the sixteenth and seventeenth centuries, with a flattened elliptical form of basket-hilt forming a complete protection to the hand, which can still move freely. In this hilt the first finger was always passed over the quillon, and the superadded guard to protect it gives the hilt an elongated form. It derives its name from the “Schiavoni,” the Doge’s guards. The illustration of this weapon here given ([Fig. 46](#)) is of a sword in the author’s collection.

Scottish broadswords with practically this hilt, although there are intermediate stages, are often erroneously called “claymores,” while, as a matter of fact, the Scottish weapon so called was a long two-handed sword, with quillons usually tending diagonally upwards, that is towards the blade; and, indeed, it is considered questionable by some authorities whether any basket-hilted sword whatever was in general use in Scotland long before the eighteenth century began. Mr. Parker Brewis, in an able paper⁴⁰ on “Four Basket-hilted Swords in the collection at the Castle, Newcastle-upon-Tyne,” writes as follows, viz.:—“This type of sword is commonly known as ‘Claymore,’ which is the English phonetic of two Celtic words, meaning ‘Great Sword.’ It was originally applied to the great two-handed swords of Scotland, but when the true claymore was gradually superseded by the basket-hilted weapon, the old name, as conveying the idea of a Highland sword, was retained, owing to long habit, notwithstanding that it was inappropriate.” The “mortuary” hilt, so named from a number of swords with this basket-hilt having been made in memory of King Charles I., was the broadsword of the Commonwealth, and the Scottish form is obviously an amalgamation of the schiavona with the mortuary. The basket-hilted sword was certainly common in England in the second quarter of the seventeenth century, and there is no reason why it should not have crossed the border long before the eighteenth century, and that it had done so is certain from the fact that mortuary hilts were largely made in the island of Islay. The ordinary Scottish basket-hilted broadsword blades bearing the name Andrea Ferara, with numerous variations, were certainly not made by the great master of Belluno, but most of these were forged in the seventeenth century. Of course, it is often the case that blade and hilt are not contemporaneous, and old Ferara and even claymore blades were frequently adapted to the newer fashion, and these cases give rise to some difficulty.

The colichemarde is a late seventeenth century fencing sword, with a blade very broad at the “fort,” and exceedingly narrow at the “foible”—the change from one to the other is very sudden. This sword was only in use for a brief period. Some of the swords of the seventeenth century were very long. The cutlass or hanger of this period is usually without quillons, but has a counter-guard.

After the commencement of the seventeenth century, it becomes more difficult to fix approximate dates for swords with any precision, and many weapons are freely attributed to that century which really belong to the eighteenth. It is the blade that bears the stamp, and many blades were transferred to other hilts; besides, the armourer was often permitted to give considerable rein to his fancy, and not unfrequently reverted to older forms. As in armour, it is an uncommon advantage to meet with weapons with the date inscribed, although, of course, many armourers’ marks serve this purpose, when they can still be deciphered; still, their presence is rarely conclusive without general characteristics being also taken into account.

The complete transformation of the sword may be said to have been effected during the eighteenth century, since which time it cannot be said to have advanced either in balance or general efficiency. Very little is known as to the early history of sword-making in England, but Sheffield was a very early centre for the industry. It was not until towards the end of the last century that English-made swords established their reputation as the best in Europe, when in an order for the East India Company, 2,650 English swords were tested in the machine already referred to in these pages, and only four failed to bear the test; while out of 1,428 German swords as many as twenty-eight were rejected.

PART XVII.

THE DAGGER.

THE dagger is a short sword in great variety of form; it is a weapon for thrusting only. We meet with it in the ages of “stone” and “bronze,” and it was in use among almost all the great nations of antiquity.

The scramasax, a short two-handed sword or dagger, is an ancient Germanic weapon of varying length. In form it resembles a single-edged cutlass. There are examples in some of the German museums; one was found in a barrow near Andernach.

Mr. John Hewitt, in his work on *Ancient Armour and Weapons*, refers to a dagger preserved in Durham Cathedral, which was supposed to have belonged to Bishop Anthony Bek in 1283, bearing the inscription “Anton Eps Dunholm.” This is doubtless the dagger now at Auckland, which was exhibited to the members of the Newcastle-upon-Tyne Society at the Castle on the 28th December, 1892. The blade, which seems originally to have been longer, is now eighteen inches in length; while the haft measures five inches. The quillons do not project far beyond the blade, and curl slightly upwards at one extremity, and downwards on the other. The authenticity of this weapon is more than doubtful, and Baron de Cosson even suspects who the forger was, and when it first appeared at Auckland. The forgery is one of the clumsiest, for it is so obvious what the hilt originally was, viz., portions of a Scotch basket-hilt.

There are representations of figures armed with the dagger in the thirteenth century, when the quillons turn up towards the blade, as is the case with most of the swords of the period. It does not appear in effigies before late in the first quarter of the fourteenth century. An anelace dagger

may be seen on the effigy of William Wenemaer, died 1325; and another on that of the second Baron Berkeley, figured in Gough, vol. i., p. 44.

The anelace dagger, which is of Italian origin, is about sixteen inches long, and derives its name from the ring which was originally attached to it, and which was connected by a light chain with a mamillière. A somewhat similar weapon was used as a dart, and often attached to the end of a staff, and then called “*langue-de-bœuf*.” An actual specimen, with the ring, was found among the *débris* at Tannenberg. This dagger is double-edged, broad in the blade, which narrows towards the point. Chaucer mentions the weapon. The larger anelace is mentioned in the notes on swords, and an illustration is given in [Fig. 43](#); the only distinction, if there be one, is that of length.

The form of the dagger is often that of the sword in miniature, and the guards, as is the case in the larger weapon, are naturally an excellent guide as to date. The guard of two knobs and the wheel-guard appear in the fourteenth century.

The poniard, with its numerous family, is shorter than the ordinary dagger.

The misericorde, an example of which is recorded as early as 1221, and which appears on the De Bohun effigy, was worn on the right side, and hooked to one of the taces. Like the stiletto, it is a short, narrow poniard; the former was used, as its name implies, by men-at-arms to give the *coup de grace* to fallen adversaries; and it was always present in jousts à outrance. The guard of the fifteenth century was usually two round knobs, but the weapon is often without any guard, and the narrow triangular blade was most effective in piercing through interstices in armour. The thumb-ring, which is above the quillons, is often met with in the fifteenth century.

The cultellus, or coutelas, as its name implies, served the purpose of both a knife and a dagger. It was the progenitor of the cutlass—coutel-hache, coutel-axe, curtle-axe, coutelace, and cutlass.



Fig. 43.—Anelace at Berlin.

The baselard, or baudelaire, is an ornamental dagger of the fifteenth century, worn by civilians in front of their persons. An example occurs on the brass of a civilian at King's Sombourne, in Hampshire (died 1380). Priests were expressly forbidden to wear the weapon.

The main gauche is an early sixteenth century weapon, and was used in conjunction with the rapier. This is the dagger that was supplied to the "schoppen" or "scabini" for the execution of the decrees of the Holy Vehme, or Vehmegericht, the secret tribunal of the middle ages prevailing in Swabia, Franconia, etc. The blade of this dagger was sometimes perforated with indentations for catching opponents' swords. Another variety was provided with a spring, which when pressed set free two extra blades, one on each side of the main blade.

The Highland dirk is in great variety of form, and usually without any guard.

It was not uncommon for dagger and sheaths to be fitted with a small knife like some of the Indian swords. During Elizabeth's reign it was common for a combatant to parry with a dagger in the left hand, when fencing with the rapier. Some representations of daggers are given in [Fig. 42](#).

PART XVIII.

THE LONGBOW.

THE longbow is a weapon of great antiquity; an example may be seen on a bas-relief in the Louvre, dated about 700 B.C. It was used by the Egyptians, Chaldæans, and Greeks; and was probably introduced into Britain by the Romans. The bow of Pandarus is related to have been made of ibex-horn, and strung with sinews. The following lines from the *Iliad* are very graphic, and descriptive of this bow and its manipulation:—

“Straight he uncased his polished bow, his spoil
Won from a mountain ibex, which himself,
In ambush lurking, through the breast had shot
True to his aim, as from behind a crag
He came in sight, prone on the rock he fell;
With horns of sixteen palms his head was crowned;
These deftly wrought, a skilful workman’s hand
Had polished smooth, and tipped the ends with gold.
He bent, and resting on the ground his bow,
Strung it anew.
His quiver then withdrawing from its case,
With care a shaft he chose ne’er shot before,
Well-feathered messenger of pangs and death.
The stinging arrow fitted to the string,
At once the sinew and the notch he drew:
The sinew to his breast, and to the bow
The iron head: then, when the mighty bow
Was to a circle strained, sharp rang the horn
And loud the sinew twanged, as toward the crowd
With deadly speed the eager arrow sprang.”

—*Iliad*, iv. 119.

An antique Greek drawing of the time of Theseus has been already referred to, whereon is an Amazon with a drawn bow, the arrow-head being barbed. Agathias, writing in the seventh century, says that the Franks did not use this weapon in war, but it is mentioned in the capitularies of Charlemagne, and there is evidence that it was not uncommon among both Anglo-Saxons and Danes. It was pre-eminently an English weapon of war, though used also in the chase in that and other countries, and was remarkable for range and sureness of aim, as well as for penetrative force. The Germanic nations applied it mainly in the chase, the Saxons especially using a short bow. An illustration occurs in a MS. in the Cotton Library.⁴¹ The English archer became justly famous under the Norman kings, and it was first under them that the bow assumed great importance as a weapon of war. Bowmen in England at this time wore a leathern jacket, which was

afterwards adopted by the French and called a “jacque d’Anglois.” On the Bayeux tapestry only one single bowman appears among the Saxon array, while there are several shown among the Norman ranks; these bows are short and thick, and arrows with barbed tips. Harold’s eye was pierced by an arrow, and but for this the Normans would hardly have won the battle. Richard I. was himself an adept in the use of the longbow, and it was the leading weapon of our armies at Crécy and Agincourt; and indeed continued to be so well into the “renaissance.” It will be remembered that at Flodden the Scottish king was killed by an arrow, and this battle may be said to have been the latest won mainly by the longbow.

The proper length of the English longbow was about the archer’s height, say between five feet six inches to six feet, with a bend of nine inches; and those made from the bough of a yew were preferred; but as yew trees were scarce, bowyers were enjoined by Act of Parliament to make four bows of “witch-hazel,” ash or elm, to one of yew; and no persons under seventeen years of age, with certain exceptions, were permitted to shoot with a yew bow, under a penalty of six shillings and eightpence. This Act of Parliament was repealed in Elizabeth’s reign. The string was either of silk or hemp, twisted or plaited, but always round where the notch of the arrow was placed. The shaft was drawn by two or sometimes three fingers to the head, and always towards the ear, when shot at short marks; but towards the breast when used at long ranges. The archer kept both eyes open, and looked only at the object aimed at, holding his weapon perpendicularly. Part of the light cavalry in the thirteenth century consisted of mounted archers. During the reign of Henry VIII. hand-guns had greatly superseded the use of the longbow, but the king himself was a skilful archer. The archer carried his sheaf of arrows, consisting of twenty-four, in his belt; the length was a clothyard shaft, feathered or plain at the base, and tipped usually with a sharp, but sometimes barbed head. These heads were of iron, pointed with steel. The archer wore a leathern wrist-guard, called a bracer, to avoid hurt by the recoil of the string. The arrow with feathers from a goose’s wing was the “broad arrow,” first used as a regal badge by King Richard I. The plain pile, without feathers, was considered to penetrate better. Henry V. enacted that the Sheriffs of Counties were to take six wing feathers from every goose for feathering arrows. Arrows of ash were preferred. They were about thirty-two inches long, and usually tipped with a sharp unbarbed head.

Any ordinary English archer would rarely miss an object the size of a man at 250 yards; and he could discharge his weapon twelve times a minute. The extreme range of a bow was “from sixteen to twenty score yards;” in fact, a “bow-shot” seems to have been used to express a distance of 400 yards, and the minimum range for archery contests was usually 220 yards.

It was the first duty of the archers in battle to send clouds of arrows against charges of cavalry, so as to disorganise their formation by killing or wounding as many of the horses of the opposing host as possible, thus causing confusion in the enemy’s ranks by rendering many riders *hors de combat*, and though rarely able to pierce a harness of proof, the arrows often found an interstice in the armour. Since the thirteenth century the armies of England maintained large numbers of mounted archers in their ranks, the complement of bowmen to a corps of fifteen hundred fully equipped lancers being from three to five thousand, while each lancer’s equipment was five or six mounted soldiers, at least two of whom were archers.

German and Italian bows rarely exceeded five feet in length. The shape of their arrow tips varied exceedingly. An ordinance of Henry I. provides that when archers were practising and any one had the misfortune to be killed or wounded by accident, it was merely to be regarded as a misadventure.

The form of the longbow of the fourteenth century was thick in the middle, narrowing towards the ends, and it was sometimes coated with paint.

The price of longbows was fixed by statute in the reign of Edward IV. at a maximum price of three shillings and fourpence each; and in order to increase the number available, every merchant vessel carrying goods to London was compelled to bring a certain number of bows in proportion to the weight of the cargo. A statute of Philip and Mary ordains that all temporal persons having estates of a thousand a year and upwards are required to furnish to the State thirty longbows and thirty sheaves of arrows.

Archers carried one or two pointed stakes as part of their equipment, for planting before them in the ground to resist cavalry; also a lead-headed

mallet, to drive them in, which was also used for despatching the enemy's wounded.

Specimens of the English longbow are of the greatest rarity. The unfortunate loss of an English war vessel, the *Mary Rose*, which sank off Spithead during the reign of Henry VIII., in 1545, furnished us with some actual specimens of the period. The whereabouts of the wreck was known, and in 1843 divers recovered several bows, a couple of which are preserved in the Tower of London; they are six feet four and a half inches long, and are made of yew.

There was a Northumberland English longbow still to the fore early in the present century, and the late Mr. Matthew Culley of Akeld, in a letter to the Newcastle Society, dated Nov. 26, 1814, wrote concerning it: "This bow had long been used by the hereditary bowmen of Wark Castle. It is described as having been formed of various coloured wood inlaid together, and of great length and strength. From the joining of different sorts of wood very valuable properties are derived, which are well known to mechanics, and more especially to ship-builders. This weapon, so dreadful in the hands of its ancient possessors, being no longer in request, was consigned to the children as a plaything." There is an English longbow at Dover Castle.

The longbow continued in use long after the introduction of firearms, but was practically superseded by the harquebus in the sixteenth century. Though used at the siege of Rochelle in 1627, its reputation had sunk so low in the reign of Charles I. that that king granted two commissions under the great seal for enforcing its use, and another to prohibit the enclosure of fields near London, which would have had the effect of interfering with the practice of archery. A curious fact in connection with the longbow is that Benjamin Franklin proposed in 1776 to equip the colonial forces with the weapon.

PART XIX.

THE CROSSBOW.

THE Latin equivalent is *arcus balistarius* or *balista manualis*. The weapon does not appear on the Bayeux tapestry, but the Princess Anna Comnena, who calls it “tzangara,” mentions it as forming part of the armament of the Crusaders, late in the eleventh century; and that it was in use by English and French soldiers in the twelfth century is shown by a bull of Pope Innocent II. in 1139, which fulminates against its barbarity, and only sanctions its use in warfare with the infidel, meaning thereby all nations still unconverted to Christianity. Such prohibitions were, however, soon brushed aside, like others of a similar character both before and since. Guillaume Guiart, writing towards the end of the thirteenth century in the *Branche des Royaumes Leguages*, mentions the weapon as being in use at the battle of “Haringues” in 1297. The first form was a simple hand crossbow, which consisted of a steel bow let into a stock which was strung for use by the action of the left foot and right hand, and discharged by a trigger, which probably gave rise to the lock of the hand-gun. During the second half of the thirteenth century various mechanical contrivances were adopted, which, while materially increasing the projective power, rendered the weapon much more unwieldy. The crossbow was in constant use during the fourteenth century, when the Genoese made it a speciality, and the services of these mercenaries were in great request in the wars of the period; it was, however, never a favourite weapon in England. At the battle of Crécy the English army used the longbow, while the French king had a body of six thousand Genoese crossbowmen in his pay, but these were unavailable by reason of the rain. The English archer could shoot twelve arrows while the crossbowman discharged his three quarrels, for it took so long to wind up the “moulinet”; the crossbow had, however, the advantage of a lower trajectory; moreover, the longbow was much lighter and more portable, besides being more easily

preserved from the action of damp, than its crossbow confrère. It does not seem that the extreme range of the crossbow has been accurately determined, but it certainly did not exceed three hundred yards. Part of the light cavalry of the thirteenth and fourteenth centuries consisted of mounted crossbowmen.

The introduction of the pavise, a large shield kept propped up before the archer, was a great protection against missiles; and a miniature from Froissart, in the Bibl. Nat. de Paris, shows crossbowmen shielded in this manner. According to a manuscript in the British Museum, the Genoese crossbowman wore a jacket with long sleeves, an iron helm, brassards, and greaves.

The steel used in the construction of crossbows was of the strongest and most elastic kind. An enactment in the reign of Henry VII. forbade the use of the crossbow under severe penalties, and in the sixteenth century crossbows were mostly used for the defence of fortresses, and on warships.

The windlass crossbow, called *à tour* by the French, was largely used at Agincourt, and the form of that time continued practically the same for centuries; indeed, up to early in the seventeenth, bows on this model were made at Malines, in Belgium, by a “confrérie de tir.” The author has one of these bows in his possession, and it is, he believes, the exact counterpart of the Agincourt bow.

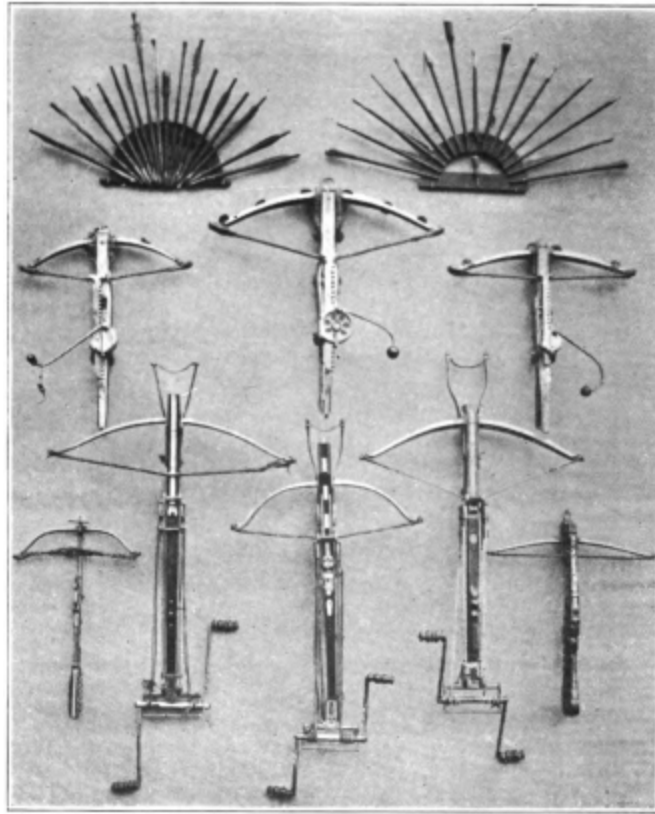


Fig. 47.—Crossbows and Quarrels.

The projectiles are usually called quarrels, and are in great variety of form, but shorter and thicker than arrows for the longbow; several specimens were found at Tannenberg, dismantled in 1399, and the complement for a crossbowman was fifty. Quarrels for the arbelest were called “*muschettæ*,” hence the word musket; but there is some doubt whether it was not the missiles of the “scorpion” that were termed thus.

A picture in the National Gallery shows how the common stirrup crossbow was bent *ad unum pedem*: the bowman places his foot in the stirrup, a cord is then fixed to the butt of the stock, the other end being attached to the waistbelt; the cord runs on a pulley, and the bow is bent by raising the body. The crossbowman wore a “brigandine” or stuff tunic, lined with strips of steel, besides his “half plates.” Illustrations of most of the varieties of the crossbow are given in [Fig. 47](#).

THE GOATSFOOT CROSSBOW.

This bow is bent by a lever of two branches, called the goatsfoot, one of which is provided with forks, which grasp the string, while the other pulls it back. It was used by horsemen.

THE LATCH CROSSBOW (ARBELETE À CRIC).

This kind, which is very heavy, was used specially in the defence of fortified places. It probably got its name from the trigger, which is formed like a latch, and is manipulated by a cog-wheel, and a notched bar called a cric. This bar has hooks at the top which grasp the string, and a handle turned by the hand of the archer winds up the “moulinet” or winch, drawing the string which bends the bow, and the tackle is slipped on to the stock from the bottom, which passes through a thick hemp or iron loop. This variety was much used by the Germans, and is probably the “latch,” although it is far from certain that the term did not apply to the “cranequin.” There are also barrel crossbows, and some with a pistol in combination.

THE WINDLASS CROSSBOW (ARBALETE À CRANEQUIN).

This bow is furnished with double cordage and a set of pulleys near the bottom of the stock, and another set placed just below the bowstring; strong cords run along the pulleys, and these are drawn taut by a small detachable windlass, which is adjustable to the bottom end of the stock, while hooks connected with the top pulleys grasp the bowstring. As soon as the bow has been bent by the action of the windlass, the tackle is removed. The top end of the stock is furnished with an iron stirrup, through which the archer thrusts his foot in order to obtain the necessary purchase for bending the bow. This type of bow was used at Agincourt, and it was greatly depended on in the defence of beleaguered places. It was also called “Arbalete à tour,” because the part to be fixed to the stock was often embattled like a tower, and the windlass was named “la clef” or “cranequin.” This bow has a much longer catch than the “goatsfoot.”

THE PRODD.

This bow is light, and was used mostly in the chase. It shot principally pebbles, but also bullets. The French called it “arbalete à jalet.” A small prodd in the author’s possession was used for shooting game, and would seem to date from late in the sixteenth or early in the seventeenth century. It takes its name from two upright pins of iron, across the top of which a thread is drawn with a bead in the centre, which required to be brought into line with the notch observable on the top of the adjustable arch placed above the trigger for sighting purposes. The cord of this bow is double, and is kept taut by beads placed there for the purpose of leaving a cavity in which to place the pebble or bullet for discharge. A vast amount of artistic skill was often applied in the decoration of crossbows, which has been specially alluded to in the opening remarks. The prodd was often used by women.

PART XX.

MACHINES FOR HURLING OR SHOOTING MISSILES, AND THE WARWOLF.

THE missile-casting engines of the eleventh and twelfth centuries are as follow, viz.:—

THE SCORPION,

named from its shape, is a machine about which there is but little reliable information; but what there is indicates it to have been a huge crossbow, the bowstring being bent on the cog principle.

THE CATAPULTA AND BALLISTA.

Their prototype was the “tormentum” of the Romans.^{[42](#)} The two machines are often confounded with each other. The catapulta was used for throwing heavy darts, while the ancient ballista threw stones only, but the mediæval variety was often arranged for both quarrels and rocks. Some ballistæ threw stones three hundred pounds in weight. The difference in the construction of these military engines from those made on the ordinary principle of the bow consisted in the addition of a mechanical force. There were also small catapultæ used like hand-guns. Remains of ballistæ were found among the *débris* of the castle of Russikon in Switzerland, which was burnt down in the thirteenth century.

Vitruvius and other writers give a full account of these machines, but the copyist, as has usually happened in all ages, made such mistakes as to render the descriptions well-nigh unintelligible, so there is still a good deal of uncertainty concerning them. In chronicles of the twelfth century crossbows are always termed “ballistæ.”

The principle applied in the ballista was that of the bow, but instead of the usual arc, with its simple directive force, a kind of double action was achieved by providing the machine with a strong rectangular frame of wood, constructed in three compartments, firmly fixed on to a stand, which was made of strong and hard wood, consisting of two uprights connected horizontally by a double crossbeam. Instead then of applying the entire arc, as in the crossbow or scorpion, and assuming such arc or bow to have been divided into four quarters, only the two end quarters were used; and in each of the outer compartments of the frame two very strong strands of twisted sinews were fixed, and through these the inner ends of the two pieces were firmly held, the bending of which gave much more elasticity and propulsive force, thus producing a recoil strong and forcible enough to project heavy missiles to a distance of as far as 250 yards. The engine was fitted with an iron groove. In sighting the machine for the discharge of a heavy stone, pieces of clay were used to keep the projectile at the necessary angle before discharge. There are four stone shot at Woolwich 15, 16, and 18 inches in diameter, supposed to be catapult balls.

The above explanation will make apparent how very difficult it is to describe even the simplest machine in mere language; besides, you have the difficulties of translation to contend with. [Fig. 48](#), from a MS. in the National Library of Paris, No. 17,339, explains the principle at a glance.

Besides these machines, there are others constructed on the sling principle, like the mangona and mangonet, from which the word “gun,” originally “gon,” is probably derived. There are two stone balls at the Rotunda, Woolwich, which are said to have been thrown from a mangonel used in the defence of Kenilworth Castle in 1266. The onager or onagre is thought by some writers to be merely the old French name for the catapult, while Grose gives a figure representing the onagre as a machine for slinging rocks. The trebuchet is a machine constructed on this principle (the swing and weighted lever), both for hurling and swinging a heavy stone against a rampart, breaching or breaking it down; it also threw barrels of Greek fire.

Matthew Paris mentions this machine as peculiarly effective. This engine seems to be the mangonel under another name. The tolleno was used in siege operations to lift soldiers up on to a wall. During the centuries immediately preceding the introduction of firearms there were many machines invented for the hurling of darts and stones, used both on land and sea—the robinet, the espringal, ribandequin, a large crossbow, etc. The missile-casting engines used on ships of war were mounted on raised platforms. The late Emperor Napoleon III. had a trebuchet constructed after an ancient inscription, and this machine is now at Vincennes.

Another called the warwolf is mentioned by several of the early writers, but they all differ considerably concerning it. Procopius describes it as a machine of the harrow family, for the defence of a gate; it seems to have been rather similar to the herse, used as a second defence after the portcullis had been forced.

The falarica was for throwing fiery darts. It was used by the Saguntines, when the shaft was wrapped round with tow steeped in oil and smeared with sulphur and resin. This was ignited and the missile launched against the “pluteus,” a machine which was the prototype of the mediæval “sow” or “cat.”

Many of these machines continued in use long after the introduction of firearms. A common feature in most ancient MSS. is that fancy names are freely applied to most of them, thus giving rise to much difficulty in their identification.

PART XXI.

MACHINES FOR ATTACKING BELEAGUERED PLACES.

THE castle of the middle ages up to the invention of the bombard was practically that of the ancient “castellum,” as far as defence was concerned, with outworks frequently of wood; and the means of attack lay in escalade, sapping and mining, the use of the battering-ram, or by a blockade.

We now touch upon the machines used in attacks on fortified places, most of which have their prototypes during ancient times in the testudo, pluteus, tenebra, etc.

The battering-ram, the tenebra of the Romans, used both on land and sea, was a heavy oak beam tapering towards the head, which was shod with iron with a point at the extremity. It was exactly the same in the middle ages as in Roman times. There is a Roman specimen in the Germanische Museum at Nuremberg, which is about a foot in diameter at the base, and about eleven feet in length. It is still shod with iron.

Sometimes in the middle ages this machine was made available for the united energies of many men, by means of beams joined together and suspended in a sling or massive trestle, whereby its force could be enormously increased. It was sometimes impelled on rollers or wheels and rapidly run forward to batter a wall. An engine similar to this is figured at Nineveh. The besieged did their best to deaden its effect by means of woolsacks or bags of hair let down from the parapet.

The “sow” or “cat,” the vinea of the Romans, is a shed on wheels, covered with raw hides, used as a cover for preparing the way for the use of movable towers and other engines. This machine is the ancient “pluteus.”

The testudo (*testa*, a shell), the more modern “tortoise,” was also a movable shed like the cat, but it contained a battering-ram for attacking a rampart.

The berefreid, beffroi or belfrey, is a movable tower used for scaling walls. It was constructed in several storeys, with intercommunications by means of ladders or staircases, and high enough to overtop the parapet of the fortress assailed; provided with a drawbridge for an assault in force, and was often rolled on wheels to the point of attack. A machine of this kind, built by order of Simon de Montfort, was used at the siege of Toulouse, and, according to the ballad of the “Albigéois,” was adapted to contain five hundred men. The last of these engines was constructed as late as the reign of Charles I., and it was taken by the parliamentary forces.

Mantlets stuck in the ground provided shelter for the archers, and other combatants, beneath the walls, against “Greek fire,” showers of rocks, and other missiles, hurled from the battlements by the defenders.

“Greek fire” was used both in attack and defence. This was a Greek invention, as its name implies, and the secret of its composition was most jealously guarded. It was known in the east of Europe as early as 673, and was for a long time regarded as supernatural by the northern nations in the “dark ages,” but the secret was discovered by the Crusaders—in fact, Philip of France brought some of it from Acre, and used it for setting fire to the English ships at the siege of Dieppe. Jesuit Petavius states on the authority of Nicetas, Theophanes, and Cedrenus, that it was invented about the year 660.⁴³ Anna Comnena gives the ingredients as bitumen, sulphur, and naphtha; and states that the Emperor Alexius discharged it at the enemy from his galleys. Others add pitch and gum to these ingredients. It was used in many ways, but its most fatal and irresistible form of application was in setting fire to fortified towns, where the wooden houses of mediæval times afforded it free scope, when inadequately guarded against by a sufficient application of raw hides to the roofs, and other means of protection. A mixture of vinegar, sand, and urine was used to put out its flames. Barrels of “Greek fire” were fired into these towns from the ancient “trebuchet,” and also by a kind of mortar; it was also freely used by the besieged for the destruction of movable towers and engines of war. Froissart, in his account

of the attack by the Black Prince on the castle of Romorantin on the Sandre, mentions an engine he calls an “aqueraux” to fire “le feu gregois.”



PART XXII.

THE SLING AND FUSTIBAL.

THESE rude missile-casting weapons, with the longbow, were greatly used by the peasantry and yeomanry of the early “middle ages.” The first-named is too familiar to need much description, and its very ancient character is universally known. The Spaniards employed it with great effect at the battle of Navarete, where, Froissart says, “they broke many helmets and skullcaps, so that they wounded and unhorsed many of their opponents.” At the Rotunda, Woolwich, are twelve sling stones of two sizes, viz., 2.35 and 1.7 inches in diameter. These stones came from Rhodes—they are pebbles covered with lead. A single slinger appears on the margin of the Bayeux tapestry; the weapon is being used by a peasant aiming at a bird.

The fustibal, or staff-sling, consists of a long pole, four feet in length, with a sling in the middle. An example is recorded in a MS., which is attributed to Matthew Paris, in Benet College Library, Cambridge, C. 5, xvi. It was wielded by both hands to cast large stones against an enemy, and was in use as late as the sixteenth century for hurling grenades. The ordinary sling was still to the fore in the fourteenth century—indeed, it was sometimes used in warfare even in the sixteenth; Grose gives an instance at the siege of Sancerre in 1572. The author saw it in Egypt, used by boys for frightening birds from the bean fields.

PART XXIII.

STAFF AND CLUB WEAPONS.

THE JAVELIN, SPEAR, AND LANCE.

THIS family of weapons is somewhat extensive, and of very great antiquity. The earliest forms were often used as missiles, and have been briefly alluded to in the introductory remarks. We have the authority of Procopius that the Frankish darts had barbed iron heads, and were used for both cutting and thrusting. Agathias refers to double axes and argones (spears). The Anglo-Saxon spear was a narrow, long-bladed weapon, while their javelin differed from that of the Normans in being shorter. The Bayeux tapestry shows Anglo-Saxons with bundles of barbed javelins in their hands. The Norman cavalry was armed with long lances, as well as swords, at the battle of Hastings.

Up to the end of the eleventh century, the lance continued of a comparatively uniform thickness about twelve feet in length, and the knight's pennon waved from it, as shown on the Bayeux tapestry, while the head was lozenge- or leaf-shape, and sometimes barbed—all these forms appear on the tapestry. The Daubernoun brass (1277) furnishes a good example of a thirteenth century lance; it is five feet long, and bears an emblazoned pennon.

The tilting-lance was from twelve to fifteen feet in its extreme length, first of uniform girth, but later thicker at the base, gradually tapering towards the point, and the swell at the grip does not occur before the fourteenth century. Ash was preferred for the shaft. The early tournament lance was required to be blunted, but owing to the many evasions of this

rule an ordinance of the fourteenth century enjoined that the head be furnished with a tip in the form of a coronal.

The length of the lance was often much reduced in the fourteenth century, and was then sometimes used as a dart, but this was considered so dangerous to the king's peace that its use in this manner was forbidden by statute. The tilting-lance of late in the fourteenth and during the fifteenth century was often made hollow, so that it was more apt to shiver at the moment of impact, and the shaft was grooved; it differs at this time in form and bulk for the different courses. Those that were used with a view to "unhorsing" were stronger, heavier, and thicker in the stem than those made with the object of being splintered; the former were provided with a pointed head, while the latter often bore a coronal. The lance used for running at the ring was shorter and much lighter than the two first-named, and was tipped with a cone; there are specimens of most of these varieties at the Tower. Froissart mentions a spear with a hook or spur at the base of the blade, used for the purpose of dragging an adversary from his saddle, but this feature might refer to one of the other weapons otherwise enumerated. A good example of the lance of the second half of the fifteenth century may be seen on "The Tapestry of Berne."

It was common for knights fighting on foot, or those dismounted by any accident, to cut down the lance to a length of five feet, for use as a spear; this was done at the battle of Poitiers.

The vamplate, a steel plate for keeping the lance in position, began as a small rondelle, but attained larger dimensions in the fourteenth century, becoming very large in the fifteenth and sixteenth centuries; the German tilting vamplate covered the shoulder and half of the arm.

The importance of the lance in battle became greatly reduced in the sixteenth century, and even earlier.

THE MACE AND MARTEL-DE-FER.

The mace is a very ancient weapon in its simple form, its use and shape having been evidently suggested by the club, and it was probably a sceptre before it became a fighting club of metal.

The type of the Bayeux tapestry, which was only used by the Saxons, is elementary and club-like, and the shape did not alter much before the beginning of the fifteenth century, when we have round, oval, cog-wheel, and dentated forms; it was sometimes provided with a short spear, welded into the top, but this was rather a French than an English form. The mace and battle-axe were the great weapons of the Plantagenets. The mace (temp. Edward I.) assumed the form of a slightly projecting cog-wheel, which became somewhat more pronounced in the next reign, as may be seen on one of the sleeping figures in Lincoln Cathedral; and the weapon was sometimes made of lead. The shape did not alter much before the beginning of the fifteenth century, when we have the round, oval, cog-wheel, and dentated forms much more pronounced than under Edward I.

Asiatic specimens are generally round in the knob, and are much lighter than European weapons. The mace hung at the saddle-bow, being passed through a socket which was attached to the saddle, and the weapon was used in the lists as well as in battle.

It survived as the weapon of the sergeant-at-arms, and fell into disuse as a weapon of war in the reign of Queen Elizabeth; after which it became a processional emblem, and was made of silver or copper-gilt, and ornamented with a crown, globe, and cross.

The small variety of mace was termed the "mazuelle." The baston (German streitkolben) is a heavy mace of hard wood, bluntly pointed, polygonal in form, thickening towards the head, while the pommel is round, and it was used in tournaments.

The martel-de-fer or pole-hammer is of ancient origin. That it was in use in the eighth century is shown by the sobriquet "Charles Martel." It was a popular weapon in the fourteenth and fifteenth centuries for both horse and foot. The Lucerne hammer is only another name for the same weapon; it is both long and short handled, while the head is either a simple war-hammer, or has a small halbard-shaped blade with a plain or dentated hammer at the opposite side, and a longer or shorter spear at the extremity.

THE BATTLE-AXE AND POLE-AXE.

The battle-axe or francisca was a leading weapon of the Franks during the Merovingian period, and it was then often used as a missile. The francisca of Childeric (457–481) was found in his tomb at Tournay, and is now in the Louvre. Procopius refers to the francisca of the sixth century as having a broad blade, sometimes double-edged, with a short haft. Roughly, the battle-axe is short in the handle, while the pole-axe, as its name implies, is long in the shaft. The former is a knightly weapon, while the latter was wielded by footmen only.

The battle-axe was greatly used by the Normans of the twelfth century. It is a weapon of the Bayeux tapestry; indeed, William the Conqueror was armed with it at Hastings—the form of the blade resembled that of an ordinary hatchet, with a curved blade.

The Anglo-Saxons used an axe, narrow-bladed and single-edged, from four to five feet long in the shaft, with great success in the battle. They first darted their javelins, and then attacked the foe with the deadly battle-axe.

The blade assumes later a great variety of forms—cleaver, cusped, etc., and the top was sometimes garnished with a hook or spear.

The pole-axe was a favourite weapon of the fifteenth century, and one of the varieties of the period combines a hatchet, a pike, and a serrated hammer: this weapon is first cousin to the halbard, and often classified as such.

The Jeddart staff is a long-shafted axe with a half-circular blade and a side spike. It is more a halbard than an axe.

The Lochaber axe, used with such telling effect at the battle of Culloden, is long-shafted; the blade and setting closely resemble that of a voulge, with its hook at the head of the staff. This hook, however, is generally absent in the voulge used in the field, and this is sometimes the case with the Jeddart staff also. There are two fine specimens of the Lochaber axe in the collection in the Castle of Newcastle-upon-Tyne.^{[44](#)}

The pole-axe, called the bardiche, is a Russian and Scandinavian weapon with a long, narrow, crescent-formed blade attached to the top of a

pole by a ringed haft, while the lower end of the blade is fastened on to the pole farther down.

The addition of a wheel-lock pistol was a feature of the pole-axe early in the reign of James I. The battle-axe, according to George Silver in his *Paradoxes of Defence*, was at the end of the sixteenth century from five to six feet long.

THE GOEDENDAG.

The late Mr. John Hewitt, in one of his contributions to the *History of Mediæval Weapons and Military Appliances in Europe*, refers to the goedendag as being a foot soldier's weapon of the thirteenth and fourteenth centuries; and he gives a drawing of a foot soldier armed with a long-shafted weapon thickening towards the head, which is surmounted by a short iron spear, firmly and thickly socketed on to the extremity.

This figure, with others, is stated by M. Felix de Vigne, in his *Recherches Historiques sur les Costumes des Gildes, etc.*, published in 1846, to have been reproduced on a drawing by himself from a fresco that had long been plastered over on a wall in an old building in Ghent, since pulled down. The soldier wears a bassinet, with camail of banded-mail overlying the surcoat, and the general aspect of the figure is that of an armed member of one of the Flemish guilds of the beginning of the fourteenth century or thereabouts. M. de Vigne claims to have established the form of the true goedendag in the weapon carried by the soldier.

The late Mr. Hermann Van Duyse in his brochure, *Le Goedendag arme Flamande sa Légende et son Histoire*, refers to the old building in which the fresco was found as by tradition a chapel of the guild of the weavers of Ghent, known as the "Leugemiete." The town records and archives of the Abbaye of St. Bavon both afford confirmatory evidence that a chapel was built very early in the fourteenth century on or near the site where the "Leugemiete" stood.

The figure mentioned by Hewitt formed one of a troop preceded by crossbowmen. The leader wears a visored bassinet, and bears a standard emblazoned with two triangular shields and five crosses argent. His sword

is long and broad, with quillons curving towards the blade. The details of the drawing point clearly to the end of the thirteenth or beginning of the fourteenth century. M. Viollet le Duc, in his *Dictionnaire du Mobilier*, defines the weapon as a variety of the voulge or fauchard, while M. Van Malderghem considers it to be a ploughshare mounted on a staff, or a sort of bill.

In a poem by W. Guiart, written in the French of the period, in the *Branche des Royaumes Leguages*, descriptive of the battle of “Haringues” in 1297, the goedendag mentioned affords many points of resemblance to the staff weapon shown on the De Vigne fresco; indeed, it can be no other.

The goedendag, whatever its form, was used with great effect at the battle of Courtray in 1302, and is called “goudendar” and “godendar” in an account of the battle in the *Grandes Chroniques*. Guiart mentions the goedendag as having been used in this battle in concert with the lance and guisarme, and the weapon is mentioned in French chronicles late in the thirteenth century.

Tradition says that the goedendag is the weapon of the fresco and poem, but garnished with spikes over the thicker portion of the staff towards the head; and there are several such weapons surviving, though this is probably a rather later variety of the weapon than that shown on the fresco, the only difference being the addition of the side spikes. Froissart mentions the weapon as being used at the battle of “Rosebecque” in 1383. Probably the true form of the goedendag is that of the poem and fresco, with or without side spikes. As to the etymology of the word itself, that is given in Guiart’s poem, where it says that it means “good day.”⁴⁵ The name doubtless took its rise from a brutal jest, as in the case of the holy-water sprinkler. The goedendag in the author’s possession has a staff seventy-five inches long, with a spike a little over seven inches at the end, and twelve short spikes dispersed in four rows round the head, projecting about one and a quarter inches from the staff, which bears the brand Z. I. In the Rotunda, Woolwich, are four similar goedendags, classed in the catalogues as “morgensterns” or “holy-water sprinklers”!

THE BILL, SCYTHER KNIFE, AND GLAIVE.

This class of weapons is often confounded with the gisarme, because they sometimes have a spur at the base. All have their prototype in the scythe of agriculture.

The bill occurs in the poem of Beowulf as part of the armament of a ship of war, and it is often mentioned in Anglo-Saxon chronicles, but it must be borne in mind that old chronicles used the phrase “bills and bows” in the sense that the former word applies generally to all long-shafted weapons. According to Silver, the bill ought not to exceed six feet in length.

Bills were in general use by footmen in the eleventh century, and indeed continued to be so until the advent of the pike. This class of weapons was largely superseded in the fifteenth century by halbards, partizans, and pikes, but the bill survived long in England. There are some particulars of this weapon in the *Brief Discourse on Warre*, written by Sir Roger Williams in 1590, in which the proper proportion of bills to pikes in battle-array is set forth as one to five. The length of the bill-shaft should not exceed six feet.

The glaive has a much larger blade than the bill. It has its edge on the outside curve, and has side branches of various sizes. The term “glaive” was often applied to the lance, and in France “le fer de glaive” denoted the sword of chivalry, as well as the headman’s blade.

The pageant glaive is a large, heavy, and usually highly decorated weapon, doubtless greatly used in processions.

THE HOLY-WATER SPRINKLER, OR MILITARY FLAIL.

This class of weapon, like several others, had its inception among the implements of husbandry; and it owes its name, like the goedendag, doubtless to a brutal jest. It is stated by Whitacre that the agricultural flail was introduced into Italy about the time of the Roman conquest of Britain. The Anglo-Saxons called it “Therscol,” or thrasher. This terrible weapon consists of a shaft of wood, garnished with iron, attached to which is a flail of iron, moving on a ring; or a chain or chains connecting the head of the shaft with a wooden or iron ball or balls at the extremity. The balls are

usually garnished with iron spikes, but this is not always the case. The holy-water sprinkler is often confounded with the “morning star,” which is a spiked mace, described under that heading.

It would appear from the Tower Survey of 1547, that the “Holy Water Sprinkler” was at that time in two varieties, viz., with long and short shafts. The above record catalogues “Holly Water Sprincles with gonnes in th’ ende. Little holly water sprincles.” Perhaps what was called the long variety was the goedendag. The author has two with short shafts, and chains at the ends, to which are attached spiked wooden balls. The MS. of Matthew Paris at Benet College, Cambridge, furnishes us with an example of the simple form.

THE MORNING STAR.

This weapon is a spiked mace, and was greatly used in Germany and Switzerland. There are both long and short shafted kinds; the latter, made of iron, is mentioned in the eleventh century, and was much used by horsemen in the fourteenth and fifteenth centuries. They were sometimes supplemented with hand-guns. This variety was called “Schiesspringel.” Several writers confound the “Morning Star” with the “Holy Water Sprinkler,” but the latter is a weapon of the flail family. The heads vary in shape, being round, square, and a half oval narrowing towards the shaft, and all are spiked.

GISARME AND VOULGES.

The gisarme is a scythe-shaped weapon, fixed on a long shaft. It is double-edged, and provided with a hook and spurs. It is often mentioned in early chronicles of the thirteenth century, and is specially alluded to by Froissart in the next century. The voulge has a broad blade, pointed at the head, and is generally square at the edge. It was usually forged with two strong iron rings, through which the head of a pole is passed. This weapon was often carried by archers. The pageant voulge is shaped very like a

Lochaber axe, with its curved, pointed, hook-like spear at the head of the shaft.

MILITARY FORKS.

These forked, trident-like weapons, of prongs of unequal length, are mentioned in records of the eleventh century. They were much used in the fourteenth century. The weapon appears in the Sloane MS., No. 346.

THE HALBARD.

The first mention of this weapon occurs in the fourteenth century. It was used by footmen only, and is somewhat varied in form. It usually has a somewhat square or crescent-shaped blade, with a sharp hook-like projection or forks on the back, and sometimes a spike from the face, but always a spear at the top. In the fifteenth century the nearly straight form prevailed, with a spur behind, while the crescent-shaped blade appeared early in the sixteenth; and the hinder spur became broader and more blade-like, and with a downward curve, while the spear at the point became much longer.

Double-bladed halbards were not uncommon.

The length at the end of the sixteenth century was about five feet, and being shorter than the pike was better adapted for hand-to-hand fighting. Silver says the length ought not to exceed five or six feet.

The halbardiers had charge of the standard.

The halbard and the partizan were the great infantry weapons before the pike came into general use. They were still to the fore in the reign of George I.

The pageant halbard is usually perforated, engraved, and otherwise ornamented.

Hewitt gives a figure, from Holbein's "Costumes Suisses," of a Swiss halbardier of the first half of the sixteenth century.

THE PIKE, PARTIZAN, SPETUM, RANSEUR, AND SPONTOON.

The pike is a footman's weapon used greatly in conjunction with the halbard and harquebus; and these three were pre-eminently the weapons of the infantry of the later "middle ages" and the "renaissance."

It was probably introduced into England in the reign of Edward III., being mentioned by Froissart, anno 1342, and did not fall into disuse much before the time of Charles II., when a writer in 1703 refers to it as a weapon "formerly" in use, the bayonet having superseded it. Viscount Dillon states in *Archæologia*, vol. li., p. 221, that "In 1515, Pasqualigo, the Venetian, writes that he had seen in the Tower pikes for 40,000 infantry, and that they have a like store at Calais, a place near Scotland!" The pike has a narrow lance-formed head, to which long strips of iron four feet in length are attached, which are screwed down the sides of a long wooden pole, the end of which is shod with iron, for fixing into the ground, to resist a charge of horsemen. There is a tassel along the shaft for easing the shoulder when the weapon is carried at the "port," and also for preventing the rain from running down the shaft.

The earlier length of the pike was ten feet, but Sutcliffe, in his *Practice of Arms*, speaks of it as up to twenty-two feet in length. A statute of 1662 fixes the length at sixteen feet. During Elizabeth's reign the cost of a pike was three shillings and eightpence, and it was "fifteene foote long besides the head." The usual length, however, was about ten feet.

It was the bayonet that deposed the pike.

The partizan, like the pike, was introduced in the reign of Edward III. The blade is long, broad, and double-edged, with hatchet-like or pointed branches at the base. It was greatly used as a pageant weapon, and much skill and taste were expended in chasing it and inlaying it with gold. The spetum is narrower and lighter, a long spear at the point, and narrow curved side branches.

The ranseur is very similar to the partizan, with a long broad blade in the centre, and projecting shorter blades on each side. It was much used in the reign of Edward IV.

The spontoon is a half pike, or something between the pike and partizan, and was carried by infantry officers.

A selection of staff and club weapons are represented in [Fig. 49](#), and most of the weapons referred to are there given.



PART XXIV.

EARLY ARTILLERY.

It is stated that some sort of cannon was known to the Moors very early, and that artillery was used in Spain during the second half of the thirteenth century in the defence of fortified places; but this is believed to be merely traditional, and that the piece of ordnance stated to be mentioned in the Archives of Ghent⁴⁶ as being in possession of that town in 1313, was probably a very rough weapon and highly tentative in character. Without wishing to cast doubt on this statement, occurring in a work published in 1843, we may remark that frequent efforts have since been made to find the passage, but without success.

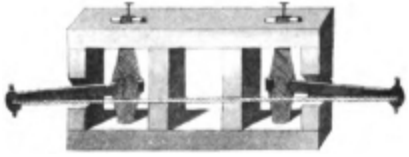


Fig. 48.—Principle of the
Ballista.

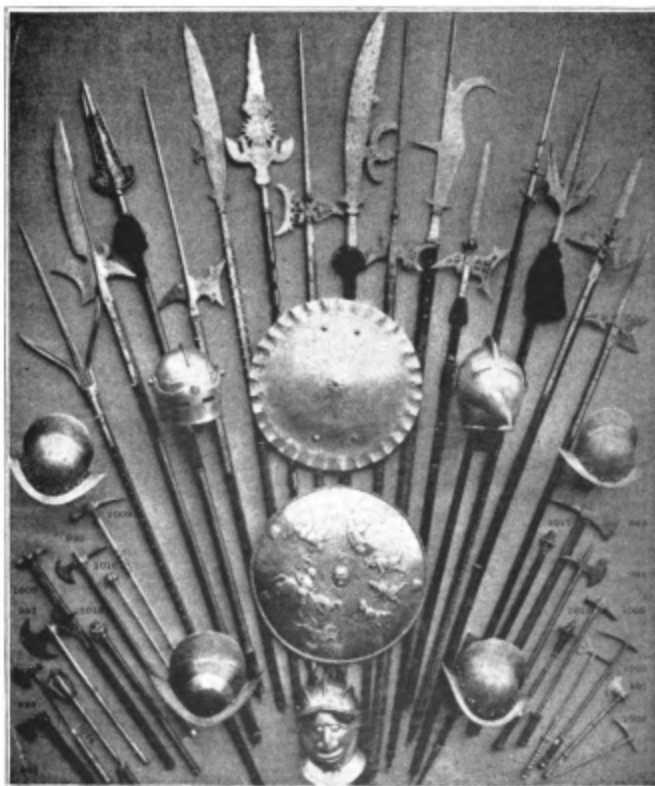


Fig. 49.—Staff and Club Weapons, etc.

The earliest firearms were only adapted for throwing fire into fortified places by means of a hollow tube, such as those described by the Princess Anna Comnena in the *Alexiad*, “tubes fixed to the prows of the Emperor’s galleys for throwing Greek fire,” and cannon discharging missiles by the agency of detonating gunpowder were probably not invented before the fourteenth century. All guns made in this century were of the crudest description, fastened on to blocks of wood, and were of wrought iron, loaded at the breech, and used principally in sieges.

There is frequent mention of firearms in German and Italian “chronicles” late in the first half of the fourteenth century, but these references are invariably characterised by extreme vagueness. Froissart frequently alludes to cannon, and says that these weapons were used by the besieged at Cambray in 1339;⁴⁷ his remarks concerning them are quite casual, and convey the impression that he attached very little importance to them. A French MS. of about 1338, in the Republican Library at Paris, mentions ordnance. This occurs in an account of the war treasurers, “To Henri de Vaumechon for buying powder and other necessities for cannon;” and a year later reference is made to cannon in the Archives of Bruges, “niewen enginen di men heet ribaude.” The statement of Villani, so often repeated, that artillery was in operation at the battle of Crécy, in 1346, is open to very considerable question, as it is tolerably certain that there were no field-pieces so early, or indeed any cannon whatever that could be moved about to any useful purpose in a battle. Froissart makes no mention of any used in campaigning; but he refers to a bombard at the siege of Oudenarde, “the noise of its discharge could be heard five leagues away,” and he also states that bombards and cannon were in operation at the siege of Quesnoy in 1340—“Those of Quesnoy let them hear their cannon,” when huge bolts were used as missiles; and that artillery was in use at the siege of Vannes, both by the besieged and the attacking English.⁴⁸ What gave rise to the tradition, if it be one, is probably the fact that Edward III. had established an ordnance factory, for siege guns, two years before the battle. Artillery of this date was quite unsuitable for field operations, and was only employed with other engines, as these examples show, in the reduction of fortified places. Demmin gives a drawing of a breech-loading cannon, open at both ends, strengthened by iron coils, which he states came from the field of Crécy, but we know not on what authority. This weapon was of forged

iron, like all the earlier ordnance. Grose, in his *History of the English Army*,⁴⁹ cites a MS., which has already been referred to in these pages, giving the force constituting the English army in Normandy and before Calais, in the twentieth year of the reign of Edward III., in which items appear for payments to gunners and artillerymen; but it would seem that their duty consisted in serving siege guns before Calais. Still, why should there be mention of what would appear to be two classes of gunners?

There was a gun foundry in France in 1346, Germany in 1378, in Switzerland in 1371. The first mention of any guns cast in England was, we believe, in 1521, when, according to Stone, brass cannon were first “cast” there; the founder’s name was Hugget, of Uckfield, in Sussex, and there are some specimens of about this date at Woolwich. Early cannon were fired by a live coal; later, by a slow match. There is nothing to indicate the date of the wooden cannon strengthened with iron coils, brought from Cochin China, and now in the Musée des Invalides at Paris. There is a mortar in the arsenal at Vienna, made in several layers of coiled hempen rope, with an outside covering of leather, which is said to have been captured from the Turks. There are also mortars made of paper, covered with leather, in the arsenal at Malta, but without any reliable record concerning their origin—doubtless they also came from the East. In Johnes’s version of *Froissart*, vol. ii., p. 252, is an account of a sea-fight between the English and Spanish fleets off Calais, King Edward commanding in person. It is there stated that the Spanish ships were amply provided with artillery, and a later passage specially mentions “cannon,”—this was probably the year after the battle of Créçy;⁵⁰ but in 1340 these weapons are referred to in connection with the naval battle of Sluys.

In 1372 some of the French ships undoubtedly carried ordnance at the battle of Rhodes; and the Venetians used bombards a few years later at the battle before Chioggia, when some of the guns burst on the first discharge; one of these weapons, which is made of leather, is still preserved at the Vienna arsenal. Leathern cannon were also used at the siege of Hohensalzburg in 1525, and by Gustavus Adolphus in 1631. We may take it that some time before this both artillery and hand-guns were regularly used in battle, but side by side with catapultæ and other engines of war, thus clearly showing that they were at this time largely experimental. They were still but sparingly found at sea in the middle of the fifteenth century, when

an English war vessel sometimes carried only one gun, and the largest ships never more than eight; and each piece of ordnance was then only provided with thirty rounds of ammunition for a month's cruise. After this time, however, the progress was rapid, and some of the Mediterranean galleys of late in the sixteenth century were armed with as many as two hundred guns. In 1377, Thomas Norbury was directed by King Richard II. to provide "two great and two less engines called cannon," to be sent to the castle of Bristol. The first reliable mention of field guns is on the occasion of a battle between the forces of Bruges and Ghent in 1382.

The first piece of ordnance was probably a mortar, the earliest form of which was a hollow tube, like an inverted cone, the butt-end being blocked with wood—they were short pieces of large bore.

The earliest artillery was breech-loading and called bombards, and some of these, towards the end of the century (the fourteenth), were capable of throwing two hundredweight shot, describing a parabolic curve of a radius of only three hundred yards, showing that the powder must have been very weak. In 1388, a stone shot, weighing 195 pounds, was discharged from a bombard called the "Trevisan."⁵¹ Drawings of these engines may be seen in MSS. 851 and 852 in the Nat. Lib., Paris. One is on a flat wooden stand, the other on a low platform with small solid wheels. [Fig. 50](#) exhibits one of these weapons. These guns, at first without trunnions, were made of bars of wrought iron, in overlapping coils or sections, welded together on a mandrel, and then hooped—in fact, similar in principle to the "Armstrong" gun. There is a breech-block in which the charge was previously laid, and fitted into the body of the piece by means of a wedge, but no apparent arrangement for sustaining the recoil. The Scottish cannon, "Mons Meg," is forged in this fashion, and a rent near the breech is instructive in laying bare the system of construction. It is of fifteenth century date, and is said to have been wrought at Mons in Flanders, but there is no evidence of this being the case—indeed, it was probably made in Scotland about the middle of the century. The calibre is 20 inches, and length 13 feet 6 inches. The projectiles used were stone shot, weighing 330 lb. The powder-chamber is less in diameter than the barrel.

Culverins were long pieces, whose projectiles were usually of lead.

Bronze bombards were made by Aran of Augsburg as early as 1378; but it was considerably later before these pieces began to be cast in iron. A very early iron specimen may be seen in the Rotunda collection at Woolwich.

Breech-loading cannon were pieces of small calibre, and were followed by those constructed on the movable chamber system, and after that by muzzle-loaders. There is an interesting piece preserved at the Artillery Museum, St. Petersburg, dating from the end of the fourteenth or beginning of the fifteenth century; it is strengthened with coils: also some good fifteenth century specimens. To judge from the quantity of old arms of all sorts found in Belgium, that country must have been as much the cockpit of Europe during the middle ages as it was in much more recent times. At the Porte de Hal Museum at Brussels are pieces of artillery of the fifteenth century, including some very early examples of considerable interest, and among these is a breech-loading cannon, mounted on a carriage with wooden wheels which are encircled by studded iron hoops. The weapon is of wrought iron, clasped round with thick iron coils—length, 0.74. There is another of similar construction and date—calibre, 0.135; length, 0.77. The carriages have been reconstructed. A bombardelle, the calibre of which is 0.13, and length, 1.30.

The muzzle-loading crapeaudeau of the first half of the fifteenth century is a small iron tube, mounted in a thick piece of wood, which stands on a small square block, with side handles for transportation—calibre, 32 mm.; it is a model executed from an old MS. A small culverin, the progenitor of the early petronel and later blunderbuss—length with mount, 1.80; barrel, 1.15; calibre, 25 mm. A breech-loading culverin of the first half of the fifteenth century—calibre, 0.065; length, 1.97. This weapon was found at Luxemburg during the demolition of part of the ramparts; it has a ring for hoisting.

There is a serpent in forged on the “Mons Meg” principle, the carriage of which is constructed from an ancient MS. ([Fig. 50](#)). A ship falconet ([Fig. 50](#)), early sixteenth century, breech-loader; turning on a pivot—calibre, 0.035; length, 1.31. The collection of early ordnance at the Königl. Zeughaus at Berlin contains some interesting specimens. Among them is an example of the short early bombard, dating from the close of the fourteenth century; and a long serpent cannon, shooting a projectile of two and a half

pounds weight, of the year 1419 (these two weapons have been constructed after contemporary drawings); two cannon, eighty-pounders; a seven-pounder bombard used by Charles the Bold, and taken by the Swiss at the battle of Nancy. There are also many others similar in character to specimens described in these pages. An interesting series of drawings of late fifteenth century artillery exists in the ordnance books of the Emperor Maximilian I., where you have examples of the bombard, serpent, snakes, falconets, mortar, and orgue. The lighter guns are mounted on rude carriages, with heavy wooden wheels encircled with iron-hooping.

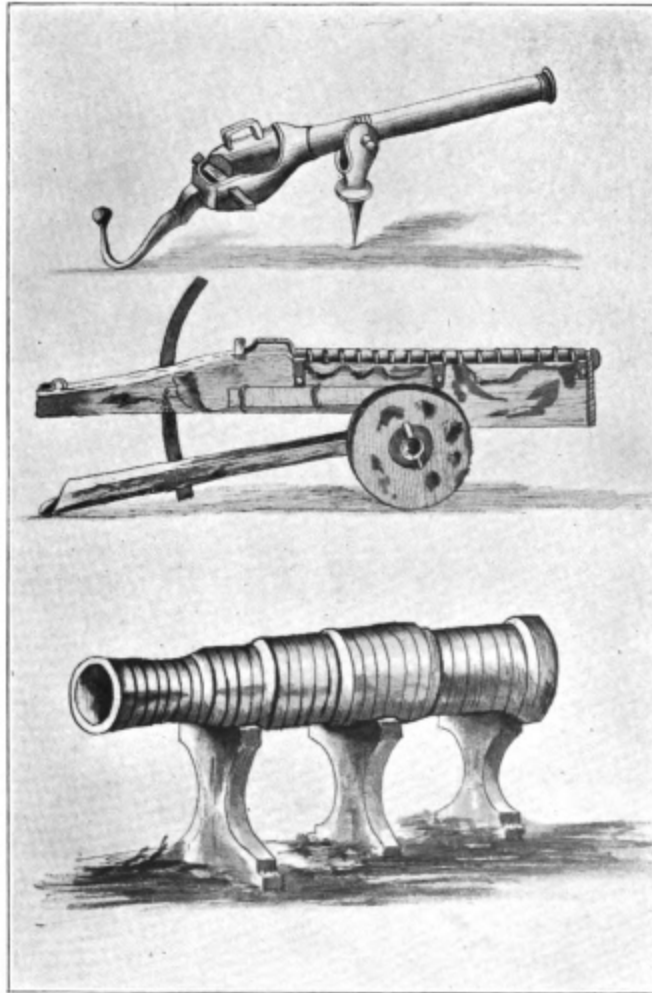


Fig. 50.—Early Artillery.

The elbow bombard, used in Italy early in the fifteenth century, was a tube fixed at right angles on to a carriage—the angle was capable of manipulation by a prop, and the breech-block is inserted in the side.

The orgue, the prototype of the modern mitrailleuse, was invented early in the fifteenth century—examples are mentioned with as many as thirty and forty barrels, and even more. There is an early specimen in the museum at Sigmaringen; and one dating from the beginning of the sixteenth century, with forty barrels, in the Imperial collection at Vienna. Another with five barrels, dating from about the end of the fifteenth century, and one a century later with sixty-four barrels; both in the collection of the Königl. Zeughaus at Berlin. A breech-loading gun of the fifteenth century may be seen in [Fig. 50](#).

The connecting link between artillery and hand-guns has been mentioned in an example at the Porte de Hal Museum, Brussels, and there are many other specimens there, called bâton à feu. Among them is a harquebus-mitrailleuse; this weapon, which is only twenty-five inches long, has nine barrels, moves on a pivot, and is fired by a wheel-lock.

The transport of the heavy and cumbrous guns of the fourteenth century was found to be attended with so much difficulty and expense that lighter cannon were introduced in the century following for field use, and rude carriages on wheels drawn by oxen were added. The bombard thus mounted was called “cerbotana ambulatoria.” Gun carriages were vastly improved during the reign of Henry VIII., when horses were employed to draw them. Means of sighting and convenience for trajectory had to be thought of, and trunnions were invented towards the middle of the fifteenth century. There was another contrivance for raising and depressing by means of a long thin prolongation, a sort of tail in fact, attached to the piece behind, and a fork was sometimes used for holding up the breech. There is a specimen with this adjustment at the Musée d’Artillerie at Paris, with an inscription bearing the date 1490. Projectiles of iron did not become common until a little later, but there was nothing specially new in a metal projectile, for such had long been used for early war engines, throwing balls both cold and hot.

The English army before Orleans in 1428 had a train of fifteen breech-loading mortars. Valturio, an Italian, writing in 1472 describes the engines

of war then in use, including cannon.

Specimens of ancient ordnance are not very numerous in England. There is a very interesting wrought-iron bombard in the collection at the Rotunda, Woolwich, dating from the commencement of the fifteenth century, or possibly somewhat earlier. It is lined with cast-iron,⁵² has a calibre of 15.1 inches; interior diameter of chamber, 14 inches; capacity of chamber, about 3.5 lb.; length of chase, 34 inches; present weight, 6 cwts. Also a wrought-iron cannon of about the same date—length, 24 inches; original calibre about 2 inches, without trunnions or cascabel, but provided with a couple of rings for transportation.

Double cannon, strengthened with coils, were common at this period, with the breech in the centre, and barrels running in two opposite directions. There are specimens at Woolwich, and at the Porte de Hal Museum, Brussels. There are several wrought-iron pieces at Woolwich, of the reign of Henry VI., and among them a serpent gun 8 feet 6 inches long, without trunnions, but provided with two rings for lifting—calibre, 4.25 inches; weight, about 9 cwts. A wrought-iron breech-loading gun with carriage was recovered from the wreck of the *Mary Rose*, sunk off Spithead in 1545, which is now at Woolwich; original calibre about 8 inches; the gun is a tube 9 feet 8 inches long, strengthened by a succession of heavy hoops, and is fixed by iron bolts to a beam of wood. The breech-block being removed for loading and charge inserted, the block is replaced and wedged, and the recoil was sustained by an upright piece of wood. There is no arrangement visible for raising or lowering the gun for taking aim. Similar guns may be seen at the Tower.

During the early days of artillery guns were constantly taken and retaken in battle after a first discharge, the process of reloading being so protracted that cavalry, or even infantry, were upon them long before the operation could be completed.

The fourteenth or early fifteenth century bombardier was clad in chain-mail, when stone shot was fired. He ignited his charge with a hot iron, guarding his face with his left hand from the sparks thrown off by the old-fashioned powder.

During the fifteenth century cannon were usually entrusted to the care of foreign mercenaries who were better disciplined than mere feudal or

communal levies, and much less liable to panic. John Jedd was appointed Master of Ordnance in England, 1483, and the office was not abolished before 1852. Hand-grenades appear in 1536. Each gun was known by a special name, of which “Mons Meg” is a familiar example. The general estimation of the use of cannon in campaigning was for long discredited by reason of the manifold imperfections of the weapons, the frequency of their capture by the enemy, and the dangers attending their discharge; they were for long employed simultaneously with the more ancient projectile engines, and the latter were preferred by many commanders to the former; but the dawn of the sixteenth century saw such manifest improvements that artillery then began to take the first place among projectile weapons. The petard was an invention of the Flemings in the sixteenth century.

Ordnance of the sixteenth century varies very much in size, cannon throwing a projectile of from thirty to forty pounds; culverins, bastard-culverins, falcons, falconets, and many other varieties discharging balls from sixteen pounds down to a single pound.

Mortars were greatly used in the middle of the sixteenth century, and howitzers for throwing hollow balls a little later.

Gunpowder first became granulated during the second half of the fifteenth century, up to which time the powder was of a fine dust, and divided from the stone projectile by a wooden wad. There were coarse and fine granulations made for charging and priming respectively. That made in the seventeenth century had become much more powerful, and a proportionate amount of metal had to be allowed in the construction of cannon. Mr. John Hewitt quotes the author of *Pallas Armata*, which states “that a culverin that shot 16 pounds of iron had but a hundred pound of metal allowed for every pound of her shot, and so she weighed then but 1,600 pounds; but now and long before this she weighs 4,300 pounds, and consequently hath the allowance of near 270 pounds of metal for every pound of shot.”

All the gunlocks we are accustomed to associate with hand-guns were used with ordnance; they were fixed to the vent-field by pins passing laterally through it, or by side screws.

The first mention of bombs occurs in 1588.

Artillery had now become an important and independent arm in all campaigning, and it will be seen how numerous cannon had become when it is stated that the train of guns attached to the army of the Emperor Ferdinand in 1556 consisted of fifty-four heavy and one hundred and twenty-seven light pieces of artillery.

Rifled cannon, the principle of which was first applied to hand-arms in Germany, were introduced in this century; examples of which may be seen in the arsenal at Berlin, and in the museums of Nuremberg and the Hague.

Viscount Dillon, P.S.A., writing in *Archæologia*, vol. li., quotes Lord Herbert of Cherbury, who consulted the records for the compilation of his history of the reign of Henry VIII. Lord Herbert writes that “great brass ordnance, as cannons and culverins, were first cast in England by one John Owen in 1535; and that about 1544 iron pieces and grenades were first cast.” Viscount Dillon remarks “that the facts as to time and place seem to be different, for in September 1516 there occurs a payment of £33 6s. 8d. to John Rutter of London, for “hurts and damages by him sustained in a tenement to him belonging wherein the king’s great gun called the ‘Basiliscus’ was cast, and for rent.” In 1532 Carlo Capello, the Venetian, writes that Henry “visited the Tower daily to hasten the works then going on there, and was founding cannon and heavy gunpowder made.” This was in anticipation of the Scottish war.

A valuable account of the guns in the Tower, numbering 64 of brass and 351 of iron, of which follow some abridged extracts, may be seen in some notes by Viscount Dillon, appearing in *Archæologia*, vol. li., pp. 223–225. He states “that there are two bronze guns, octagonal externally, with bores $2\frac{1}{2}$ and $2\frac{3}{4}$ inches, corresponding in form with types of 1500–1530, presumably of Venetian make. The ‘Brode Fawcon, shooting iij shotte,’ is rectangular externally, has three bores side by side, and the three spaces for placing the three chambers, as in early breech-loading cannon. The ‘French gonnes of Brasse’ may have been part of the spoils of Boulogne in 1554, or else the work of the same Peter Bawde who cast brass guns for King Henry at Houndsditch as early as 1525.” His lordship is of opinion that the seventeen “Scottishe gonnes of Brasse” would include some of the pieces taken at Flodden, which, according to Hall, consisted of “5 great curtalls, 2 great culverynges, 4 sacres (hawks), and 5 serpentynes, etc.” Viscount Dillon mentions in his notes that the Scotch made cannon in 1460, and that

the iron guns in the Tower comprise eleven of the numerous varieties in use in Henry VIII.'s time, and he gives the names of makers of that period, both English and foreign. These notes, of which this is but a very imperfect outline, should be read *in extenso* by all specially interested in the subject.

PART XXV.

EARLY HAND-GUNS.

THE invention, or at all events the first application of these weapons for the purposes of warfare, in the sense of the use of detonating gunpowder for the discharge of projectiles, in contradistinction to those applied merely for setting fire to buildings, is probably due to the Flemings or Italians, but the approximate date of their introduction is very difficult to trace, as early writers on the subject so often confound hand-guns with cannon, and *vice versâ*; besides, some of the earlier guns were innocent of any projectile whatever, being simply used for frightening horses, an office at that time far from being contemptible in repelling an onset of men-at-arms. The earliest mention of hand-guns occurs in connection with Perugia as early as 1364,⁵³ and an inventory of Nuremberg, of 1388, refers to forty-eight of these weapons as being in the possession of that city. There are other examples of the use of what would appear to be hand-guns occurring in Italian, French, and German manuscripts of the last quarter of the century, but it is rarely absolutely clear whether artillery or hand-guns are meant, especially when the word “bombard” or “bombarde” is used, unless, as in the case of Perugia, where the dimensions are given. In German MSS. the use of the word “handbüchsen” is, of course, conclusive; and such a case occurs in connection with Ratisbon in 1379. These early “handbüchsen” or “handbombards” could not be very heavy, as there exist several “illuminations” at Vienna, where one of the two gunners who served the piece holds the weapon with his right hand, with the round thin stock against his breast; his colleague stands apart with the ramrod in his hand, apparently after having loaded the piece. One of these “illuminations” shows that the charge is being ignited near the mouth of the piece, which might go to show that the gun was innocent of projectile. These pictures would seem to date very early, probably not later than 1350–60. Juvenal des Ursins mentions a hand-gun as being in use in 1414. A Florentine writer states that these weapons were used at the siege of Lucca, in 1430; and what is still more to the

point is that an actual and early specimen, made of brass, was found among the *débris* at Tannenberg, a castle besieged and demolished in 1399: this weapon was probably of as early make as the Nuremberg guns. It was only with great difficulty that the early rough hand-guns made their way at all against those weapons where manual or mechanical force was used. Both the longbow and crossbow were infinitely superior to the clumsy tube stuck on to the end of a stick, not only in regard to precision of aim, but also in the number of missiles that could be discharged within a given time, and it was principally on this account that these firearms are so rarely mentioned by mediæval writers. Actual specimens preserved are few and far between, and this is not surprising when one considers how very soon the weapons became obsolete in the rapid improvements that took place.

There is a connecting link between early artillery and hand-guns in various weapons from the small elementary semi-portable cannon fixed to the end of a long wooden shaft, and fired from a forked support or from a wall; and later, large models of guns of the *harquebus* type manipulated in the same manner. The latter form was the “*arquebus à croc*,” weighing up to sixty pounds, and was from five to six feet long. This class of weapon was much used in sieges, and they were sufficiently portable to be carried and worked by three or four men. Most national collections contain specimens of these firearms.

Mr. John Hewitt figures an early hand-gun, taken from the Burney MS., which is simply a replica of the weapon found at Tannenberg. Hand-cannon were being made at Augsburg in 1381. An early weapon of this kind is figured on a piece of tapestry in the church of Notre-Dame de Nantilly, Saumur. The piece is served by two soldiers, one holding it with both hands, while his comrade applies a hot coal. The form of the visored *bassinet* worn by these soldiers would fix the date as being late in the fourteenth century, and actual specimens of this time may be seen at the Historische Museum at Berne, and at the Germanische National Museum at Nuremberg.

In the collection at the Königl. Zeughaus, Berlin, is a hand-gun dating from late in the fourteenth century or early in the fifteenth, consisting of a stock and barrel. The former is rudely cut for the shoulder, like the butt of a crossbow, while the latter is a tube between three and four feet in length, with a touch-hole on the right side; calibre, 16mm. Some drawings of about 1430 in the Hauslab Library show similar pieces. This weapon is to all intents and purposes the prototype of the modern hand-gun, and is, in fact, a very early form of “*Hakenbüchse*,” one of the many names for the *harquebus*.

Late in the fourteenth, or early in the following century, hand-guns like small culverins, with a touch-hole on the right side, were in use and discharged from the shoulder. The weapon was fired by applying a match to the touch-hole, and the soldier had to find his way to it while he took aim. Like the Berlin example, this class of weapon was rudely fashioned to the shoulder. The hand-cannon consists of a small bombard fixed to a wooden shaft, and fired by means of a match. The following items occur in a roll of purchases of the Castle of Holy Island, in Northumberland, for the year 1446:—

“Bought ij hand-gunnes de ere iiijs.
Item, gonepowder iiijs.”

Demmin gives a drawing from a manuscript dated 1472, and Herr Wendelin Boeheim that of a petronel (*poitrine*, the chest), a kind of hand-bombard, fired by a horseman from a forked rest fixed on to the saddle. The author has a specimen of this kind of support in his possession, which is hollow, and combines a long dagger screwed in at the top; but this accessory points to a rather later period than that of the hand-gun in question. It is an early form of linstock. The hand-gun when not in use hung suspended from the rider's neck; it was attached by a ring to a necklace, and fired from the breast, and the left arm sustained the petronel, while the right hand manipulated the match-cord. The character of the armour on the figure would indicate a date in the second half of the century (the fifteenth), and the weapon is the prototype of the modern blunderbuss. The figure is taken from Victor Gay's work. A still earlier example, but very similar, appears in one of a series of “notes” of great ability and industry, by Major Sixl in the *Zeitschrift für historische Waffenkunde*, and the features of both correspond very closely. The hand-gun of the earlier example is provided with a “hac” or spur; the horse on which the gunner is seated is unbarded, excepting for a crinet with a long spear springing from between the ears like a unicorn, while the horse of the later figure is barded, and the bassinet visored.

The first person of note that we hear of as having been killed by a hand-gun was the Earl of Shrewsbury at Châtillon in 1453.⁵⁴

The type of weapon used by a contingent of three hundred Flemings in the ranks of the army of Edward IV. in 1471 was the hand-culverin; and the English Yeomen of the Guard were armed with it in 1485, as also was the Swiss contingent of six thousand men at the battle of Morat in 1476. These

hand-culverins were each served by two men, one for holding the gun, and the other for applying the match, etc.; they were fired by a fuse-cord.

By the end of the fifteenth century the priming was held in a pan at the side of the barrel, and the pan was protected by a lid, which moved on a pivot. The next improvement was the attachment of the pan to the plate, and the stock was more bent. These weapons, the length and weight of which varied greatly, were in general use; the bore was usually about half an inch. Examples may be seen at the Musée des Invalides, Paris, and in many other national collections. A hand-gun of the harquebus type is figured in "The Triumph of Maximilian"; the stock is straight, and almost square. The figure bearing it wears a bandolier collar! A similar weapon, with a primitive form of serpentin, is figured in one of the books of Maximilian I., about 1500.

These early hand-guns were full of drawbacks and imperfections; an uncertain aim and form of ignition, whereby the weapons often missed fire; the long time required for loading; the cumbersome accessories, such as bullets, rest, and match; besides one granulation of powder for charging and another for priming, all combined to discredit the value of these weapons as against bills and bows; the effect of which was much more rapid in action. So much so was this the case, that owing to their dilatory habit both hand-guns and ordnance were frequently captured in battle after a first discharge, and their servers rendered *hors de combat*. They had practically nothing with which to defend themselves. The long dagger screwed into the butt of the rest was no match at all as against long-handled weapons, such as the gisarme, halbard, and bill. All this taxed the ingenuity of the time for the production of a surer and more reliable weapon with more simplicity of action. Here, as in the case of early crossbows, mechanical appliances came to the aid of the human arms and fingers, making the manipulation of hand firearms somewhat less cumbersome and dilatory.

The hakenbüchse, hagbut, hackbutt, hackenbuse, hequebutte, arquebus, and harquebus, are all names for the same kind of weapon, which is merely a development from the ruder forms, with a smaller calibre than the hand-culverin; but the great distinction generally observable between it and older forms is the presence of a pair of movable nippers called "serpentin," the prototype of the "cock," a primitive example of which has been already referred to. Hand-guns of this type, however, existed before the appearance of the serpentin; and the word "haken," with variations, as a matter of fact refers to the "hac or haken," which is a projecting spur of iron placed on the bottom

side of the stock, near the head; the object of which was to deaden the recoil by placing the spur against a stone rampart. There are many examples in the Königl. Historische Museum at Dresden. A very early instance of the use of the "hac" occurs on a hand-gun preserved at Berne, and there are drawings in the University Library at Heidelberg of several examples of the harquebus of the fourth quarter of the fifteenth century, with the "hac," but of course without serpentín. The oscillatory movement made in applying fire with the hand naturally caused the weapon to swerve, thus interfering greatly with the accuracy of aim; and at length the earliest form of lock called the serpentín was invented, the object of which was to let down the match mechanically. Thus we have the earliest form of matchlock, and the stock became shaped for the shoulder. Harquebuses with the serpentín gave victory to the Spaniards at the battle of Pavia. Philip de Commines mentions the weapon towards the end of the fifteenth century as a new invention.

The serpentín is adjusted on a pivot through the stock, and forms a lever for the fingers beyond it. Then, holding a match, it is brought into contact with a slow match in a holder on the barrel and ignited; then by raising the lever, it is forced into the flashpan and touch-hole, where the priming is placed, and the gun discharged. This movement is in three varieties: the earliest moves towards the pan from the stock, while later it was fixed in the opposite direction; in the third it is propelled by a snap. First manipulated by the hand, then with a lever, and afterwards by a crank in connection with the trigger. The idea of the serpentín goes back to the fourteenth century, for the Froissart preserved in the town library at Breslau shows a drawing of a hand-bombard with an elementary form of triggered serpentín; and the same adjustment occurs in representations of these primitive weapons on a drawing preserved in the Hofbibliothek at Vienna. The mainspring was a further simplification of procedure in lending a more direct action to the serpentín, which fell with greater force, and obviated the necessity for blowing on the match.

The harquebus was of several kinds and sizes, some fired from a rest, others from the shoulder or breast. There was also the heavy semi-portable weapon already referred to, served by three or four men; used both for field and fortress work. The length of the hand harquebus ranged from two and a half feet and upwards; barrels are both muzzle and breech-loaders; bores are of various sizes, sometimes very wide and bell-mouthed. The great disadvantage of the matchlock was the trouble and uncertainty experienced in retaining fire, and in it being necessary always to have a lighted match, or means of striking a light. This was especially felt in the chase, and the wheel-lock, which is said to

have been invented by Johann Kiefuss of Nuremberg, in 1517, provided a much needed improvement on the older method; there is, however, at least one earlier example of this lock with the date inscribed. It did not, however, displace the matchlock for war purposes, owing to the greater cheapness and simplicity of the latter, which continued in use up to the eighteenth century. There is an example of a regimental matchlock musket at the Rotunda, Woolwich, dating about 1700—barrel, 46 inches long; calibre, 0.540 inch; steel mounts. The main principle of the wheel-lock is to generate the spark which is to ignite the powder for firing the shot in a self-acting manner, in contradistinction to the principle of the matchlock, where the ignition was served by a match which required to be kept constantly burning.

The costliness of the wheel-lock, which was made in as many as ten separate pieces, greatly restricted its use as regards hand-guns, but it was applied generally to pistols, and pieces for the hunting field. Cavalry used weapons with this lock, as it was very inconvenient to manage the match-cord on horseback, especially as it required regulating with every shot fired. Ignition was accomplished by sparks which were caused by the friction of a steel wheel, notched long and crosswise, rubbing against a flint, or by the striking of the wheel against a cube of solid pyrites. The lock was wound up by a spanner, which hung at the soldier's belt. The main details of this lock are as follow, viz.:—A serrated wheel, connected to the backplate by a chain and spring, forming with the backplate the bottom of the flashpan, and wound up by a spanner. With the wheel-barrel is connected one end of a strong spring, by a chain, which winds round the barrel when the wheel is turned, tightening the spring until the catch of a bar drops into a corresponding notch of the wheel, thus holding spring and wheel cocked. After winding up, the trigger is pressed, releasing the wheel, which revolves round with great energy, by means of the accumulated force lent it by the winding, and coming into contact with the pyrites in the cock produces the sparks that ignite the priming in the flashpan trough, and fires the piece. Various improvements in the mechanism of this lock took place from time to time.

There are examples of wheel-lock weapons in the Tower of London dating from about the middle of the sixteenth century; a breech-loading harquebus, with a lock of something like the same date, is in the Musée d'Artillerie at Paris. A harquebus revolver, with seven barrels, may be seen in the Hohenzollern collection at Sigmaringen, and there are countless examples existing among the museums of Europe, and notably at Dresden.

During the sixteenth century, and especially in the later half, the footman wore half-armour, and usually discharged his weapon from a prop.

In a matchlock the match is lit at both ends.

The air-gun was invented in Germany in 1560. In this weapon the bellows are wound up against a spring, which is released by pulling the trigger; the receiver is in the stock, and filled by a pump.

The principle of rifling barrels was certainly applied as early as 1510, and there are very early examples of revolvers. There is one in the Tower of London with a matchlock, dating from about the middle of the sixteenth century. A patent for rifling barrels was taken out in London in 1635. It is said that the invention of grooved arms is due to Gaspard Kollner of Vienna, in 1498; other writers attribute it to August Kollner of Nuremberg, early in the sixteenth century; but whether the grooves were straight or spiral, or when they became the latter, is not so obvious; at all events, the principle was not much adopted for military arms before the seventeenth century.

The caliver is a harquebus or light musket of a standard calibre, introduced into England during Elizabeth's reign; it was four feet ten inches long, discharged without a rest, and the fire was much more rapid than that of its predecessors, and had the great advantage of uniformity of projectile. Edmund Yorke, writing in Queen Elizabeth's reign, says: "Before the battle of Mounghuntur, the princes of the religion caused several thousand harquebusses to be made, all of one 'calibre,' which was called 'Harquebuse de calibre de Monsieur le Prince.'"⁵⁵

Hand-grenades of the sixteenth century were made of very coarse glass, almost slag or pottery; they were nearly three and a half inches in diameter, holding from three to seven ounces of powder.

The snaphance was the immediate precursor of the flintlock, and was a German invention of the second half of the sixteenth century, fired through the medium of sulphurous pyrites. This lock forms the connecting link between the wheel-lock and flintlock, there being a hammer instead of a wheel; the pan is the same, but the cover was moved back by a spring, leaving the powder clear for the action of the sparks. A fine collection of these weapons may be seen in the Dresden Museum.

The method of extracting fire by means of flint and steel is an ancient one, being mentioned by both Virgil and Pliny. The credit of the invention of the

familiar flintlock is claimed by France, anno 1640, but an actual specimen in the Tower armoury, dated 1614, effectually disposes of this pretension. The French claim that the improvements of the screw-plate, “à miquet,” led to the mechanism of the flintlock; but it was long before the system displaced that of the old matchlock. The musketeer continued to carry his matchlock gun up to the beginning of the seventeenth century, and even later, while the flintlock continued in use until long after Waterloo; indeed, matchlock, wheel-lock, and flintlock weapons were all to the fore together for a part of the seventeenth century.

Wheel-lock pistols formed part of the equipment of the Reiters or Pistoliers of the second half of the sixteenth century. Hefner says pistols were common in Germany in 1512, before the invention of the wheel-lock. The pistol of the Reiters, who usually wore blackened demi-armour, are very easily recognisable by the round pommel.

The pistol was often combined with other weapons, both for battle and for the chase, and such combinations are often met with in the axe, mace, and even sword; while there are instances of pistols with two and even three locks. The introduction of these weapons produced great changes in warlike tactics. The etymology of the word is uncertain, some maintaining that the name arose from the weapon having been invented in Pistoja; others believe that the word originated from a coin of the time, the pistole, from the fact, if it be one, that the bore of the weapon had the same diameter as the coin.

Hand-guns of the later middle ages and the “renaissance” may be divided into plain weapons for the ordinary soldiers, and decorated guns for leaders and parade, besides hunting purposes. Brescia was a great centre for their manufacture. Numbers of these guns were fired without touching the shoulder, the recoil being provided for by placing the thumb firmly against the nose.

The musket (muchite, so named from the sparrow-hawk), which was longer and more powerful than the harquebus, though similar in construction and mechanism, appears in the third quarter of the sixteenth century, and St. Remy refers to it as being in use about the end of the seventeenth century. It was first fired from the breast, then from a long-forked rest, furnished with a spike at the end for sticking into the ground; but this fell into desuetude in the seventeenth century. It was found very difficult to keep the powder dry in the bandoliers,⁵⁶ which were cases of wood or tin, each containing a charge of

powder, and strung round the neck; and powder flasks began to be used about 1540, the bullet-bag being carried on the soldier's right hip.

Powder flasks appear very early in the sixteenth century, with the well-known arrangement for the measured charge; early examples are given in the arsenal books of Maximilian I. They were first very small, but gradually increased in size as the century wore on, mostly circular in form, but later they are often three-cornered, and frequently made of horn, wholly or in part. Cartridges superseded their use about the middle of the seventeenth century, and the bayonet is first mentioned about the same time.

Arrows or quarrels were often used as projectiles for the musket, but this happened mostly at sea.

The harquebusier of the seventeenth century carried a weapon two and a half feet long.

The carbine or caraben is a gun with a wide bore, first used in Queen Elizabeth's reign.

The decoration of many of the hand-guns of the sixteenth century was of a most artistic character, the barrels being often enriched with chasings, fine metal incrustations, or damascened, while the stocks were curiously and delicately carved and inlaid. It is generally assumed that the material usually used for inlaying is ivory, but it is really bleached stagshorn, and inlaying with tortoise-shell was also not uncommon.

A great amount of decorative skill was also expended on powder flasks.

There were several diminutives and combinations of the leading hand-guns referred to. Examples of early hand-guns are given in [Fig. 51](#).

* * * * *

It is well to furbish up bygone things and ages, and to remember now and then what we owe to cumulative history. Master Wace, the chronicler of the Norman Conquest, says in his retrospections: "All things hasten to decay; all fall; all perish; all come to an end. Man dieth, iron consumeth, wood decayeth, towers crumble, strong walls fall down, the rose withereth away, the war-horse waxeth feeble, gay trappings grow old, all the works of men perish. Thus we are taught that all die, both clerk and lay; and short would be the fame of any after death if their history did not endure by being written in the book of the clerk."

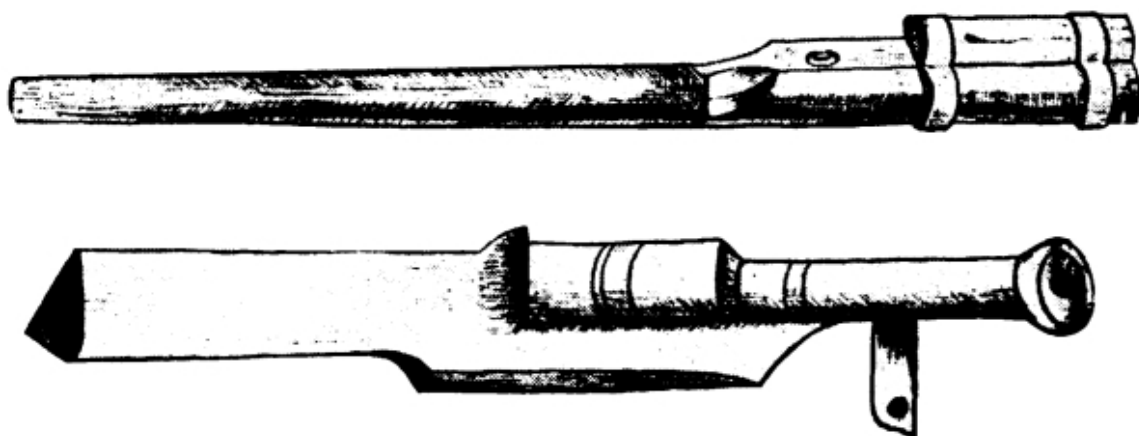


Fig. 51.—Early Hand-guns.

FOOTNOTES

¹ Dalstrom's *Illustreret Verdenshistorie*, vol. i., p. 122.

² A similar fragment was found at Cataractonium (see *Archæological Journal*, vol. iii., p. 296).

³ *Proc. Soc. Antiq. Newc.* (O.S.), p. 155.

⁴ In old German "Brunne."

⁵ *Zeitschrift für historische Waffenkunde*, vol. i., p. 288.

⁶ Where the rings have been hammered flat a decidedly double appearance is given to the mail.

⁷ Demmin.

⁸ These two figures are given in Hewitt.

⁹ Since writing the above, I see from Mr. J. Starkie Gardner's work that Mr. J. G. Waller, F.S.A., considers the insertion of the thong to constitute what is known as "banded mail," and this would quite account for the appearance it presents on effigies. If this be so, there is an actual specimen at Woolwich, which has already been mentioned.

¹⁰ On the brass of William de Aldeburgh, in Aldborough Church, Yorkshire.

¹¹ A kind of cloth.

- ¹² The words “helm” and “var-helm” appear repeatedly in the epic poem of Beowulf.
- ¹³ The first attempt at a movable visor seems to have occurred in France, during the reign of Louis le Gros.
- ¹⁴ This helm was given to Sir S. Rush Meyrick by the Dean: a flagrant instance of how such trust property was treated in his day.
- ¹⁵ The term “men-at-arms” was often applied to knights on foot or on horseback, but its early significance was heavy-armed infantry. The grades mentioned in the army of Philip Augustus were: bannerets, knights, squires, and “men-at-arms.”
- ¹⁶ “Notes on the Hanseatic League,” by the writer. *Proceedings of the Society of Antiquaries, Newcastle-upon-Tyne*, 1893–94.
- ¹⁷ These pieces will be referred to fully under the section “Maximilian Armour.”
- ¹⁸ Johnes’s *Froissart*, vol. iii., p. 23.
- ¹⁹ *Archæologia*, vol. li., p. 250.
- ²⁰ Nithard, the nephew of Charlemagne. Book III.
- ²¹ The illustration occurs in “Paul Lacroix.”
- ²² Vol. i., p. 169.
- ²³ The lance of the thirteenth century was always sharp, and, as mentioned more particularly earlier in this chapter, the coronal was a contrivance of the fourteenth century. The word “stechen” means to pierce, so its very designation carries the course back possibly to the thirteenth century.
- ²⁴ Very often the Grandguard and Volant-piece are screwed together.

- ²⁵ The piece usually called Pass-guard is the projecting guard over the shoulders for stopping pike-thrusts, but we have Viscount Dillon's authority that the Tilting Elbow-guard is really the Pass-guard.
- ²⁶ *Helmets and Mail*, p. 84.
- ²⁷ The Redmarshal effigy is in the County of Durham, and the Downes effigy is in the North Choir Aisle of Macclesfield Church, in Yorkshire.
- ²⁸ Hewitt.
- ²⁹ Like many classifications of the kind, this is rather arbitrary, as we have many late instances of "bear-paw" sollerets.
- ³⁰ Caylus figures a Roman caltrop (*Recueil* iv., Pl. 98).
- ³¹ The designation "Gothisch" (Gothic) seems as ridiculous and inappropriate when applied to armour as to architecture.
- ³² The mentonnière is throughout referred to as the combined piece of gorget and chin-piece as used with the sallad.
- ³³ A specimen of the work of this great artist may be seen on a sword-hilt in the Armeria Reale, Turin.
- ³⁴ In the early chronicles "bills and bows" are often mentioned. It must be borne in mind, however, that the word "bills" often covered all long-handled weapons.
- ³⁵ In the Republican Library at Paris, a MS. written by Marcus Græcus, called "Liber Ignium." It is dated 846.
- ³⁶ *Proceedings*, vol. v., p. 26.
- ³⁷ Demmin.
- ³⁸ Boenheim of Vienna says that he was born in 1530, and died about 1583.

- ³⁹ The father of Lucio, the great armour-smith.
- ⁴⁰ *Archæologia Æliana*, vol. xxii., pp. 1 *et seq.*
- ⁴¹ Greener on the Gun, p. 3.
- ⁴² This class of machine was termed “tormenta,” from the twisting of the ropes which supplied the propelling force.
- ⁴³ *Encyclopædia Britannica*, under “Fire.”
- ⁴⁴ Others in the author’s collection are stated to have been used by the Town Guard of Newcastle-upon-Tyne.
- ⁴⁵ See *Proceedings of Newcastle-upon-Tyne Society of Antiquaries*, vol. ix., in a paper by the writer.
- ⁴⁶ Rénard, Liège.
- ⁴⁷ Johnes’s *Froissart*, vol. i., p. 145.
- ⁴⁸ Johnes’s *Froissart*, vol. i., p. 190.
- ⁴⁹ Vol. i., p. 278.
- ⁵⁰ This occurs in a part taken from two MSS. in the Hafod Library, “not in any of the printed copies.”
- ⁵¹ Meyrick.
- ⁵² The rough projectiles of this period would doubtless soon cause damage to the interior of cannon, necessitating its frequent renewal.
- ⁵³ The archives run: “Il nostre comune di Perugia fece fare ... 500 bombarde, una spanne lunghe,” etc. General Kohler mentions this in his book.
- ⁵⁴ Hollinshed.
- ⁵⁵ Maitland’s *History of London*.

[56](#) Small cases for holding measured charges of powder ready for loading.

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Transcriber's Notes

Simple typographical errors were corrected.

Inconsistent hyphenation was not changed. Ambiguous hyphens at the ends of lines were removed only when other, mid-line examples were found in the text. One Index entry was changed to match the page it referenced.

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The index was not checked for proper alphabetization or correct page references.

Figure 43 is out-of-sequence, as indicated in the original List of Illustrations.

Page [80](#): “Beauté” was misprinted as “Beaulte”; changed here.

Page [123](#): “Grünewalt” was printed as “Gruenwalt” on this page, but as “Grünewalt” in the Index. Using the umlaut seems to be the correct spelling.

Footnotes originally were at the bottoms of pages, but in this eBook, they have been sequentially renumbered, collected, and placed just before the Index.

[Footnote 46](#), originally on page [205](#): “Liège” was misprinted as “Liége”; changed here.

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